applied sciences

MDPI

Review

Thin-Film Coating Technologies for Energy-Efficient Glazing: Materials, Deposition Systems, Methods of Analysis, and Functional Performance

Ana Tufescu [1], Corneliu Munteanu [1,2,*], Florin Brinza [3,*], Viorel Paleu [1], Daniela-Lucia Chicet [1], Bogdan Istrate [1] and Fabian-Cezar Lupu [1]

[1] Faculty of Mechanical Engineering, Gheorghe Asachi Technical University of Iasi, Blvd. Mangeron, No. 61, 700050 Iasi, Romania; ana.tufescu@academic.tuiasi.ro (A.T.); viorel.paleu@academic.tuiasi.ro (V.P.); daniela-lucia.chicet@academic.tuiasi.ro (D.-L.C.); bogdan.istrate@academic.tuiasi.ro (B.I.); fabian-cezar.lupu@academic.tuiasi.ro (F.-C.L.)
[2] Technical Sciences Academy of Romania, 26 Dacia Blvd., 030167 Bucharest, Romania
[3] Faculty of Physics, Alexandru Ioan Cuza University of Iasi, Blvd. Carol I, No. 11A, 700506 Iasi, Romania
* Correspondence: corneliu.munteanu@academic.tuiasi.ro (C.M.); fbrinza@uaic.ro (F.B.); Tel.: +40-744-647-991 (C.M.); +40-721-510-863 (F.B.)

Abstract

Low-emissivity (low-E) coatings are among the most effective thin-film technologies for reducing radiative heat losses and controlling solar heat gain in buildings, which account for approximately 30–40% of global primary energy consumption. This expanded review follows the technological evolution of low-E glazing from early transparent-conductor "heat mirrors" to modern multi-silver dielectric/metal/dielectric (D/M/D) architectures and emerging functional coatings. Four complementary perspectives are addressed: (i) the materials employed, from silver-based multilayers and transparent conducting oxides (ITO, FTO, AZO, GZO) to seed, blocker, and protective dielectric layers; (ii) the deposition systems, contrasting on-line pyrolytic/CVD "hard" coatings with off-line magnetron-sputtered "soft" coatings, together with ALD, sol–gel, and evaporation routes; (iii) the methods of analysis used to correlate microstructure, composition. and interfaces with optical, electrical, and thermal behaviour (XRD, XRR, SEM/TEM, AFM, XPS, SIMS, spectrophotometry, ellipsometry, emissivity, and U-value metrology according to EN 410/EN 673 and ISO 9050); and (iv) the functional performance of low-E stacks in insulating glass units, vacuum glazing, retrofit films, and smart-window systems across climate zones. Persistent research gaps are identified in long-term durability and ageing, indium-free scalable materials, standardized accelerated testing, and multi-objective design of thinner, more selective, and more robust stacks.

Keywords: low-E; emissivity; silver multilayer; transparent conducting oxides (TCO); thin-film characterization; EN 410; EN 673; solar heat gain coefficient; smart windows

Check for updates

Academic Editor: Andrea Li Bassi

Received: 22 July 2026
Revised: 9 August 2026
Accepted: 12 August 2026
Published: 17 August 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Windows are widely recognized as the thermally weakest components of the building envelope: buildings are responsible for roughly 30–40% of global primary energy use and a comparable share of energy-related CO_2 emissions, while windows may account for up to 40% of the heat dissipated through the envelope of highly glazed buildings [1–4]. The term "low-E" (low-emissivity) denotes coatings that reduce the emissivity of the glass surface in the thermal infrared (approximately 5–50 μm), thereby suppressing the radiative

component of heat exchange between indoor and outdoor environments. Uncoated float glass exhibits a hemispherical emissivity of about 0.84, whereas commercial low-E coatings reach values below 0.1 for pyrolytic products and as low as 0.02–0.05 for sputtered multi-silver stacks [1,5,6]. In practice, low-E coatings deliver their full benefit when integrated into insulating glass units (IGUs) or vacuum glazing, where the reduced emissivity lowers the radiative conductance of the gas or vacuum cavity and thus improves the overall thermal transmittance (U-value) of the glazing [6–8].

The performance of low-E glazing is conventionally reported through a small, consistent set of metrics: the U-value (thermal transmittance, calculated according to EN 673 [9] or measured in guarded hot boxes), the visible transmittance T_v (weighted according to EN 410/ISO 9050 [10,11]), the solar factor g (or solar heat gain coefficient, SHGC), the thermal-infrared emissivity ε, and derived indices such as the light-to-solar-gain ratio (LSG = T_v/g), for which spectrally selective products target values above 2 [12–14]. These quantities link the physics of the thin-film stack, such as the free-carrier (Drude-type) reflection in the infrared combined with interference-based antireflection in the visible, to the energy balance of the building.

Historically, the field emerged from the convergence of two ideas developed in the heat-mirror and transparent-conductor communities: surfaces with low thermal-IR emissivity, intended to suppress radiative exchange, and spectrally selective films that remain transparent in the visible while reflecting near- and thermal-infrared radiation [5,12,15]. Early work on conducting micro-grids and ultrathin noble-metal films demonstrated the feasibility of transparent IR reflectors, while the commercial introduction of low-E glazing at the end of the 1970s transformed the concept from a laboratory heat-regulation strategy into an architectural product category.

The 1980s and 1990s consolidated the scientific basis of the field. Chopra et al. systematized transparent conducting oxides (TCOs) based on non-stoichiometric or doped oxides of Sn, In, Cd, and Zn, establishing the electronic and optical framework required for transparent conductors [15]. At the same time, demand for large-area architectural coatings accelerated the development of high-rate sputtering and robust optical-stack design. By the late 1990s, industrial overviews of low-E coatings on architectural glass had already identified high-throughput magnetron sputtering, stable reactive processes, and chemically stable stacks as the decisive enabling technologies for large panes and insulating glass units [16].

Industrial development then crystallized into two complementary product families. Hard-coat products, typically fluorine-doped tin oxide (FTO) deposited pyrolytically on the float line at approximately 600 °C, gained market share because they are mechanically and chemically durable, can often withstand handling and heat treatment, and can be used in more exposed positions. Their limitations are higher emissivity and lower spectral selectivity than sputtered stacks. Soft-coat products, deposited off-line in vacuum, embed one or more ultrathin Ag layers in dielectric/metal/dielectric architectures; they deliver the lowest emissivity and the highest light-to-solar-gain ratios, but require careful protection inside IGUs or equivalent barrier systems [1,6,16,17].

A second historical shift occurred after 2000, when single-silver products evolved toward double-, triple-, and, in niche cases, quadruple-silver solar-control stacks. Heat-treatable double-silver coatings became widespread in the early 2000s, followed by triple-silver products around the mid-2000s, as façade design increasingly required high visible transmittance combined with low solar heat gain [1,6,18]. The literature of this period also shifted from purely optical selectivity to full-system performance, linking thin-film properties to U-value, SHGC/g-value, daylight availability, and climate-dependent energy demand [13,14,19–23].

Recent reviews mark a further expansion of the low-E paradigm. Since roughly 2010, the research agenda has incorporated vacuum glazing, smart windows, retrofit films, first-surface low-E concepts for automotive and polymer glazing, and durability studies focused on corrosion, Ag migration, white-dot defects, humidity resistance, and accelerated ageing [7,8,17,24–31]. Consequently, low-E coatings are no longer treated only as passive IR mirrors; they are now part of multifunctional fenestration systems that must balance spectral selectivity, durability, manufacturability, embodied impact, and occupant comfort over long service lives.

This review is organized along the four axes announced in its title. Section 2 discusses the materials used in low-E stacks and the deposition systems used to produce them. Section 3 surveys the methods of analysis, such as structural, compositional, optical, environmental, and thermal, that connect processing to performance. Section 4 examines the functional performance of low-E coatings in complete glazing systems across climates, and Section 5 outlines future trends. Table 1 summarizes the historical co-evolution of materials, coating types, deposition routes, characterization methods, and performance targets.

Table 1. Historical evolution of materials for low-E glazing.

Period	Dominant Materials	Coating Type	Deposition Methods	Analysis Methods	Targeted Performance
1976	Conductive micro-grids; thin Au/Ag/Cu films	Early "heat mirrors"	Micro-grid fabrication; thermal evaporation; early sputtering	UV–Vis/IR spectroscopy; sheet resistance	Transparent surfaces with high thermal-IR reflection
1979	Early low-E coatings on glass or polymer film	First commercial low-E glazing	Industrial on-line/off-line coating routes	Emissivity, U-value and solar-optical metrics	Beginning of market adoption; heat-loss reduction
1983	Doped/non-stoichiometric Sn, In, Cd, and Zn oxides; metals	TCO "heat mirrors"	Spray pyrolysis, CVD, evaporation, and sputtering	XRD, optical spectra, Hall/sheet resistance	Framework for visible transparency combined with IR reflection
1995	Ag/dielectric stacks; FTO and other TCOs	Large-area coated architectural glass	High-rate sputter coating; pyrolytic/CVD routes	Spectrophotometry, emissivity, U/g-value metrics	Scale-up driven by regulations and demand for coated façades
1997	Metal/dielectric low-E stacks	Industrial soft-coat D/M/D systems	Large-area high-rate magnetron sputtering; AC magnetrons	Optical/thermal coefficient and process-control metrology	Transparent, chemically stable stacks targeting k <= 1 W m^{-2} K^{-1}
2000	ITO, SnO_2:F and ZnO-based TCOs	TCO layers for low-E and electrochromic windows	Industrial thin-film routes, especially sputtering/CVD	Electrical and optical characterization	TCO platform for low-E glazing and smart-window electrodes
2007	Noble-metal films, n-doped oxides, and chromogenic layers	Spectrally selective transparent conductors	Sputtering, CVD, sol–gel, and related thin-film routes	Spectral/angular selectivity; electro-optical properties	Linking low-E, solar-control, vacuum glazing, and smart windows

Table 1. *Cont.*

Period	Dominant Materials	Coating Type	Deposition Methods	Analysis Methods	Targeted Performance
2008–2010	Low-E coatings for VIG; electro-/thermo-/gasochromic materials	Low-E integrated with vacuum and smart glazing	Sputtering/CVD for low-E layers; device fabrication routes	U-value, optical modulation, durability and reliability	Very low U-values plus dynamic daylight/solar-energy control
2012–2017	Ag stacks, TCOs, solar-control, and smart layers	Advanced fenestration and façade systems	Sputtering, CVD, retrofit films, and thin-film devices	Building simulation with optical/thermal testing	Climate-specific energy, daylight, and comfort optimization
2018	THR/THM materials; Ag, TCOs, VO_2, and dielectrics	Transparent heat-regulating/heat-mirror coatings	Scalable thin-film routes; interface-engineered stacks	Microstructure, interfaces and optical/thermal spectra	Microstructure and interfaces recognized as dominant performance factors
2022	Solar-control retrofit films and low-E glazing packages	Retrofit-oriented solar-control coatings	Applied films and coated-glazing packages	Field/simulation assessment; VLT, SHGC, and U-value	Retrofit energy savings and façade-level performance assessment
2024	Ag-based soft low-E stacks; blockers, barriers, and topcoats	Durable soft low-E and first-surface concepts	Magnetron sputtering combined with protective/barrier strategies	Corrosion, humidity/UV/cyclic tests; SIMS, XPS, and SEM	Durability mechanisms and protection strategies for soft low-E coatings

This review was initiated as the scientific foundation for an applied project aimed at developing energy-efficient low-E-coated glass panes through a cost-effective and scalable manufacturing route. The preliminary technology assessment showed that existing reviews provide valuable but largely segmented perspectives, focusing separately on transparent conducting oxides, Ag-based multilayers, smart-window materials, deposition technologies, or building-level energy performance. However, a unified analysis tracing the chronological and scientific evolution of the materials used specifically in low-E glazing, while linking material selection and interface design to deposition constraints, characterization evidence, durability, and product-level performance, remains limited. The present review therefore aims not merely to catalogue coating compositions and published results, but to establish an integrated materials-to-performance framework. Particular attention is given to conclusions consistently supported across the literature, results that depend strongly on processing or testing conditions, and questions that remain unresolved for industrial implementation. This applied perspective also motivates the emphasis on scalable deposition, critical-material use, heat-treatment compatibility, long-term stability, and the balance between optical-thermal performance and manufacturing cost.

2. Materials and Methods

2.1. Materials Used for Low-E Glazing

Material selection for low-E glazing cannot be separated from multilayer architecture, interface design, deposition conditions, and the intended position of the coating within the final glazing system. The relevant materials do not perform a single function independently; instead, they form an interacting system in which infrared reflection, visible-light

transmission, electrical conductivity, optical interference, adhesion, chemical protection, and thermal stability must be achieved simultaneously. Three principal material platforms define the present low-E design space: Ag-based dielectric/metal/dielectric multilayers used predominantly in high-performance soft-coat products; transparent conducting oxides used as durable hard coatings, transparent electrodes, or alternative heat mirrors; and adaptive or multifunctional materials that introduce thermochromic, electrochromic, protective, or other additional functions [16,18,24,26–29,32–41].

These platforms occupy different technological and performance domains rather than representing directly interchangeable solutions. Ag-based multilayers provide the lowest emissivity and the highest spectral selectivity, but their performance depends critically on metal continuity, interface quality, protection against oxidation and diffusion, and compatibility with heat treatment [16,18,35–37,41–48]. Pyrolytic FTO offers lower spectral selectivity but substantially greater mechanical and chemical robustness, whereas sputtered ITO remains a conductivity–transparency benchmark whose cost and dependence on indium have stimulated the development of AZO, GZO, and doped SnO_2 alternatives [24,32–34,38,49–52]. Thermochromic, electrochromic, and hybrid oxide systems extend low-E glazing towards dynamic solar control and multifunctionality, although transition temperature, colour, cycling stability, weather resistance, and industrial scalability remain important limitations [26–29,40,51,53–60]. The material families are also not mutually exclusive: TCOs can act as electrodes in electrochromic devices, dielectric layers can simultaneously provide optical adjustment and chemical protection, and barrier or seed layers can determine whether an ultrathin Ag film develops the morphology required for low emissivity.

Accordingly, the following discussion is organized around shared scientific and technological questions rather than as a sequential account of individual studies. Section 2.1.1 examines Ag continuity, nucleation, and interface stability in metal-based multilayers. Section 2.1.2 analyses the conductivity–transparency–durability trade-off in TCO-based coatings and indium-free alternatives. Section 2.1.3 considers adaptive and multifunctional material systems in relation to their additional functions and level of technological maturity. Finally, Section 2.1.4 compares the material platforms and identifies the principal selection criteria and unresolved questions for industrial low-E glazing.

2.1.1. Ag-Based Multilayers

Silver is the reference metallic component of high-performance soft-coat low-E glazing because it combines very high free-carrier reflectance in the near- and thermal-infrared with comparatively low absorption in the visible range. Gold and copper were investigated in early heat-mirror structures, but their intrinsic visible absorption and colour make them less suitable for architectural glazing [15,16,18,43]. Consequently, modern soft-coat products employ one or more ultrathin Ag layers embedded in dielectric/metal/dielectric architectures. The Ag layer provides the primary low-emissivity function, whereas the surrounding layers control nucleation, optical interference, chemical stability, and compatibility with subsequent processing [16,18,35–37,41–46].

The central scientific problem is that a Ag layer with a nominal thickness of only a few nanometres does not immediately behave as a continuous metal film. During the initial growth stage, Ag generally forms isolated islands, which subsequently develop into percolating networks and finally into a continuous conducting layer. Below the continuity threshold, the coating exhibits high optical absorption, poor infrared reflectance, unstable sheet resistance, and increased emissivity. Increasing the Ag thickness improves continuity and infrared reflection, but beyond an application-dependent optimum, it reduces visible transmittance and may alter the reflected and transmitted colour. Kim et al.

reported an increase in near-infrared reflectance at 1500 nm from 18% to 86% when the Ag thickness in $TiO_2/Ag/TiO_2$ multilayers increased from 8 to 14 nm, with 12 nm providing the best balance between visible transmittance and infrared reflection for the particular stack investigated [37]. This result illustrates the general thickness trade-off, but it should not be interpreted as defining a universal optimum for all Ag-based low-E architectures.

The literature broadly agrees that the effective Ag continuity threshold is determined not only by nominal thickness but also by the surface on which the metal is deposited. Surface energy, roughness, chemical termination, contamination, deposition rate, working pressure, substrate temperature, and energetic-particle bombardment can all influence nucleation density and island coalescence. Seed or wetting layers are therefore used to promote earlier formation of a continuous Ag film. ZnO is the most established seed material in commercial low-E stacks. Álvarez et al. showed that Ag wets and coalesces more effectively on ZnO than on SnO_2 under the investigated conditions, resulting in improved continuity at lower nominal thickness [35]. Other studies have demonstrated similar improvements using ultrathin metallic or oxide seed layers [18,42,43]. However, the reported percolation thicknesses and the magnitude of the seed-layer benefit vary between studies because the deposition conditions, substrate roughness, measurement methods, and criteria used to define continuity are not identical. Apparent differences in the literature therefore reflect both genuine interface effects and limited experimental comparability.

Once a continuous Ag layer has been obtained, its properties must be preserved during deposition of the upper part of the stack. Reactive sputtering of an oxide or nitride directly above Ag can expose the metal to oxygen, energetic ions, or plasma-induced damage. Thin blocker layers based on Ti, NiCr, or substoichiometric TiO_x are commonly inserted to limit Ag oxidation, improve adhesion, and reduce interdiffusion during deposition and heat treatment [1,18,36,41–44]. Their design introduces another compromise: an insufficient blocker may fail to protect the Ag layer, whereas an excessively thick or incompletely oxidized metallic blocker increases visible absorption. The most suitable blocker cannot therefore be selected independently of the upper dielectric, reactive-gas exposure, deposited energy, and subsequent thermal-processing route.

The dielectric layers surrounding Ag also perform several coupled functions. They protect the metal mechanically and chemically, provide interference-based antireflection that recovers visible transmittance, and tune the solar factor and colour of the glazing. SnO_2, ZnO, TiO_2, Si_3N_4, and SiO_2 are among the most frequently employed materials [16,18,36,40,46,48]. High-index TiO_2 can provide strong antireflection with relatively thin layers, but its optical advantage must be balanced against interface and thermal stability. Kulczyk-Malecka et al. observed Ag agglomeration and diffusion in $TiO_2/Ag/TiO_2$ coatings after annealing at 600 °C, demonstrating that a stack optimized in the as-deposited state may not remain optimal after tempering [36]. Si_3N_4 and SiO_2 are attractive where diffusion resistance or chemical robustness is prioritized, although the use of lower-index materials can require different layer thicknesses and optical designs. Thus, no dielectric can be identified as universally superior without considering the complete stack and its processing history.

Evidence from commercial coatings confirms that Ag-based low-E glazing should be treated as an interface-engineered multilayer system rather than as an isolated metal film. ToF-SIMS profiling of a commercial double-silver product identified a complex sequence containing SnO_2-based dielectrics, ZnO seed layers, TiO_x blockers, Ag layers, and additional base and protective layers [41]. Each layer fulfils a specific function, but its behaviour depends on the neighbouring materials. The lower dielectric influences Ag nucleation; the seed layer controls wetting and continuity; the blocker protects Ag during reactive deposition; and the upper dielectric and topcoat determine optical interference, chemical re-

sistance, and compatibility with handling, tempering, lamination, and insulating-glass-unit assembly. This functional partition is summarized in Figure 1.

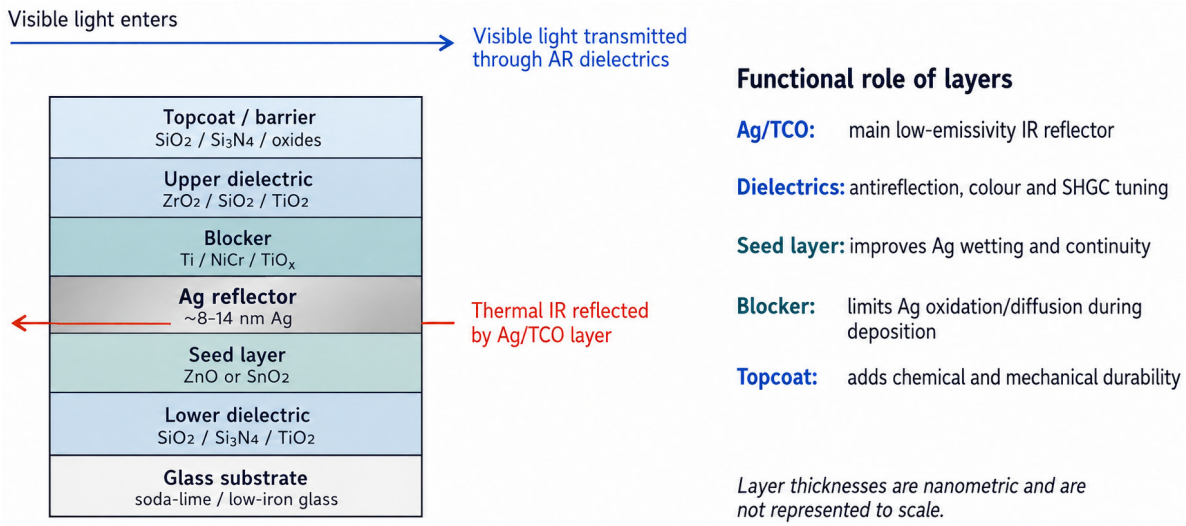


Figure 1. Schematic of a generic soft-coat low-E stack showing the functional position of the glass substrate, lower dielectric, seed layer, Ag layer, blocker, upper dielectric, and topcoat.

Commercial single-, double-, and triple-Ag coatings are based on the same dielectric/metal/dielectric principle, but the higher-order architectures are not obtained simply by increasing the total Ag thickness. Each additional Ag layer requires its own optically and chemically coordinated sequence of seed, blocker, and dielectric layers, while the thicknesses of the complete stack must be re-optimized to preserve constructive visible-light transmission and increase reflection in the solar near-infrared and thermal-infrared regions. Single-Ag products provide the simplest compromise between high visible transmittance, low emissivity, solar gain, and production cost and are widely used in residential and heating-dominated applications. Double-Ag coatings provide lower solar heat gain and a higher light-to-solar-gain ratio at comparable visible transmittance, making them particularly suitable for solar-control façades. Triple-Ag products extend this selectivity further and are used where stringent cooling-load, daylight, and façade-performance requirements justify greater coating complexity and cost [1,6,17–23,30,41].

From an industrial perspective, each additional Ag layer increases the number of cathode stations, deposited interfaces, target materials, and interacting process variables. Double- and triple-Ag products therefore require tighter control of the deposition rate, residual reactive gases, Ag continuity, blocker oxidation, dielectric optical thickness, cross-pane uniformity, and reflected colour. Their heat-treatable versions impose additional constraints because every Ag-containing sub-stack must remain stable during tempering without excessive diffusion, agglomeration, or optical shift. The greater number of interfaces also increases the number of possible defect, diffusion, or corrosion-initiation sites, although it does not necessarily imply lower durability when suitable blockers, protective layers, and process control are employed. The principal industrial penalty is therefore a narrower processing window, greater monitoring and qualification requirements, higher material and equipment utilization, and a potentially lower production yield. Moreover, once one continuous Ag layer has already reduced cavity radiation substantially, the main benefit of adding second and third Ag layers is often improved solar selectivity and control of SHGC rather than a proportional reduction in the glazing U-value. Consequently, the choice between single-, double-, and triple-Ag products is governed by the required balance between visible transmittance, emissivity, solar gain, colour, climate-specific energy

performance, manufacturability, durability, and cost rather than by the number of Ag layers alone [1,6,17,18,30,41,48].

A consistent conclusion across the literature is therefore that the performance of an Ag-based low-E coating is controlled primarily by continuity and interface quality rather than by the bulk optical properties of Ag alone. Nevertheless, quantitative comparisons remain difficult because studies use different substrates, deposition systems, layer sequences, thermal treatments, spectral ranges, and definitions of film continuity. Important unresolved questions include the reproducible formation of continuous Ag at progressively smaller thicknesses over architectural areas, the development of low-absorption blockers that remain effective during tempering, and the correlation between initial interface quality and long-term environmental durability. Addressing these questions requires studies that report the complete stack architecture and processing history and that correlate morphology, sheet resistance, optical spectra, emissivity, and post-ageing structural analysis on the same samples.

2.1.2. TCO-Based Coatings

Transparent conducting oxides constitute the principal non-metallic material platform for low-E and related functional glazing. Their optical and electrical behaviour results from the combination of a wide band gap, which limits absorption in the visible range, and a sufficiently high free-carrier concentration to provide electrical conductivity and increased reflectance at infrared wavelengths. However, these requirements are intrinsically coupled: increasing carrier concentration generally improves conductivity and shifts the plasma response towards shorter wavelengths, but it may also increase free-carrier absorption and reduce visible or near-infrared transmittance. Consequently, TCO development is governed by a conductivity–transparency–selectivity trade-off rather than by the maximization of a single material property [5,18,32–34].

Fluorine-doped tin oxide (FTO) is the most mature TCO platform for hard-coat low-E glazing. It can be deposited by pyrolytic or atmospheric-pressure chemical vapour deposition directly onto the hot float-glass ribbon, typically at temperatures close to 600 °C. The resulting coating is chemically bonded to the glass and can withstand handling, storage, tempering, bending, and, in suitable configurations, exposure in monolithic glazing. This robustness is its principal industrial advantage. Its main limitation is optical-thermal performance: typical FTO coatings exhibit higher emissivity, commonly in the range of approximately 0.15–0.20, and lower spectral selectivity than optimized sputtered Ag multilayers [1,6,12,24,49]. FTO and Ag-based soft coatings should therefore not be regarded as direct substitutes. FTO is favoured when process integration, durability, and ease of handling dominate, whereas Ag-based stacks are preferred when the lowest emissivity and highest light-to-solar-gain ratio are required.

Indium tin oxide (ITO) remains the reference sputtered TCO for the combination of low electrical resistivity and high visible transmittance. Its well-established deposition and optical modelling make it a useful performance benchmark, particularly for transparent electrodes and electrically controlled glazing [5,32–34]. Nevertheless, the cost, supply risk, and critical-material status of indium have motivated sustained research into alternative TCO compositions. These concerns do not invalidate the technical performance of ITO, but they reduce its attractiveness for very large coated areas when less critical materials can provide adequate, even if not identical, performance.

Doped ZnO materials, particularly Al-doped ZnO (AZO) and Ga-doped ZnO (GZO), are among the most extensively investigated indium-free alternatives. They can achieve resistivities of the order of 10^{-4} Ω·cm under optimized deposition conditions, while retaining high visible transmittance [24,33,34]. Their constituent elements are generally

more abundant than indium, and the films can be deposited using DC or RF magnetron sputtering, chemical vapour deposition, and related atmospheric or solution-assisted routes. Ga-doped ZnO produced by aerosol-assisted chemical vapour deposition illustrates the potential of non-vacuum processing for energy-efficient glazing, while also demonstrating that dopant concentration affects not only carrier density but also film growth, morphology, and the resulting optoelectronic properties [50].

The literature nevertheless does not establish AZO or GZO as universally equivalent replacements for ITO. The lowest resistivity values are often obtained for particular thicknesses, substrate temperatures, atmospheres, or post-deposition treatments that may not be compatible with all glazing products. Minami noted that achieving ITO-like stability becomes increasingly difficult for doped ZnO films at thicknesses below approximately 30 nm [33]. Moreover, comparisons between published values are complicated by differences in film thickness, crystallinity, dopant activation, substrate temperature, measurement geometry, and environmental history. A low resistivity measured on a relatively thick laboratory film does not necessarily demonstrate that the same material will retain its conductivity, transparency, and uniformity as an ultrathin coating deposited over architectural-scale glass.

Doped SnO_2 and TiO_2 systems broaden the TCO design space. Tantalum-doped tin oxide has been investigated as a selectively solar-transmitting, high-temperature-stable coating, illustrating the possibility of adjusting carrier concentration and spectral response without using indium [38]. Nanoscale Ta_2 layers have also been explored as transparent conductive components for smart-glass applications [52]. These results are promising, but their relevance to conventional architectural low-E products must be assessed cautiously. Performance demonstrated for solar-thermal or smart-glass applications does not automatically imply equivalence to a commercial low-E coating because the required spectral range, emissivity target, film thickness, environmental exposure, and integration route may differ.

Across the TCO literature, there is broad agreement on the fundamental conductivity–transparency compromise and on the industrial maturity of pyrolytic FTO. There is also general agreement that ITO provides a demanding optoelectronic benchmark and that AZO, GZO, and doped SnO_2 offer strategically important indium-free routes. The apparent inconsistencies concern the degree to which these alternatives approach ITO performance and the conditions under which that performance remains stable. Much of this variation can be attributed to differences in deposition method, thickness, dopant concentration, thermal treatment, substrate, and evaluation criteria rather than to irreconcilable conclusions about the materials themselves.

The principal unresolved challenge is therefore not simply to obtain a low resistivity or high transmittance in an isolated laboratory film, but to retain the required combination of conductivity, visible transparency, infrared response, colour neutrality, uniformity, and durability over large glass areas and throughout subsequent processing. Comparative studies should report film thickness, deposition and annealing conditions, carrier concentration and mobility, sheet resistance, spectral transmittance and reflectance, emissivity, and environmental stability on a common basis. Until such evidence becomes more widely available, FTO remains the established durable hard-coat solution, ITO the principal TCO benchmark, and AZO, GZO, and other doped oxides promising but application-dependent alternatives rather than universal replacements for either ITO or Ag-based multilayers.

2.1.3. Adaptive and Multifunctional Material Systems

Adaptive and multifunctional coatings extend the role of glazing beyond the static reduction in radiative heat transfer. Whereas conventional Ag- and TCO-based low-E coatings are designed to maintain an approximately fixed spectral response, adaptive materials

modify their optical properties in response to temperature, an applied voltage, or another external stimulus. Multifunctional coatings may additionally provide ultraviolet filtering, antireflection, photocatalytic activity, surface protection, or selective solar control. These functions can improve the overall performance of a glazing system, but they should not be treated as equivalent to achieving very low thermal-infrared emissivity. Their relevance depends on whether the design objective is static insulation, dynamic solar control, occupant comfort, surface functionality, or a combination of these requirements [26–29].

VO_2 is the most extensively investigated thermochromic material for passive solar-control glazing. Its semiconductor-to-metal transition produces a temperature-dependent change in near-infrared transmittance while preserving partial transparency in the visible range. This passive response is attractive because it does not require electrical wiring, control electronics, or direct user intervention [28,29,55,56]. However, the literature also identifies a persistent set of limitations: the transition temperature of unmodified VO_2 is generally higher than the range preferred for building operation; the material can exhibit a noticeable colour; visible transmittance and solar modulation are difficult to maximize simultaneously; and long-term stability under humidity, ultraviolet radiation, and repeated thermal cycling remains insufficiently documented for many proposed compositions.

The reported performance of VO_2-based coatings varies considerably among studies. These differences are not necessarily contradictory because transition temperature, hysteresis, luminous transmittance, and solar modulation depend on doping, crystallinity, film thickness, morphology, substrate, antireflection layers, and the spectral range used in the calculation [28,55,56,59]. Building-level conclusions are also climate- and control-dependent: a reduction in solar gain may be beneficial during cooling periods but may penalize useful passive heating in colder conditions. Consequently, the performance of a thermochromic material cannot be assessed from the magnitude of its optical transition alone. It must be evaluated together with visible-light admission, switching temperature, climatic context, coating position, and the thermal characteristics of the complete glazing system [55–57,59].

Electrochromic systems provide active rather than passive modulation. Devices based on WO_3, NiO, or related oxides use transparent conductive electrodes and an ion-conducting medium to reversibly change their optical state under an applied voltage [26,27,53,54]. Their principal advantage is controllability: solar and visible transmittance can be adjusted in response to weather conditions, building loads, daylight requirements, or occupant preferences. This flexibility comes at the cost of increased system complexity. Electrochromic glazing requires transparent electrodes, electrolyte or ion-conducting layers, charge-balancing materials, electrical connections, sealing, and an appropriate control strategy. Switching speed, colour neutrality, optical memory, uniformity over large areas, and degradation over repeated cycles therefore become as important as the initial modulation range.

There is broad agreement that electrochromic devices offer greater operational control than passive thermochromic coatings, while thermochromic systems offer simpler autonomous operation. Neither conclusion establishes universal superiority. The preferred approach depends on climate, façade orientation, required switching frequency, acceptable control complexity, and the relative importance of cooling, heating, daylight, and glare. Published results can appear inconsistent when material-level optical modulation is compared directly with device-level or building-level energy performance. Differences in the device area, switching protocol, initial optical state, weather data, control logic, and evaluation period must therefore be considered before comparing reported energy savings or comfort benefits [26,27,29,53–57].

Hybrid and sol–gel-derived coatings represent a different form of multifunctionality. Mixed SiO_2-, TiO_2-, and ZnO-based systems can combine antireflection, ultraviolet protection, surface modification, and moderate infrared-control functions using comparatively low-capital wet-chemical processes [40,51]. A TiO_2-ZnO-SiO_2 coating has been reported to combine UV protection and infrared reflection with an emissivity of approximately 0.38 [40]. This result is relevant as evidence of multifunctional behaviour, but it remains substantially above the emissivity achieved by optimized multi-Ag soft coatings. Moreover, the cited study is currently available as a non-peer-reviewed preprint, so its quantitative performance should be regarded as preliminary until independently validated. Other SiO_2/TiO_2 and hybrid nano-ceramic coatings demonstrate useful optical or protective functions, but their application objectives and performance metrics are not always those of conventional low-E glazing [51,60].

Emerging materials and deposition concepts further broaden the research space, but technological feasibility should be distinguished from demonstrated glazing performance. For example, atomic layer deposition of Ag halides establishes the possibility of producing conformal AgX thin films with controlled composition [39]. However, successful film deposition alone does not demonstrate low emissivity, reversible optical switching, environmental durability, or suitability for large-area architectural glazing. Similar caution is required when results obtained for photovoltaic enhancement, solar-thermal control, or other specialized optical applications are transferred to low-E window design. These studies may identify useful materials or interface strategies, but they do not by themselves establish equivalence to a qualified low-E product.

The literature therefore supports a common conclusion: adaptive and multifunctional materials currently extend rather than replace established Ag- and FTO-based low-E platforms. Their principal value lies in dynamic solar management or in combining several surface functions, whereas their principal limitations concern optical trade-offs, durability, control complexity, and industrial scalability. Apparent disagreements between studies frequently arise because different authors optimize different quantities (for example, transition temperature, visible transmittance, solar modulation, switching speed, emissivity, or simulated annual energy use) and because material-level results are not always translated consistently to complete glazing systems.

Most importantly, smart glazing should not be regarded as a separate alternative to low-E coatings, but as a dynamic extension of their spectral-selective function. A conventional low-E layer continuously limits long-wave radiative heat transfer, whereas an electrochromic, thermochromic, or other responsive component can adjust solar and visible transmittance according to outdoor conditions and building requirements. During sunny winter periods, a clear or high-transmittance state can admit useful passive solar gains, while the low-emissivity function reduces heat loss during cold evenings, nights, and overcast periods. Under strong summer irradiation, a tinted or near-infrared-rejecting state can limit solar heat gain and help preserve a cooler indoor environment. During cloudy conditions, or at times of lower solar exposure, a clearer state may maintain daylight availability and reduce artificial-lighting demands. Morning and evening operations may also differ according to the façade orientation and solar incidence angle. Therefore, a well-designed dynamic low-E system can alternate between solar-heat admission, heat retention, and solar-heat rejection rather than maintaining a fixed optical state throughout the day and year. However, the resulting energy benefit depends on the climate, season, façade orientation, occupancy, switching thresholds, response time, and integration with HVAC, shading, and lighting controls [26,27,29,53–57,61–65].

The unresolved questions are consequently both scientific and technological. They include the simultaneous optimization of visible transparency and solar modulation, adjustment of the thermochromic transition to a building-relevant temperature range, colour neutrality, cycling and weathering stability, compatibility with large-area and cost-effective deposition, and the development of comparable evaluation protocols. Future studies should distinguish clearly between low-emissivity performance and supplementary functionality and should report the optical state, spectral range, switching conditions, complete device architecture, durability history, and glazing-level consequences. Such reporting would allow adaptive and multifunctional systems to be compared on a common basis and would clarify the applications in which their additional complexity produces a genuine advantage over static low-E coatings.

2.1.4. Comparative Synthesis and Material-Selection Criteria

The comparison of low-E material platforms does not identify a single universally optimal material. Instead, it reveals a set of application-dependent compromises between emissivity, visible transmittance, solar control, colour, durability, process compatibility, scalability, and cost. Material selection must therefore begin with the intended glazing product and its operating conditions rather than with the highest performance value reported for an isolated film. The relevant questions include whether the coating will be protected inside an insulating glass unit or exposed in a monolithic product, whether tempering or bending is required, whether the priority is minimum heat loss or reduced solar gain, and whether the optical response must remain static or change dynamically [1,6,16,18].

For high-performance soft-coat glazing, the literature consistently identifies Ag-based dielectric/metal/dielectric multilayers as the preferred platform when the primary objectives are very low emissivity and high spectral selectivity. Their advantage results from the strong infrared reflectance of continuous ultrathin Ag combined with interference-based recovery of visible transmittance. However, this performance depends on the precise control of nucleation, interfaces, blocker thickness, and dielectric architecture. It also requires protection against oxidation, interdiffusion, handling damage, and environmental exposure [16,18,35–37,41–48]. The lowest-emissivity material platform is therefore also the platform most dependent on deposition precision and product-level encapsulation.

Pyrolytic FTO occupies a different region of the design space. Its emissivity and light-to-solar-gain ratio are generally less favourable than those of optimized multi-Ag coatings, but its mechanical and chemical robustness, compatibility with tempering, and integration with the float-glass production line make it preferable when durability and simplified handling dominate [1,6,12,24,49]. The comparison between FTO and Ag-based coatings is consequently not a disagreement in the literature but a difference in optimization priorities. Soft-coat products maximize optical-thermal performance under protected conditions, whereas hard-coat products accept lower selectivity in exchange for greater process and service robustness.

ITO, AZO, GZO, and other doped oxides provide additional options, particularly for transparent electrodes, hybrid heat mirrors, smart-window devices, and indium-free material strategies. ITO remains the optoelectronic benchmark, while AZO and GZO reduce dependence on indium and can achieve competitive conductivity under favourable deposition conditions [5,24,32–34,50]. Nevertheless, reported laboratory performance does not yet establish universal equivalence at an architectural scale, especially for ultrathin films subjected to heat treatment, humidity, and long service periods. Doped SnO_2 and TiO_2 systems further broaden the available material palette, but their suitability must be assessed against the spectral and durability requirements of the intended glazing application rather than inferred from conductivity alone [38,52].

Adaptive and multifunctional systems introduce another selection criterion: the value of an additional function relative to the complexity required to obtain it. Thermochromic materials offer passive temperature-dependent modulation, whereas electrochromic devices provide controllable optical states but require electrodes, ion-conducting layers, electrical connections, sealing, and control logic [26–29,53–57]. Hybrid and sol–gel-derived coatings can add ultraviolet protection, antireflection, surface protection, or moderate infrared control through potentially lower-capital deposition routes [40,51,60]. These systems may offer clear advantages in specific applications, but they generally extend rather than replace conventional low-E platforms when the principal target is minimum thermal-infrared emissivity.

Table 2 summarizes the functional position, advantages, and principal limitations of the material families used in low-E and related glazing systems. The comparison shows that individual materials must be evaluated as components of a stack. Ag or a TCO provides the main infrared-reflecting or electrically conducting function; seed layers determine nucleation and continuity; blockers and barriers limit oxidation and diffusion; dielectrics tune optical interference and protect the active layer; and adaptive materials introduce an additional response to temperature or electrical control. The properties of the complete coating therefore cannot be predicted reliably from the bulk properties of any single constituent.

Table 2. Functional material families used in low-E glazing stacks and their design roles.

Material Family	Examples Used in Low-E Glazing	Main Position/Function in the Stack	Advantages	Limitations/Design Risks	Ref.
Noble-metal IR reflectors	Ag; historically Au and Cu	Main optically active low-emissivity layer in soft-coat stacks	Very low emissivity; strong IR reflection; high selectivity when embedded in dielectrics	Island growth below percolation; corrosion; diffusion/agglomeration during heat treatment	[16,18,35–37,42,43,45–47]
Seed/wetting layers	ZnO, SnO_2, Zn-containing layers	Below Ag, controlling nucleation and continuity	Lower Ag coalescence thickness; improved sheet resistance and emissivity at low Ag thickness	Interface chemistry and roughness can still degrade Ag continuity	[35,41,44,48]
Blocker/barrier layers	Ti, NiCr, TiO_x, oxide/nitride barriers	Above or around Ag during reactive deposition and heat treatment	Protect Ag against oxidation and interdiffusion; improve adhesion	Metallic blockers absorb visible light if too thick; barriers add process complexity	[1,18,36,42,44,47]
Antireflection dielectrics	SnO_2, ZnO, TiO_2, Si_3N_4, SiO_2	Lower/upper dielectrics and topcoats around Ag	Recover visible transmittance; tune colour and solar factor; add mechanical/chemical protection	High-index oxides require careful thermal stability control; low-index layers may increase thickness	[16,18,36,46,48,66]

Table 2. *Cont.*

Material Family	Examples Used in Low-E Glazing	Main Position/Function in the Stack	Advantages	Limitations/Design Risks	Ref.
TCO heat mirrors	FTO, ITO, AZO, GZO, Ta:SnO_2, Ta:TiO_2	Hard-coat low-E layers, electrodes, smart-window components	Durable hard coatings; transparent conductivity; compatible with electrochromic/smart functions	Higher emissivity than Ag stacks; indium cost for ITO; stability challenges in ultrathin AZO/GZO	[24,32–34,38,50–52]
Adaptive/smart materials	VO_2, WO_3/NiO, photochromic/thermochromic films	Dynamic modulation layers combined with transparent electrodes	Temperature- or voltage-controlled solar/daylight management	Transition temperature, colour, cycling durability, and weathering remain key issues	[26–29,53–56,59]
Hybrid/sol–gel protective coatings	SiO_2/TiO_2, TiO_2-ZnO-SiO_2, nano-ceramic coatings	Additional UV-protective, antireflective or protective layers	Low-capital processing; tunable surface functionality; retrofit potential	Usually not a substitute for multi-Ag stacks when very low epsilon is required	[28,40,51,57,58,60]

Figure 2 expresses this relationship as a materials function map linking active materials, interface layers, and protective components to visible transmittance, emissivity, solar heat gain, colour, and durability. The map synthesizes evidence concerning Ag continuity and seed-layer effects, thermal diffusion in dielectric/Ag/dielectric stacks, the chemical complexity of commercial multi-Ag coatings, TCO-based alternatives, and thermochromic materials [28,35–37,41,45,50,55,66]. Presenting these findings by function rather than by individual study makes clear that similar materials may perform different roles depending on their position in the stack. ZnO, for example, may act as a TCO, a seed layer below Ag, or an optical dielectric, while TiO_2 may provide antireflection, protection, or an active contribution in a multifunctional coating.

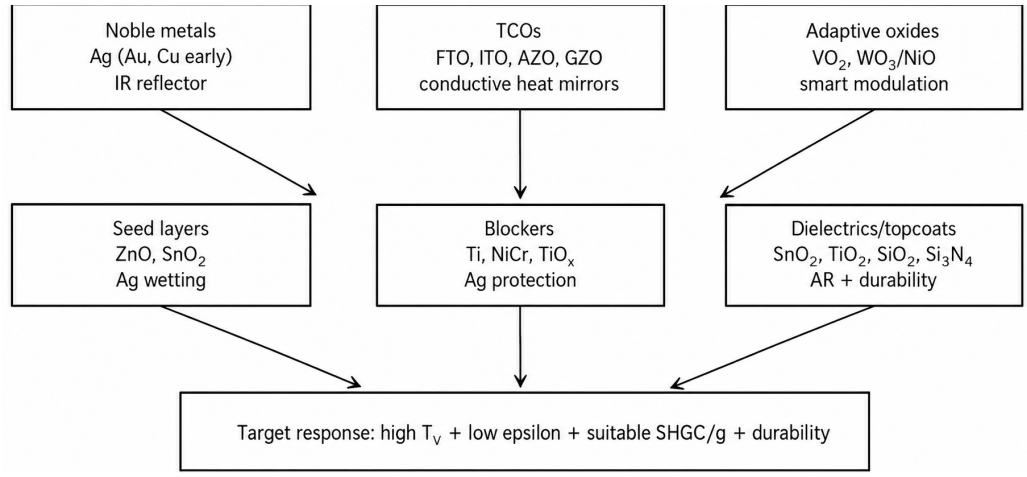


Figure 2. Materials function map for low-E glazing: IR reflectors, TCOs, adaptive oxides, seed layers, blockers, and dielectric/topcoat packages act together to determine visible transmittance, emissivity, SHGC/g-value, and durability.

The strength of the available evidence also differs between material platforms. Ag-based soft coatings and pyrolytic FTO are supported by extensive industrial experience, product-level measurements, and established processing routes. ITO is a mature transparent-conductor benchmark, but its use over very large areas is affected by indium availability and cost. AZO, GZO, thermochromic oxides, hybrid sol–gel coatings, and emerging functional materials often show promising film-level performance, but the evidence is less consistent regarding long-term durability, large-area uniformity, processing yield, and integration into qualified glazing products [24,26–29,33,34,40,50–60,67]. A technically rigorous comparison must therefore distinguish between commercial validation, device-level demonstration, laboratory proof-of-concept, and preliminary results reported in preprints.

Economic comparisons require similar caution. A process described as low-cost at laboratory scale is not necessarily economical for architectural glass once precursor utilization, coating speed, area uniformity, thermal treatment, quality control, rejection rate, encapsulation, and service life are included. Conversely, a high-capital vacuum process may achieve a low cost per coated area when operated at high throughput and may provide performance that cannot be reached by a simpler wet-chemical route. Published studies rarely report these factors on a common basis, so material cost alone cannot establish the most economical platform [1,6,16,18,40,50,51]. The appropriate comparison is the cost of obtaining and maintaining the required product performance over the intended service life.

The main areas of agreement are therefore clear: continuous Ag remains the preferred active material for the lowest-emissivity soft coatings; FTO remains the established durable hard-coat platform; ITO is the TCO performance benchmark; and adaptive or hybrid materials provide additional functions rather than direct equivalence to conventional low-E coatings. The principal uncertainties concern the industrial durability of indium-free TCOs, the reproducibility of progressively thinner Ag layers, the long-term stability of adaptive systems, and the relationship between laboratory performance, manufacturing costs, and service-life energy benefits.

Material selection should consequently be based on a common evidence set comprising complete stack architecture, deposition and thermal history, spectral transmittance and reflectance, emissivity, colour, sheet resistance where relevant, environmental durability, process scalability, and cost per qualified coated area. This framework connects the material-level comparison developed in Section 2.1 with the deposition-route analysis presented in Section 2.2 and provides a basis for distinguishing genuine material advantages from results that depend primarily on experimental or product configuration. To facilitate a quantitative comparison while avoiding the direct juxtaposition of non-equivalent specimens, Table 3 distinguishes between product-class benchmarks and representative experimental coatings. The values are reported exactly for the sample configuration and measurement conditions specified in the cited studies. Where standardized emissivity or sheet resistance values were unavailable, the original infrared, solar, or resistivity metric was retained rather than estimated. The quantitative data summarized in Table 3 are visualized as cross-study performance maps in Figure 3.

The comparison reveals two recurrent limitations in the available literature. First, many studies described as relevant to low-E applications report visible transmittance and reflectance at one or several NIR wavelengths, but do not provide standardized thermal-infrared emissivity. Second, electrical, environmental, and thermal measurements are rarely reported together for the same coating. Consequently, apparently superior values obtained for an individual indicator do not necessarily demonstrate superior product-level performance. A more reliable comparison requires the combined reporting of T_v, spectral

transmittance and reflectance, hemispherical emissivity, sheet resistance, colour, haze, and changes in these quantities after standardized thermal and environmental exposure.

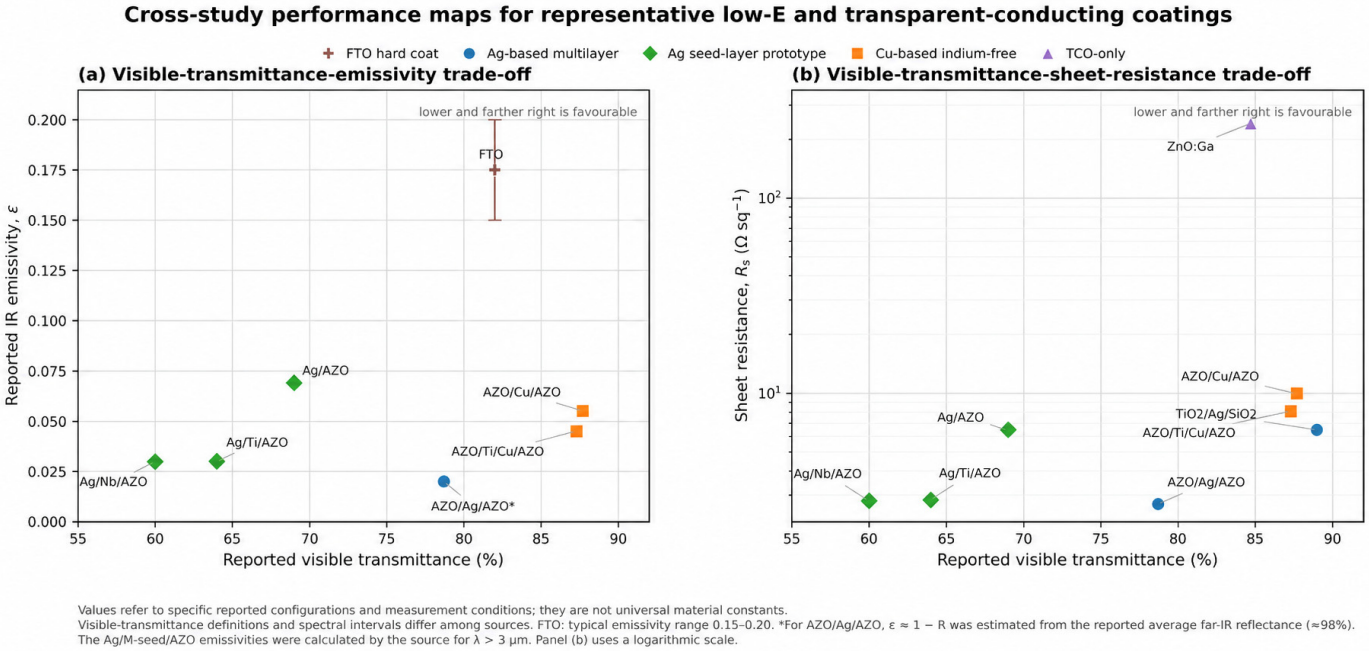


Figure 3. Cross-study quantitative performance maps for representative low-E and transparent-conducting coatings: (**a**) Reported visible transmittance versus infrared emissivity and (**b**) reported visible transmittance versus sheet resistance. Values refer to the specific configurations and measurement conditions reported in the cited studies and should not be interpreted as universal material constants. The visible transmittance definitions and spectral intervals differ among sources; the figure therefore represents a performance map rather than a ranking under identical test conditions. The FTO point shows the typical product-class emissivity range of 0.15–0.20. For AZO/Ag/AZO, emissivity was approximated as $\varepsilon = 1 - R$ from the reported average far-infrared reflectance of approximately 98%. The emissivities of the Ag/M-seed/AZO prototypes were calculated by the source for wavelengths above 3 μm. Panel (**b**) uses a logarithmic sheet-resistance scale. Sources: Refs. [1,6,12,16,50,66,68–71].

Table 3. Representative optical, electrical, and stability data reported for static low-E and TCO-based coating systems.

Material System and Representative Configuration	Deposition Route and Conditions	Visible Transmittance	Emissivity or Reported IR/Solar Indicator	Electrical Performance	Thermal Stability and Durability Evidence	Critical Interpretation	Ref.
Commercial multi-Ag soft-coat systems; one or more ultrathin Ag layers embedded between dielectric and interface-control layers	Off-line magnetron sputtering; product-specific multilayer architecture	Product-dependent	$\varepsilon \approx 0.02 - 0.05$ for optimized multi-Ag products	NR as a standardized product-level value	Ag requires dielectric, blocker, and topcoat protection. Resistance to heat treatment is strongly dependent on the suppression of oxidation, diffusion, and agglomeration; Ag redistribution has been observed in insufficiently protected $TiO_2/Ag/TiO_2$ stacks after annealing at 600 °C	Provides the lowest emissivity and highest spectral selectivity, but performance depends on precise interface engineering and protected placement within an IGU	[1,5,6, 16,18, 36,42, 43]
$TiO_2/Ag/TiO_2$ multilayer; Ag thickness varied from 8 to 14 nm	DC/RF magnetron sputtering	A 12 nm Ag layer was identified as the best compromise between visible transmission and NIR reflection; standardized (T_v) NR	Reflectance at 1500 nm increased from 18% to 86% when Ag thickness increased from 8 to 14 nm	NR	No dedicated accelerated durability test reported	Demonstrates the Ag-thickness trade-off: increasing continuity improves IR reflection, but excessive Ag thickness penalizes visible transmission. Single-wavelength NIR reflectance should not be interpreted as hemispherical emissivity	[37]
$TiO_2/Ag/SiO_2$, approximately 25/10/70 nm	E-beam evaporation with ion-assisted deposition at room temperature	Average transmittance of approximately 89% over 400–700 nm	Standardized emissivity NR	(R_s = 6.5 Ω/square	No visible transmittance attenuation after exposure to 55 °C and 95% relative humidity for 6 h according to ISO 9211 [72,73]; the upper SiO_2 layer acted as a moisture barrier	Combines high visible transparency and low sheet resistance. However, the short humidity exposure does not establish long-term outdoor durability or resistance to thermal cycling	[66]

Table 3. *Cont.*

Material System and Representative Configuration	Deposition Route and Conditions	Visible Transmittance	Emissivity or Reported IR/Solar Indicator	Electrical Performance	Thermal Stability and Durability Evidence	Critical Interpretation	Ref.
Optimized $SnO_x/Ag/SnO_x$ trilayer on glass	DC sputtering at room temperature	Luminous transmittance $\tau_L = 81\%$ including the glass substrate	Solar factor (g = 0.55); standardized emissivity NR	NR	No standardized ageing or thermal-cycling result reported for the optimized configuration	Demonstrates favourable visible-transmission/solar-gain control, but the absence of emissivity, sheet-resistance, and durability data prevents a complete comparison with commercial low-E products	[45]
Pyrolytic FTO hard coat; commercial K-glass used as an optical benchmark	On-line APCVD/pyrolytic deposition on hot float glass, typically near 600 °C	(T_v = 82%) for the K-glass benchmark	Typical FTO emissivity $\varepsilon \approx 0.15 - 0.20$ IR reflectance of 63% at 2500 nm reported for the K-glass benchmark	NR in the cited benchmark comparison	High mechanical and chemical durability; compatible with handling and subsequent heat treatment	Lower spectral selectivity and higher emissivity than optimized Ag stacks, but substantially stronger evidence of industrial maturity, large-area uniformity, and exposed-surface durability	[1,6,12,16,50]
ITO benchmark TCO films	Commonly magnetron sputtered; properties depend strongly on oxygen content, thickness, and post-treatment	Commonly above 80% in the visible for optimized films	No universal emissivity value; free-carrier IR reflection depends on carrier concentration and thickness	Resistivity commonly of the order of (10^{-4} Ω cm); (R_s) is thickness-dependent	Generally stable under established TCO processing conditions, although performance depends on composition and microstructure	Remains the TCO benchmark for the transparency–conductivity balance, but indium availability and cost limit its attractiveness for large-area architectural use	[5,32–34]

Table 3. *Cont.*

Material System and Representative Configuration	Deposition Route and Conditions	Visible Transmittance	Emissivity or Reported IR/Solar Indicator	Electrical Performance	Thermal Stability and Durability Evidence	Critical Interpretation	Ref.
Sputtered AZO and GZO indium-free TCOs	DC or RF magnetron sputtering; process- and dopant-dependent	Commonly above 80% for optimized films	No universal emissivity value reported	Resistivity can reach the order of (10^{-4} Ω cm (R_s) and depends on thickness	Stability comparable to ITO becomes more difficult to maintain at thicknesses below approximately 30 nm	Offers an indium-free route with competitive laboratory conductivity, but ultrathin-film stability, environmental robustness, and large-area process reproducibility remain less firmly established	[24,33,34]
ZnO containing 3.0 at% Ga on glass	Single-source AACVD at 350 °C	Average transmittance of 84.7% over 380–780 nm	IR reflectance of 48.9% at 2500 nm; standardized emissivity NR	R_s = 241.9 Ω/□; ρ = $1.16 \cdot 10^{-2}$ Ω·cm	No accelerated ageing or thermal-cycling test reported		[18]
TiO_2/Ag/TiO_2 multilayer; Ag thickness varied from 8 to 14 nm	DC/RF magnetron sputtering	A 12 nm Ag layer was identified as the best compromise between visible transmission and NIR reflection; standardized (T_v) NR	Reflectance at 1500 nm increased from 18% to 86% when Ag thickness increased from 8 to 14 nm	NR	No dedicated accelerated durability test reported	Demonstrates the Ag-thickness trade-off: increasing continuity improves IR reflection, but excessive Ag thickness penalizes visible transmission. Single-wavelength NIR reflectance should not be interpreted as hemispherical emissivity	[37]

Table 3. *Cont.*

Material System and Representative Configuration	Deposition Route and Conditions	Visible Transmittance	Emissivity or Reported IR/Solar Indicator	Electrical Performance	Thermal Stability and Durability Evidence	Critical Interpretation	Ref.
TiO_2/Ag/SiO_2, approximately 25/10/70 nm	E-beam evaporation with ion-assisted deposition at room temperature	Average transmittance of approximately 89% over 400–700 nm	Standardized emissivity NR	R_s = 6.5 Ω/□	No visible transmittance attenuation after exposure to 55 °C and 95% relative humidity for 6 h according to ISO 9211; the upper SiO_2 layer acted as a moisture barrier	Combines high visible transparency and low sheet resistance. However, the short humidity exposure does not establish long-term outdoor durability or resistance to thermal cycling	[66]
Optimized SnO_x/Ag/SnO_x trilayer on glass	DC sputtering at room temperature	Luminous transmittance ($\tau_L = 81\%$), including the glass substrate	Solar factor (g = 0.55); standardized emissivity NR	NR	No standardized ageing or thermal-cycling result reported for the optimized configuration	Demonstrates favourable visible-transmission/solar-gain control, but the absence of emissivity, sheet-resistance, and durability data prevents a complete comparison with commercial low-E products	[45]

NR: not reported for the stated sample configuration in the cited source. The reported values should not be interpreted as universal material constants. They depend on layer thickness, substrate, deposition conditions, post-treatment, spectral weighting, and measurement methods. Values of infrared reflectance at a single wavelength, solar factor, and thermal emissivity describe different physical quantities and are therefore retained separately. The typical emissivity ranges assigned to commercial multi-Ag and FTO product classes originate from review-level evidence, whereas the other numerical values refer to the specific experimental configurations indicated in the table.

2.2. Deposition Methods

The deposition route is not merely a manufacturing choice; it determines the admissible material palette, the substrate temperature, the achievable tolerance in layer thickness, and the final position of the coating in the glazing system. Low-E products are therefore divided into two industrial logics: on-line pyrolytic/CVD hard coatings, which are produced directly on the float line and tolerate handling, tempering, and outdoor exposure, and off-line vacuum soft coatings, where magnetron-sputtered Ag-based multilayers provide the lowest emissivity and highest spectral selectivity but require protection inside an IGU. Research-scale routes such as evaporation, ALD, and sol–gel complement these families by enabling model stacks, diffusion barriers, smart-window layers, and retrofit or low-cost oxide coatings [6,12,16,33,38–40,49,66,74–94].

2.2.1. Magnetron Sputtering for Soft-Coat Low-E Stacks

Large-area magnetron sputtering remains the central process for high-performance soft-coat glazing, as presented in Figure 4. In an industrial horizontal in-line coater, washed jumbo glass plates pass through load locks and a sequence of vacuum chambers, each dedicated to a specific metal, oxide, or nitride layer. Ag is usually sputtered from metallic targets in Ar to minimize oxidation, while ZnO, SnO_2, TiO_2, Si_3N_4, or related dielectrics are deposited either from ceramic targets or by reactive sputtering from metallic targets in Ar/O_2 or Ar/N_2. Reactive sputtering is especially sensitive because the target can move between metallic and poisoned states; partial-pressure control, plasma monitoring, pulsed/mid-frequency power, and dual-magnetron AC configurations are therefore used to stabilize the deposition rate, optical constants, and refractive-index uniformity [16,33,37,38,75–80].

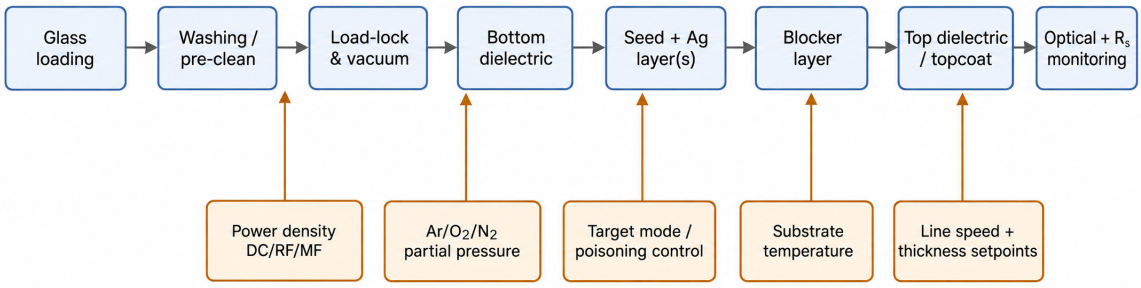


Figure 4. Schematic of an off-line large-area magnetron sputtering route for soft-coat low-E stacks.

From a materials viewpoint, the sputtering sequence must preserve the continuity and smoothness of Ag. The bottom dielectric and seed layer define the nucleation density and wetting of silver; the Ag layer thickness controls the transition from island growth to a continuous conductive film; and the blocker layer must protect Ag during subsequent reactive oxide deposition without adding excessive visible absorption. Process variables such as working pressure, ion energy, substrate temperature, residual water, and deposition order are therefore linked directly to sheet resistance, thermal-IR reflectance, visible colour, and long-term corrosion behaviour [18,35–37,42–45,75–82].

2.2.2. Pyrolytic, APCVD, and Aerosol-Assisted CVD Routes

Hard-coat low-E products are commonly produced by pyrolytic/APCVD deposition on the hot float-glass ribbon at temperatures of about 600 °C. Volatile tin or dopant precursors react at the glass surface to form chemically bonded, mechanically robust TCO layers, especially fluorine-doped tin oxide (FTO). This integration with the float line provides high throughput, low post-processing sensitivity, and compatibility with tempering, bending, and monolithic glazing. The limitation is that the high process temperature and atmospheric chemistry strongly restrict stack complexity: thick oxide layers and relatively

simple TCO architectures are favoured, while ultrathin Ag-based D/M/D stacks cannot be processed in the same way [6,12,49,83–86,92–94].

Aerosol-assisted CVD occupies an intermediate position between classical APCVD and solution processing. It allows metal–organic or salt precursors to be delivered as droplets and can produce doped ZnO or related oxides for energy-efficient glazing at more flexible process conditions than float-line APCVD. The key research questions are precursor decomposition, dopant incorporation, haze, growth morphology, and whether the electrical/optical performance can approach sputtered TCOs without the capital costs of a vacuum coater [24,50,83–86,92–94].

2.2.3. Evaporation, ALD, and Sol–Gel as Complementary Routes

Thermal and electron-beam evaporation remain useful for model D/M/D stacks and for low-temperature multilayers on substrates that cannot tolerate aggressive plasma conditions, as presented in Figure 5. When combined with ion-assisted deposition, evaporation can improve the density and adhesion of dielectric layers and has produced transparent $TiO_2/Ag/SiO_2$ electrodes with low sheet resistance and high visible transmittance. Its main limitation for architectural glass is scale: the technique is well suited for controlled laboratory stacks and specialized products, but less attractive than magnetron sputtering for high-throughput jumbo-pane production [66,74].

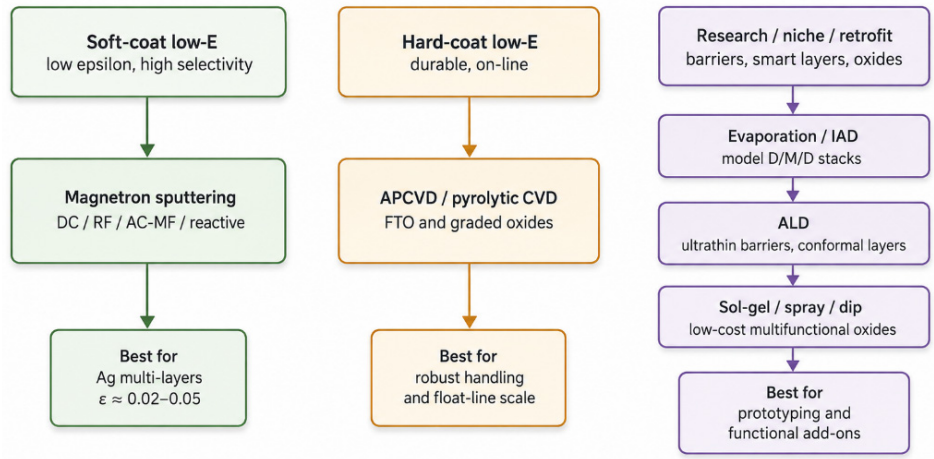


Figure 5. Deposition-route map for low-E and related functional glazing coatings. The scheme summarizes how sputtering, APCVD/pyrolytic CVD, evaporation/IAD, ALD, and sol–gel routes fit within the coating-design space.

ALD is not yet the workhorse route for complete commercial low-E stacks, but it is highly relevant for ultrathin barriers, seed layers, conformal protective oxides, and novel functional layers. Because ALD proceeds through self-limiting surface reactions, it offers sub-nanometre-thick control and conformality that are difficult to achieve with line-of-sight PVD. This makes it attractive for diffusion barriers that suppress Ag migration, for encapsulation of first-surface coatings, and for future hybrid stacks combining low-E function with switchable or phase-change behaviour [39,87–89].

Sol–gel, dip-coating, spin-coating, and spray-derived processes form the low-capital, wet-chemical side of the deposition toolbox. They are particularly useful for SiO_2, TiO_2, ZnO, and mixed-oxide coatings, including antireflective, UV-blocking, photocatalytic, or multifunctional layers. For classical low-E performance, they usually cannot match the emissivity of sputtered multi-silver coatings, but they are attractive for retrofit products, large-area oxide coatings, hybrid stacks, and multifunctional glazing where moderate emissivity reduction is combined with low cost or extra optical functions [40,51,90,91].

Figure 5 synthesizes information from several groups of sources. The industrial sputtering line and chamber sequence are based on Schaefer et al. [16], Kelly and Arnell [75], and Greene [86]. The simplified reactive-sputtering control loop, including O_2/N_2 partial pressure, target poisoning, and deposition-rate control, is based on Berg and Nyberg [76] and Sarakinos et al. [80]. The conceptual relationship between pressure, temperature, ion bombardment, and film morphology draws on Anders [77], building on the Thornton-style structure-zone background of the sputtering literature, allows a conceptual relation between pressure, temperature, ion bombardment, and film morphology; Zhao et al. [49], Gordon [92], and Choy [93] for a schematic of APCVD precursor flow over hot float glass and graded-oxide growth; George [88], Puurunen [87], and Heidari et al. [39] for the ALD cycle and its application to ultrathin barriers and topcoats; and Hench and West [90], Brinker et al. [91], and Eltohamy et al. [40], who describe the sol–gel chain from sol preparation through coating and drying to annealing and oxide-network formation.

2.2.4. Scale-Up Criteria and Process Selection

Tables 4 and 5 compare the principal deposition routes using a common set of technological and manufacturing criteria. Table 4 addresses the material and process window, deposition productivity, coating uniformity, and industrial scalability, whereas Table 5 extends the comparison to achievable layer complexity, relative process-cost drivers, environmental considerations, strengths, limitations, and the current role of each method in large-area manufacturing. The two tables should therefore be read together.

For product development, the deposition route should be selected by matching the required function to the process window. A low-emissivity winter product with high visible transmittance usually favours a single-Ag-sputtered stack; a solar-control façade product often requires double- or triple-Ag-sputtered stacks; a durable monolithic or exposed coating favours pyrolytic FTO; and adaptive glazing may combine TCO electrodes with electrochromic or thermochromic layers. The most important scale-up criteria are layer-thickness reproducibility across square metres of glass, colour uniformity at normal and oblique incidence, process stability over long runs, compatibility with tempering/lamination, environmental durability, target utilization, and cost per coated area [1,6,16,17,30,75–94].

The comparison included in Tables 4 and 5 explains why commercial production is concentrated around two process families. Magnetron sputtering dominates soft-coat manufacturing because no other established large-area technique provides the same combination of dynamic productivity, nanometre-scale control, material flexibility, across-pane uniformity, and multilayer complexity. Its high capital cost is justified when the equipment operates continuously at high yield and produces differentiated single-, double-, or triple-Ag products. APCVD, by contrast, dominates hard-coat production because its direct integration into the float line eliminates a separate vacuum-coating stage and produces chemically bonded coatings that tolerate handling, tempering, storage, and, in suitable configurations, exposed-surface use. Its commercial advantage is therefore derived from process integration and robustness rather than from the lowest achievable emissivity.

The remaining routes occupy complementary positions. Evaporation enables well-controlled model structures but is less suited to high-throughput jumbo panes; conventional ALD provides exceptional interface and barrier control but is limited by cycle time and precursor utilization; and sol–gel or spray-derived processes reduce equipment complexity but introduce wet-film uniformity, curing, defect-control, and durability challenges. Consequently, the selection of a deposition route should be based on the required product architecture and service environment rather than on the deposition rate or equipment cost alone.

Table 4. Comparison of deposition routes for low-E, solar-control, and related functional coatings on glass: materials, substrate temperature, deposition rate, and industrial scalability [1,6,12,16,33,38–40,49,66,74–96].

Method	Typical Materials	Substrate Temp.	Deposition Rate and Effective Throughput	Coating Uniformity and Industrial Scalability
Magnetron sputtering (DC/RF/AC-MF, reactive)	Ag, NiCr/Ti blockers; ZnO, SnO_2, TiO_2, Si_3N_4; ITO/AZO/GZO	RT−300 °C	High in continuous dynamic operation, although strongly material-dependent. Metallic Ag layers can be deposited rapidly, whereas reactive oxides and nitrides may become rate-limited by target poisoning and process stabilization. Industrial productivity is governed by pane speed, cathode length, number of coating chambers, and equipment uptime	High and industrially demonstrated for jumbo glass panes. Moving substrates, extended cathodes, gas distribution control, and optical monitoring provide good across-pane reproducibility. Uniformity can nevertheless be affected by target erosion, reactive-gas gradients, arcing, and chamber history
Pyrolytic/APCVD on float line	FTO/SnO_2:F, doped tin oxides, graded oxides	~550–650 °C	Very high effective throughput. Film formation occurs during the passage of the hot-glass ribbon beneath successive precursor-delivery zones and is therefore integrated into continuous float-glass production	Very high scalability and good width uniformity when precursor flow, reactor geometry, temperature, and boundary-layer transport are controlled. The method is commercially proven for continuously produced architectural glass
Aerosol-assisted CVD/spray-CVD	Ga:ZnO, Al:ZnO, SnO_2, and mixed oxides	~300–600 °C	Moderate. Film formation can be relatively rapid, but total productivity is influenced by aerosol generation, droplet transport, precursor decomposition, solvent evaporation, and post-deposition treatment	Potentially scalable, but large-area uniformity is less mature than industrial APCVD or sputtering. Thickness, roughness, haze, and composition may vary with aerosol distribution and substrate temperature
Thermal/e-beam evaporation + IAD	TiO_2/Ag/SiO_2, simple D/M/D heat mirrors	Low to moderate	Moderate for individual layers. In the representative TiO_2/Ag/SiO_2 system, reported rates were approximately 0.3 nm s^{-1} for TiO_2 and 0.5 nm s^{-1} for Ag. Effective throughput decreases when several sources, vacuum cycles, and layer changes are required	High uniformity on small and medium substrates, but more difficult over jumbo panes because of line-of-sight transport, source geometry, thickness gradients, and material-distribution efficiency
ALD	Al_2O_3, TiO_2, ZnO, SnO_2, ultrathin barriers, Ag-halide prototypes	~100–300 °C	Low for conventional temporal ALD. Growth is typically expressed per reaction cycle; for Al_2O_3, approximately 0.11–0.12 nm per AB cycle has been reported. Purge and precursor-exposure times strongly limit total throughput. Spatial ALD can increase productivity but is not yet established for complete commercial low-E stacks	Excellent thickness uniformity and conformality at the substrate scale, including non-planar surfaces. Scaling reactors to jumbo panes while preserving precursor separation, short cycle times, and economical precursor use remains challenging

Table 4. *Cont.*

Method	Typical Materials	Substrate Temp.	Deposition Rate and Effective Throughput	Coating Uniformity and Industrial Scalability
Sol–gel/spin/dip/spray	SiO_2, TiO_2, ZnO, TiO_2-ZnO-SiO_2 mixed oxides	RT + anneal	Rapid coating application per pass, but moderate or low effective throughput when hydrolysis, ageing, drying, repeated coating, and annealing are included. A single growth rate in nm s^{-1} is therefore not directly meaningful	Geometrically compatible with large substrates and comparatively simple equipment. Uniformity depends strongly on solution viscosity, withdrawal or spray speed, evaporation, wetting, drainage, edge effects, and particulate

Table 5. Comparison of deposition routes for low-E, solar-control, and related functional coatings on glass: complexity, costs, environmental considerations, strengths, limitations, and large-area manufacturing role [1,6,12,16,33,38–40,49,66,74–96].

Method	Achievable Layer Complexity	Relative Process Costs and Principal Cost Drivers	Principal Environmental Considerations	Strengths	Limitations	Large-Area Manufacturing Role and References
Magnetron sputtering (DC/RF/AC-MF, reactive)	Very high. Sequential chambers allow complex stacks containing single, double, or triple Ag layers, seed layers, blockers, oxides, nitrides, and protective topcoats, with nanometre-scale thickness control	Very high capital costs; medium-to-high operating costs, but competitive costs per square metre at high production volumes. Main drivers are vacuum equipment, cathodes and power supplies, targets, pumping, cooling, maintenance, chamber cleaning, and yield losses during process drift	Dry process with little liquid waste, but electricity consumption for vacuum generation, cooling, and plasma operation can be important. Target manufacture, incomplete target utilization, coating of chamber walls, and the use of scarce materials such as Ag or In contribute to the material footprint	Dominant soft-coat route; precise multilayer thickness; scalable jumbo panes; lowest epsilon with multi-Ag stacks	High capital costs; vacuum maintenance; Ag soft coatings require IGU protection; reactive processes need closed-loop control	Dominant commercial route for high-performance soft-coat low-E and solar-control glazing. Selected when minimum emissivity, high spectral selectivity, and complex optical design are required [16,18,35–37,43,74–82,95]

Table 5. *Cont.*

Method	Achievable Layer Complexity	Relative Process Costs and Principal Cost Drivers	Principal Environmental Considerations	Strengths	Limitations	Large-Area Manufacturing Role and References
Pyrolytic/ APCVD on float line	Low to moderate. Multiple reaction zones are possible, but high temperature, atmospheric chemistry, precursor compatibility, and limited residence time restrict the material sequence and favour comparatively simple oxide stacks	Extremely high infrastructure requirement when considered as part of a float line, but low incremental costs per square metre at full production scale. Major drivers are precursor delivery, exhaust treatment, process control, and the need for integration with float-glass manufacture	Does not require vacuum and uses the existing thermal state of the float ribbon, reducing separate reheating. Environmental burdens arise from precursor manufacture, incomplete precursor conversion, gaseous byproducts, corrosive or hazardous chemicals, and exhaust-abatement requirements	Hard coatings chemically bonded to glass; high throughput; robust handling, tempering, and single-pane exposure	Higher emissivity and weaker spectral selectivity than sputtered Ag; restricted materials palette and stack design	Dominant commercial route for FTO and related hard-coat products. Favoured when productivity, mechanical robustness, heat-treatment compatibility, and exposed-surface use are more important than achieving the lowest emissivity [6,12,49,83,86,92–94]
Aerosol-assisted CVD/spray-CVD	Moderate. Doped oxides and mixed compositions are readily accessible, but abrupt nanometre-scale interfaces and very complex multilayers are more difficult than in vacuum sputtering	Low-to-medium capital costs compared with vacuum coating; operating costs are precursor- and solvent-dependent. Additional drivers include aerosol-generation efficiency, precursor utilization, drying, exhaust handling, and post-annealing	Avoids high-vacuum pumping but may consume organic solvents and metal–organic precursors. Potential burdens include VOC emissions, aerosol losses, precursor residues, and energy for heating and post-treatment	Flexible precursor chemistry; atmospheric processing; useful for doped oxides and research-to-scale bridges	Control of haze, droplet transport, morphology, and dopant activation can be difficult over architectural areas	Promising research-to-pilot route for indium-free TCOs and mixed oxides, but not yet equivalent to float-line APCVD or magnetron sputtering in architectural-scale process maturity [24,50,83,92–94]
Thermal/e-beam evaporation + IAD	Moderate. Suitable for simple dielectric/metal/dielectric heat mirrors and specialized optical stacks, but less flexible than multi-cathode sputtering for long industrial layer sequences	Medium-to-high capital and operating costs. Cost drivers include high-vacuum equipment, electron-beam or thermal sources, crucibles, source replenishment, material losses, substrate fixtures, and limited production area per cycle	Mainly a dry process, with little liquid waste, but vacuum pumping and source heating require energy. Material deposited on shields and chamber walls reduces utilization and generates cleaning waste	Clean model stacks; low substrate temperature; IAD improves film density and adhesion	Less suitable for high-throughput jumbo glass; line-of-sight deposition; scale and uniformity challenges	Useful for research, model multilayers, low-temperature substrates, and specialized optical products; limited compatibility with high-throughput jumbo-glass manufacture [66,74]

Table 5. *Cont.*

Method	Achievable Layer Complexity	Relative Process Costs and Principal Cost Drivers	Principal Environmental Considerations	Strengths	Limitations	Large-Area Manufacturing Role and References
ALD	Very high interface complexity and sub-nanometre control, particularly for ultrathin barriers, seed layers, nanolaminates, and conformal encapsulation. Thick layers or complete architectural stacks become time-consuming	High operating costs for conventional ALD. Principal drivers are specialized precursors, purge gases, long cycle times, precursor-delivery efficiency, reactor utilization, and exhaust treatment	Precise thickness control can reduce excess material, but long operating times, purge-gas consumption, precursor synthesis, low precursor utilization, and energy-intensive equipment may produce a substantial burden per deposited thickness. Spatial and continuous ALD may reduce some of these burdens	Atomic-scale thickness control; conformal encapsulation; excellent for blockers, diffusion barriers, and first-surface protection.	Slow for thick layers; precursor cost; not yet a primary route for complete large-area low-E stacks	Most attractive as a complementary route for diffusion barriers, seed layers, first-surface encapsulation, and specialized functional layers rather than as the primary process for a complete low-E stack [39,87–89,96]
Sol–gel/spin/dip/spray	Low to moderate. Multicomponent oxides are easily formulated, but repeated high-quality nanometre-scale layers require separate drying or curing stages and may accumulate roughness or stress	Low to moderate. Multicomponent oxides are easily formulated, but repeated high-quality nanometre-scale layers require separate drying or curing stages and may accumulate roughness or stress	Low to moderate. Multicomponent oxides are easily formulated, but repeated high-quality nanometre-scale layers require separate drying or curing stages and may accumulate roughness or stress	Low capital costs; easy oxide chemistry; useful for antireflective, UV-blocking, photocatalytic and retrofit coatings.	Moderate emissivity reduction; shrinkage/cracking and durability issues; less precise than sputtered multi-Ag stacks.	Low to moderate. Multicomponent oxides are easily formulated, but repeated high-quality nanometre-scale layers require separate drying or curing stages and may accumulate roughness or stress

The terms high, moderate, and low indicate relative industrial trends rather than universal numerical ratings. Absolute deposition costs cannot be transferred reliably between processes without specifying equipment scale, energy prices, labour, substrate dimensions, production yield, maintenance, and product-qualification requirements. Likewise, the environmental considerations identify principal process hotspots and do not constitute a harmonized life-cycle ranking. Such a comparison would require a common functional unit, preferably one square metre of qualified coated glass delivering a specified optical-thermal performance over a defined service life.

3. Methods of Analysis

3.1. Structure, Morphology, and Composition

Because the functional response of a low-E stack is set by layers of nanometric thickness and by their interfaces, structural and compositional analyses are the indispensable links between deposition parameters and optical-thermal performance, as synthesized in Table 6. X-ray diffraction (XRD) identifies the crystalline phases and preferred orientation of the layers, such as continuous crystalline Ag, wurtzite ZnO, rutile, or anatase phases in Sn/Ti oxides, and yields crystallite sizes from line broadening; grazing-incidence geometries are preferred for thin stacks, since in symmetric geometry, the amorphous glass substrate produces a high background that can mask weak film reflections [36,40,97]. X-ray reflectometry (XRR) complements XRD by providing layer thickness, density, and interface roughness, and is particularly suited to periodic multilayers. Cross-sectional transmission electron microscopy (TEM) directly verifies the layered architecture, interface abruptness, and local crystallinity, and is the arbiter when diffusion or intermixing is suspected [36].

Scanning electron microscopy (SEM) documents film morphology and continuity: for silver on oxide surfaces, it captures the three-stage growth sequence, such as isolated islands, percolating networks, and continuous film, that governs the onset of metallic infrared reflectance [35,37,98]. Atomic force microscopy (AFM) quantifies surface topography and RMS roughness, a predictor of optical scattering, of Ag wetting quality and of subsequent durability; sub-nanometre roughness values (e.g., $R_a \approx 0.39$–0.75 nm for TiO_2 and ITO films) are routinely resolved [97,98].

Compositional analysis spans several complementary probes. Energy-dispersive X-ray spectroscopy (EDS/EDX), usually coupled to SEM, provides approximate elemental composition and mapping (for example, confirming silver agglomeration in annealed $TiO_2/Ag/TiO_2$ stacks [36]). X-ray photoelectron spectroscopy (XPS) resolves surface composition and chemical states, distinguishing metallic Ag^0 from oxidized silver and quantifying dopant chemistry in TCOs [66]. Auger electron spectroscopy (AES) offers surface and depth-profile analyses with a high lateral resolution, while secondary-ion mass spectrometry (SIMS) delivers the highest depth-profiling sensitivity for trace elements and diffusion studies; time-of-flight SIMS combined with unsupervised machine learning has recently been used to reconstruct the complete 3D chemical architecture of a commercial double-silver low-E coating, identifying every functional layer at single-voxel resolution [41]. Rutherford backscattering spectrometry (RBS) provides composition and thickness measurements without reference standards and is used in advanced research settings [38]. Table 7 summarizes the toolbox.

Thickness and interface metrology are particularly important because the visible colour and IR emissivity of a low-E stack can change measurably with nanometre-scale deviations. X-ray reflectometry (XRR) provides layer thickness, density, and interface roughness for relatively smooth stacks, while cross-sectional TEM directly verifies layer sequence, crystallinity, intermixing, and diffusion after heat treatment. These methods are most powerful when combined with optical models: measured layer thicknesses can be inserted into transfer-matrix calculations, and discrepancies between simulated and measured spectra then reveal absorption, roughness, or composition errors [13,14,36,74,99].

Depth-resolved chemical analysis has become increasingly relevant for commercial multi-layer products. XPS and AES resolve surface chemistry and oxidation states, but their sampling depth is limited; SIMS and ToF-SIMS provide trace sensitivity and three-dimensional depth profiling, which is essential for identifying ultrathin blockers, seed layers, contaminants, and diffusion pathways. Recent ToF-SIMS work on commercial double-silver coatings also demonstrates how machine learning workflows can help classify complex depth profiles and relate them to stack functions [41].

The selection of a characterization method should be guided by the suspected performance-limiting mechanism rather than by the routine application of individual techniques. No single method can establish a complete causal relationship between processing, structure, degradation, and low-E performance. For Ag continuity, plan-view SEM and AFM reveal island formation, percolation, pinholes, and surface roughness, but morphological evidence should be confirmed by sheet-resistance and infrared-reflectance measurements; XRD alone is generally insufficient because discontinuous ultrathin Ag may produce weak or broadened signals superimposed on the glass background [35–37,98,100–104]. Interface diffusion and Ag agglomeration are most directly examined locally by cross-sectional TEM or STEM-EDS/EELS and over greater depth ranges by SIMS or ToF-SIMS, while XRR can detect changes in average interface roughness and density [36,41,105–117]. Oxidation of Ag or blocker layers is primarily a chemical-state problem and therefore requires XPS or AES, preferably complemented by depth profiling and ToF-SIMS detection of oxygen, chlorine, sulphur, or other corrosion-related species [41,66,114–117]. Degradation mechanisms should be investigated by comparing identical locations before and after controlled exposure and by correlating microscopy and chemical analysis with changes in sheet resistance, emissivity, visible transmittance, haze, and colour [17,48,118–134]. Finally, optical-performance changes should be diagnosed by combining UV–Vis–NIR and FTIR measurements with ellipsometry, XRR-derived thicknesses, electrical measurements, and transfer-matrix modelling. This correlative approach can distinguish whether an observed spectral deviation originates from thickness drift, altered optical constants, Ag discontinuity, interface roughening, diffusion, or chemical degradation [13,14,18,74,99,135–154].

Table 6. Principal structural, morphological, and compositional analysis methods applied to low-E thin-film stacks depend on the question in diagnosis.

Question in Low-E Stack Diagnosis	Preferred Methods	Information Obtained	Typical Cautions	Refs.
Has Ag or a blocker layer undergone oxidation or chemical attack?	XPS, AES, XPS/AES depth profiling, ToF-SIMS	Ag chemical state; Ti, Ni, or Cr blocker oxidation; oxygen penetration; Cl-, S-, or carbon-containing contaminants	XPS is surface-sensitive; Ag chemical-state shifts may be small; ion sputtering can reduce oxides or broaden interfaces	[41,66,114–117]
Which mechanism controls degradation after environmental or thermal exposure?	Paired before/after SEM, AFM, TEM, XPS, SIMS, sheet resistance, UV–Vis–NIR, FTIR/emissivity	Corrosion, pinhole growth, Ag agglomeration, interface diffusion, delamination, roughening, or loss of electrical continuity	Degradation may be spatially localized; exposure conditions and measurement positions must be reported; accelerated tests do not directly establish service life	[17,48,118–134]
What caused the measured optical or thermal performance change?	UV–Vis–NIR spectrophotometry, FTIR, ellipsometry, XRR, sheet resistance, transfer-matrix modelling, supported by SEM/TEM when required	Separation of thickness drift, optical-constant changes, increased absorption, roughness, loss of Ag continuity, and interface degradation	Optical inverse solutions may be non-unique; measurements should use the same sample history, spectral interval, and spatial position	[13,14,18,74,99,135–154]

3.1.1. Crystallinity, Thickness, and Interface Roughness

For multilayer low-E coatings, structural analysis must be treated as a depth-resolved problem rather than as a single-film measurement. Conventional theta-2theta XRD is useful for detecting crystalline Ag, ZnO, SnO_2, or TiO_2 phases, but grazing-incidence XRD/GIXRD is usually more informative because it suppresses the amorphous-glass background and increases the contribution of the coating. In practice, Ag texture, oxide crystallinity, and heat-treatment-induced phase changes are interpreted together with linewidth broadening, although Scherrer-type grain-size estimates must be treated with caution because strain, instrumental broadening, and the very small illuminated volume can all distort the results [105–108,155,156], as presented in Figure 6.

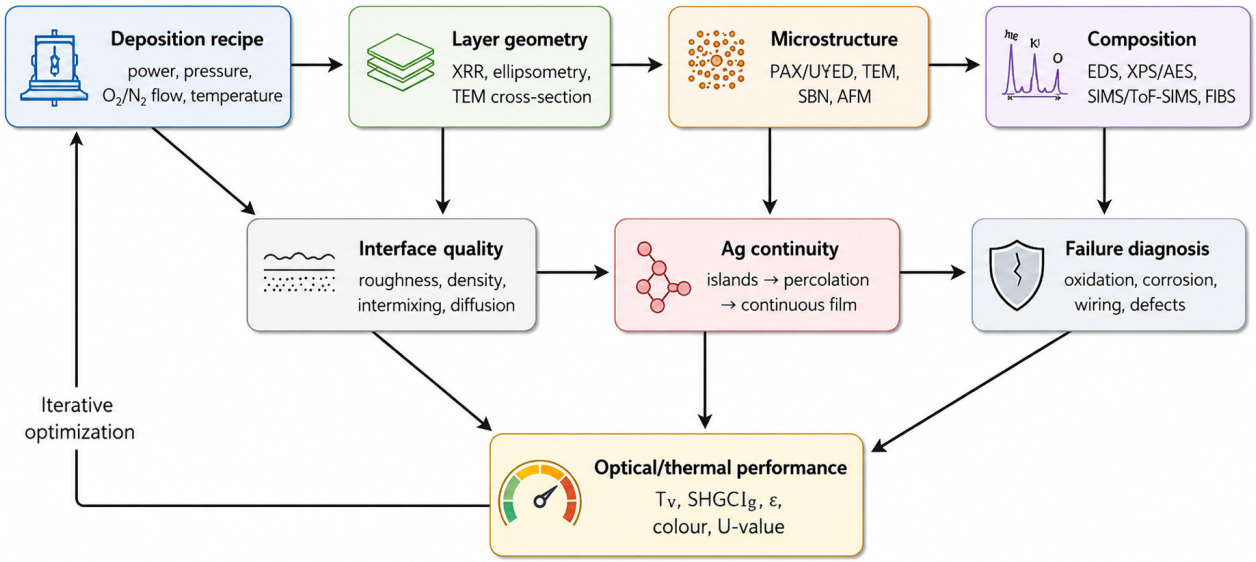


Figure 6. Correlative analysis workflow for low-E stacks. The scheme links deposition variables to thickness/roughness, crystallinity, morphology, depth-resolved composition, and final optical-thermal performance; a redrawn conceptual figure based on the cited literature in this subsection.

X-ray reflectometry (XRR) is particularly valuable for low-E stacks because it exactly measures the quantities that determine optical interference: layer thickness, electron density, and interface roughness. The Parratt formalism and the Nevot-Croce roughness correction remain the standard basis for fitting specular reflectivity curves, while modern multilayer fitting tools make it possible to compare alternative stack models and to test whether an apparent optical shift is caused by thickness drift, density variation, or roughening. XRR should ideally be cross-checked against cross-sectional TEM for complex multi-silver products since strong parameter correlations can produce multiple mathematically acceptable fits [105–108].

3.1.2. Morphology, Silver Continuity, and Surface Topography

The most important morphological question in Ag-based low-E coatings is whether the metal layer has passed from isolated islands to a continuous conducting film. SEM provides plan-view evidence for island coalescence, pinholes, and corrosion defects, while AFM quantifies roughness parameters such as Ra, RMS roughness, peak-to-valley height, and lateral correlation length. For ultrathin Ag, these measurements are not merely descriptive: small changes in roughness alter optical absorption, scattering, sheet resistance, and the stack's susceptibility to humidity-induced degradation. When AFM is used, filtering, plane subtraction, and scan-size selection should be reported because they can substantially alter numerical roughness values [100–103].

Morphology should also be linked to the wetting layer because a low-roughness top surface does not guarantee a continuous Ag layer (Figure 7). Therefore, plan-view SEM/AFM should be combined with electrical sheet resistance, optical NIR/IR reflectance, and, where possible, cross-sectional TEM (Figure 7). The complementary SEM and AFM evidence is illustrated in Figures 7 and 8. The practical acceptance criterion is not simply "smoothness", but a continuous Ag morphology with limited grain-boundary pathways, protected interfaces, and no local thickness collapse after heat treatment or environmental exposure [35–37,98,100–104].

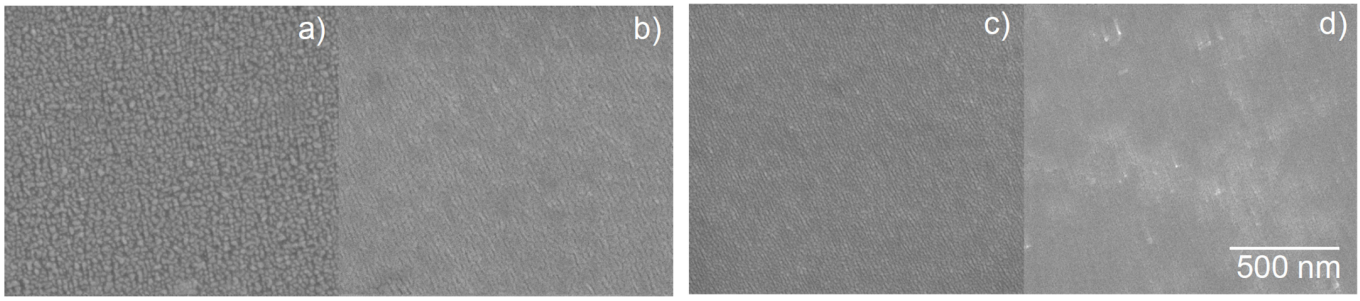


Figure 7. SEM top-view images of sputter-deposited Ag thin films with and without an Al seed layer. The original figure compares non-seeded Ag layers with nominal thicknesses of 3 nm (**a**) and 6 nm (**c**) with seeded layers of the same nominal thickness (**b**,**d**); the Al seed layer has a nominal thickness of 0.5 nm [42]. © 2022 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) licence (https://creativecommons.org/licences/by/4.0/).

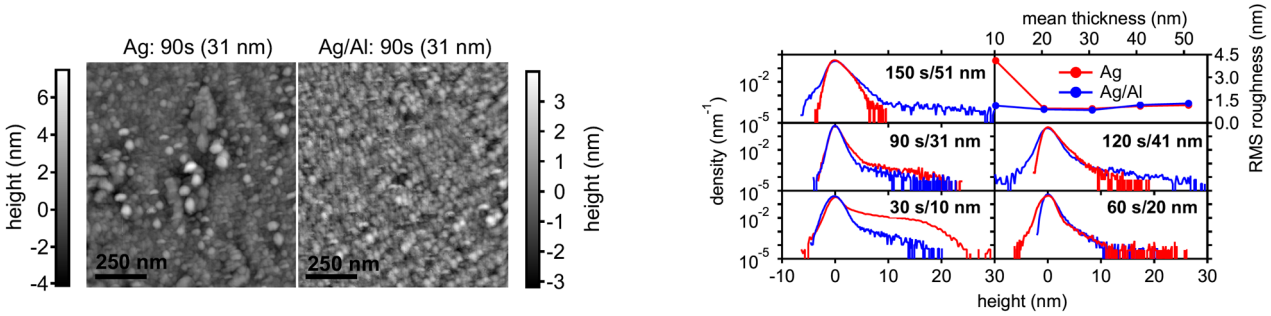


Figure 8. AFM height profiles and roughness distribution for non-seeded and Al-seeded Ag thin films. The source figure reports height distributions measured over 5 μm × 5 μm areas, RMS average roughness as a function of mean film thickness, and representative 1 μm × 1 μm AFM height profiles for thicker Ag films [42]. © 2022 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) licence (https://creativecommons.org/licences/by/4.0/).

Table 7. Expanded structural, morphological, and compositional analysis matrix for low-E thin-film stacks [35–37,41,97,98,100–117,155,156].

Question in Low-E Stack Design	Preferred Method(s)	Information Obtained	Typical Artefacts/Cautions	Refs.
Is each layer present with the target thickness?	XRR, ellipsometry, cross-sectional TEM	Layer thickness, density, periodicity, interface quality	XRR fits can be non-unique; TEM lamella preparation may introduce curtaining or local damage	[105–112]

Table 7. *Cont.*

Question in Low-E Stack Design	Preferred Method(s)	Information Obtained	Typical Artefacts/Cautions	Refs.
Is the Ag layer continuous?	Plan-view SEM, AFM, sheet resistance, TEM	Island/percolation/ continuous morphology, roughness, grains, pinholes	SEM contrast may confuse topography and composition; AFM scan size affects roughness metrics	[35,37,98, 100–104]
What crystalline phases and textures are present?	XRD, GIXRD, SAED/TEM	Ag texture, $ZnO/SnO_2/TiO_2$ phases, crystallite-size trends	Glass background and strain broadening can distort weak-film peaks	[36,97,105, 155,156]
Are blockers, seeds, and dielectrics chemically distinct?	XPS, AES, ToF-SIMS, STEM-EDS/EELS	Oxidation state, dopant distribution, buried-layer chemistry	XPS is surface-sensitive; sputter profiling can reduce oxides or mix interfaces	[41,44,113–117]
Has heat treatment caused diffusion or agglomeration?	TEM/STEM-EDS, SIMS, XPS depth profiling	Ag redistribution, blocker failure, intermixing at oxide/metal interfaces	Depth resolution depends on sputter rate and matrix effects; TEM is local	[36,41,109–117]
Can structure explain optical/thermal deviations?	Correlative XRR/XRD/AFM/TEM + optical modelling	Mechanism linking deposition drift to colour, absorption, and emissivity shifts	Requires consistent sample history and measurement locations	[13,14,18,74, 99–117,155,156]

3.1.3. Cross-Sectional Microscopy and Elemental Mapping

Cross-sectional TEM and STEM are the most direct methods for validating the physical layer sequence in a low-E stack. Bright-field TEM can reveal the continuity of the Ag layer and the sharpness of dielectric/metal interfaces, while high-angle annular dark-field STEM gives Z-contrast that is useful for separating Ag-rich layers from oxide layers. Selected-area electron diffraction or nanobeam diffraction can confirm local crystallinity, and EELS can help identify light elements or oxidation states where conventional EDS is less sensitive [109–113].

Elemental mapping must be interpreted with an awareness of spatial resolution and sample-preparation artefacts. Focused-ion-beam lamella preparation can cause redeposition, curtaining and Ga implantation; long electron-beam exposure can also induce migration or crystallization in sensitive oxides. For this reason, EDS maps should be acquired at a sufficiently low dose to avoid damage, quantified only when standards or validated models are available, and compared with non-destructive or less localized probes such as XRR, XPS, RBS, or SIMS [104,109–113]. The complementary depth-resolved structural and compositional workflow is summarized in Figure 9.

3.1.4. Surface Chemistry, Oxidation States, and Depth Profiles

Surface and near-surface chemistry are decisive for low-E durability because the optically active Ag layer is protected by nanometre-thick blockers and dielectrics. XPS identifies metallic Ag versus oxidized Ag species, Ti/Ni/Cr blocker oxidation, Zn/Sn/Ti oxide chemistry, and carbon or chloride/sulphur contamination. AES can complement XPS for high lateral resolution, while sputter-depth profiling extends both methods into buried layers. However, sputtering can preferentially remove oxygen, reduce oxides, or broaden

interfaces, so XPS/AES depth profiles should be treated as semi-quantitative unless sputter rates and matrix effects are carefully controlled [114–117,156].

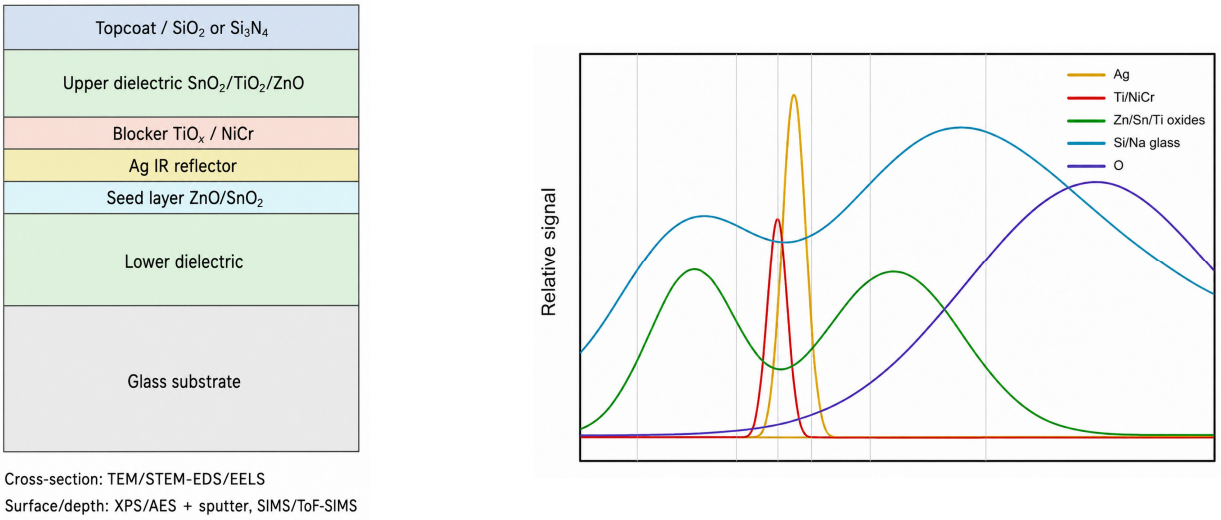


Figure 9. Original conceptual scheme for depth-resolved structural and compositional analysis of a multilayer low-E coating. The left panel represents a cross-section observed by TEM/STEM-EDS/EELS, while the right panel illustrates how XPS/AES/SIMS/ToF-SIMS depth profiles can distinguish Ag, blockers, seed layers, dielectrics, and the glass substrate.

SIMS and ToF-SIMS are especially useful when the question concerns trace contaminants, interdiffusion, and multilayer identity. Their strength is sensitivity and depth resolution rather than straightforward quantification: Ag, Zn, Sn, Ti, Si, Na, O, Cl, and S signals can reveal layer order, diffusion paths, and corrosion precursors, but ion yields are matrix-dependent. Recent ToF-SIMS work on commercial double-silver coatings shows why multivariate and machine learning workflows are becoming attractive for low-E products: they can classify complex depth spectra and associate chemical signatures with stack functions or failure mechanisms [41,115–117].

Key classical and recent sources for these characterization methods include the following: Parratt [105], Nevot and Croce [106], and Holy et al. [107] provide the XRR reflectivity formalism with Kiessig fringes and rough-interface models used to explain the extraction of thickness, density, and roughness; Smilgies [155], Scherrer [156], and Ali et al. [97] support the GIXRD/XRD discussion of peak broadening, preferred orientation, and grain size, linking crystallinity to Ag/TCO performance; Binnig et al. [100], Necas and Klapetek [101], and Smith et al. [98] cover AFM topography and the roughness-quantification workflow; Newbury and Ritchie [104] and Pazzaglia et al. [113] clarify EDS mapping principles and the boundary between qualitative and quantitative analysis for thin films; Egerton [109], Midgley and Dunin-Borkowski [110], and Haider et al. [111] illustrate cross-sectional TEM/STEM and EELS diagnostics for validating layer order and intermixing; Seah and Dench [114], Scofield [115], and Greczynski and Hultman [116] frame the XPS chemical-state workflow and the charge-referencing cautions relevant to Ag/blocker oxidation states; and McDonnell and Heeren [117], together with Bamford et al. [41], exemplify SIMS/ToF-SIMS depth profiling and multivariate classification for tracing diffusion, contaminants, and layer identity.

3.2. Optical Properties

Optical characterization begins with the spectral transmittance T(λ) and reflectance R(λ), from which the spectral absorptance follows as A(λ) = 1 − T(λ) − R(λ). Measurements are performed with UV–Vis–NIR spectrophotometers over the full solar range (typically

300–2500 nm), commonly following ASTM E424 [157] or the European framework [13,14,97]. From the spectra, the standardized integrated quantities are computed: visible transmittance T_v (illuminant D65, photopic observer), solar transmittance T_{sol} and reflectance R_{sol}, and the solar factor g (SHGC), which adds to the direct solar transmittance the fraction of absorbed energy re-emitted inward; EN 410 and ISO 9050 define the weighting functions and calculation procedures, and spectral selectivity is expressed through the T_v/g (LSG) ratio [12–14]. Because coated glazing is strongly angle-dependent, inter-laboratory comparisons of angle-resolved transmittance and reflectance have been essential to establish measurement comparability across instruments [13].

Spectroscopic ellipsometry determines the optical constants n(λ) and k(λ) and the thicknesses of the individual layers, typically over 190–1700 nm, feeding transfer-matrix models used to design and to reverse-engineer stacks; Drude parametrization of the Ag and TCO layers links the optical response to carrier density and mobility, which can be validated independently by Hall and four-point-probe measurements [5,14,18,38,99]. In the thermal infrared, Fourier-transform infrared (FTIR) spectrometry measures the reflectance from which the spectral emissivity is obtained via Kirchhoff's law; integration against the 283 K blackbody function according to EN 12898 [158] yields the corrected emissivity used in U-value calculations [1,14]. Good practice reports both the near-normal and hemispherical emissivity, since the conversion factor depends on the coating's optical constants [1].

The optical data must be processed in a manner consistent with glazing standards. EN 410 and ISO 9050 define the weighting functions used to derive visible transmittance, solar transmittance, solar reflectance, and solar factor (g-value) from measured spectra; EN 673 then relates these optical and emissivity data to the thermal transmittance of insulating glass units. For complex or scattering glazing, measurements often require integrating spheres, angle-dependent transmittance/reflectance, and careful separation of specular and diffuse components [13,14].

For low-E stacks, spectrophotometry and ellipsometry answer different but complementary questions. Spectrophotometry directly evaluates product performance over UV–Vis–NIR ranges and, with FTIR, the thermal-IR region; ellipsometry estimates n, k, and thickness for each layer, enabling model-based diagnosis. Transfer-matrix calculations are then used to design dielectric thicknesses that recover visible transparency, tune colour neutrality, and minimize solar heat gain without sacrificing IR reflectance. In practice, a good optical design is not the one with the lowest emissivity alone, but the one that reaches the required combination of T_v, SHGC/g-value, colour, angular stability, and manufacturability [12–14,19].

3.2.1. Spectral Response, Weighting Functions, and Performance Metrics

A more detailed optical analysis of low-E coatings starts from the distinction between measured spectral functions and standardized performance quantities (see Figure 10). The raw curves T(λ), R_f(λ), and R_b(λ) carry the physical information about interference, Ag/TCO free-carrier reflection, and parasitic absorption, while T_v, T_{sol}, R_{sol}, colour coordinates, and g/SHGC are weighted quantities that depend on illuminant, observer function, solar spectrum, and inward-flow assumptions. Envelope-type methods can be used as a rapid first estimate of optical constants and thickness for weakly absorbing films, but multilayer low-E stacks normally require simultaneous treatment of transmittance, front/back reflectance and, where possible, ellipsometric Ψ/Δ data because the same spectrum can be reproduced by different combinations of n, k, roughness, and thickness if the model is under-constrained [135–137].

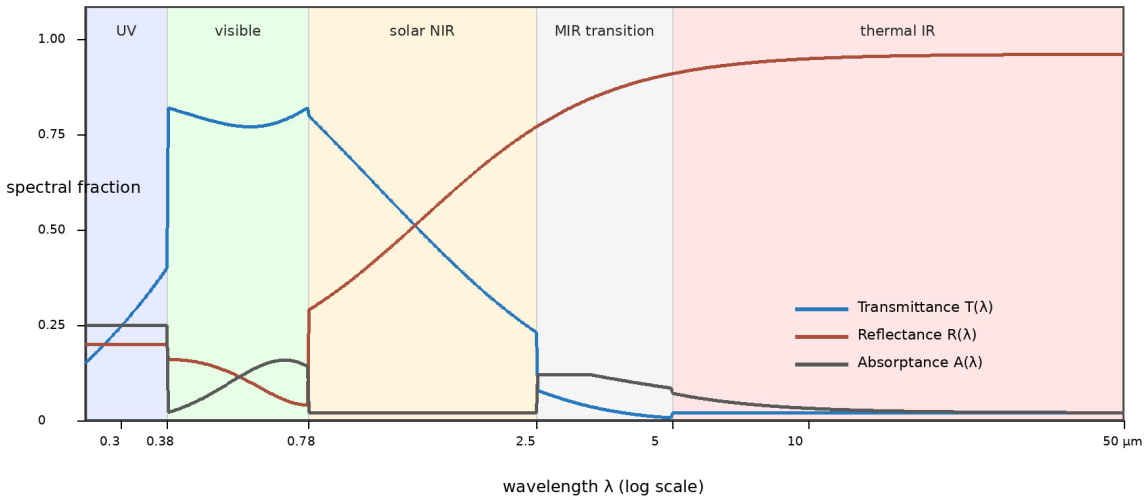


Figure 10. A high-performance low-E stack should transmit strongly in the visible region, reduce selected near-infrared solar gain, and reflect thermal infrared radiation, resulting in a low long-wave emissivity. The curves are conceptual rather than measured data. The scheme illustrates the complementary use of UV–Vis–NIR/FTIR spectra and weighted metrics such as T_v, Tsol, Rsol, g/SHGC, and emissivity. The figure is an explanatory scheme together with measured UV–Vis–NIR/FTIR spectra and weighted metrics such as T_v, T_{sol}, R_{sol}, g/SHGC, and emissivity [135–154].

The practical consequence is that optical characterization must report enough information for both building-performance use and thin-film diagnosis, as presented in Table 8. For architectural glazing, reporting only a single visible transmittance or U-value is insufficient: the spectral shape determines daylight colour, solar heat admission, glare, plant/occupant perception, and compatibility with photovoltaic, electrochromic, or thermochromic functions. In multilayer stacks, a small change in Ag thickness or dielectric optical thickness can move the interference extrema, change exterior reflected colour, and alter LSG, even when the nominal emissivity is almost unchanged [140–143,148–154].

3.2.2. Ellipsometry and Optical Modelling

For research samples and reverse-engineering of commercial coatings, optical modelling is usually formulated as a layered stack whose calculated T, R, and ellipsometric Ψ/Δ are fitted to measurements. The transfer-matrix formalism remains the standard calculation engine because it treats constructive/destructive interference, absorption, and complex refractive indices layer by layer. For real glazing, the model should distinguish coherent nanometre-scale coatings from incoherent millimetre glass panes, laminated interlayers, and air/gas cavities; mixed coherent/incoherent formulations are therefore preferable to a purely coherent thin-film model when the whole glazing unit is simulated [140–143], as synthetized in Figure 11.

The choice of dielectric function is equally important. Transparent oxides and nitrides can often be represented with Cauchy, Sellmeier, or Tauc–Lorentz-type functions in the transparent/near-band-edge region, while Ag and TCO layers require Drude or Drude–Lorentz terms to describe free-carrier absorption and infrared reflection. For doped oxides such as ITO, AZO, GZO, or Ga-doped ZnO, changes in carrier concentration move the plasma edge and therefore alter the balance between visible transparency and NIR/thermal-IR reflectance. This is why ellipsometric n,k fitting should be interpreted together with sheet resistance/Hall measurements and structural evidence of roughness, grain boundaries, and density [138,139,144–147].

Table 8. Expanded optical-property matrix for low-E glazing and related functional coatings [135–154].

Optical Quantity	How It Is Obtained	Physical Meaning in Low-E Stacks	Design or Diagnostic Use
T(λ), Rf(λ), Rb(λ), A(λ)	UV–Vis–NIR spectrophotometry; A = 1-T-R for the measured side.	Spectral balance between visible transparency, NIR rejection, and absorption losses.	Checks whether colour neutrality and solar-control targets are met; identifies parasitic absorption by blockers or discontinuous Ag [135–137].
T_v, T_{sol}, R_{sol}, colour	Spectral weighting with illuminant/observer/solar spectrum according to glazing standards.	Transforms spectra into comparable product metrics.	Allows comparison between products and climate-specific selection of glazing [148–154].
g/SHGC and LSG = T_v/g	Combination of direct solar transmission with inward-flow fraction from absorbed energy.	Links optical absorption and solar admission to building loads.	Separates daylight benefit from cooling penalty; essential for hot-climate solar-control glazing [148–154].
n(λ), k(λ), layer thickness d	Spectroscopic ellipsometry, transfer-matrix fitting and/or envelope methods.	Layer-resolved optical constants and optical thickness.	Reverse-engineering of stack design; sensitivity analysis for Ag, TCO, and dielectric thickness [138–147].
Thermal-IR emissivity ε	FTIR reflectance and Kirchhoff relation; hemispherical correction when needed.	Long-wave radiative exchange at room temperature.	Input for U-value calculations and comparison of hard-coat vs. soft-coat products [144,147].
Angular T/R and diffuse/specular split	Goniometric spectrophotometry or integrating-sphere measurements.	Incidence-angle dependence, scattering, and haze.	Critical for façades, tilted roof glazing, laminated/scattering films, and comfort analysis [140–143,148–154].

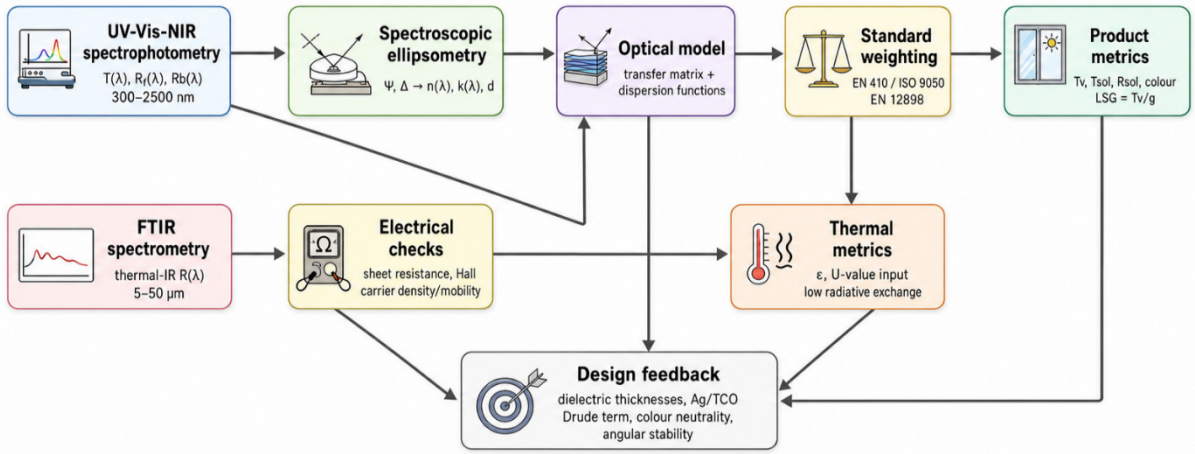


Figure 11. Optical characterization and modelling workflow. UV–Vis–NIR spectrophotometry, FTIR, ellipsometry, and electrical data are combined through transfer-matrix and dispersion-function models to obtain product metrics (Tv, Tsol, Rsol, colour, LSG) and thermal metrics (emissivity and U-value input). The diagram is a schematic synthesis of procedures discussed in [135–154].

3.2.3. Angle Dependence, Colour, and Uncertainty Reporting

Because façades are rarely illuminated at normal incidence, optical data should not be limited to near-normal spectra. Low-E and solar-control coatings can show strong angular shifts in reflectance colour and solar transmittance because interference optical thickness scales with incidence angle and polarization. Angle-dependent measurements

are especially important for highly glazed office façades, rooflights, tilted BIPV glazing, laminated glass, and retrofit films, where real solar exposure includes a broad angular distribution. For complex or scattering systems, integrating-sphere measurements and a separation of specular and diffuse components are necessary to avoid overestimating clear-view transmittance or underestimating glare and haze [140–143,148–154].

A good optical-properties subsection should therefore state the spectral range, incidence angle, reference side of the coating, detector/sphere configuration, baseline correction, polarization assumptions, wavelength resolution, and uncertainty. When modelled values are presented, the layer sequence, fitted thicknesses, dispersion functions, parameter bounds, and goodness of fit should be reported. This is particularly important for low-E stacks because several physically different structures can produce similar visible spectra but very different emissivity, durability, and heat-treatment behaviour [135–154].

More useful information can be retrieved from Manifacier et al. [135] and Swanepoel [136] for envelope-method diagrams showing rapid extraction of n, k, and thickness from the transmission extrema of weakly absorbing single films; Harbecke [140], Katsidis and Siapkas [141], and Centurioni [142] for transfer-matrix and coherent/incoherent multilayer calculation schemes, which explain why nanometre-scale coatings and millimetre-thick glass panes require different optical-coherence treatment; Baenas and Machado [143] for glazing-level optical models of multilayer, laminated, and PV/glass structures that bridge thin-film modelling and full window products; Fujiwara and Kondo [144] and Rakić et al. [147] for ellipsometry-derived dielectric functions and the Drude free-carrier response that link carrier density and metal optical constants to NIR/TIR reflections; Jelle [149], Goia et al. [150], and Ichinose et al. [151] for spectral and angle-dependent glazing-property plots and weighting-function examples connecting spectra to Tv and solar factors; and Alghamedi et al. [153], together with Gorantla et al. [154], for spectrally selective glazing examples showing how optical properties influence both daylight and cooling-load indicators.

3.3. Corrosion and Weather Resistance

The Achilles' heel of soft-coat low-E products is the environmental sensitivity of the ultrathin silver layer. Exposure to humidity, atmospheric chlorides, and sulphur compounds produces localized corrosion ("white-dot" defects), dewetting, and agglomeration, all of which raise emissivity and create visible haze [1,18,36]. Thermal loads are equally critical: during tempering or bending (620–700 °C), silver can diffuse into adjacent dielectrics and agglomerate, as demonstrated by EDS-mapped annealing studies of $TiO_2/Ag/TiO_2$ at 600 °C [36]; heat-treatable stacks therefore rely on diffusion barriers and sacrificial blockers. For this reason, soft coatings are almost always applied to a cavity-facing surface (#2 or #3) of an IGU, and the overall durability of the product is co-determined by the edge seal and spacer system that keeps the cavity dry [159].

Accelerated environmental testing subjects coated samples to cyclic thermal stress (typically −20 to +80 °C), condensation–humidity cycles, salt-spray, and UV irradiation (e.g., 550 $W \cdot m^{-2}$ according to DIN EN ISO 4892-2 [160], with adhesion and climate protocols per ISO 9142 [161]), while standards of the EN 1096 [162] family define durability classes for coated glass [1,163]. Long-term outdoor performance depends measurably on local atmospheric conditions: multi-year estimates of low-E solar performance show site-dependent degradation kinetics, underscoring the gap between laboratory ageing protocols and service life [164]. Several state-of-the-art reviews identify exactly this point (the scarcity of published ageing data and of microstructure-resolved degradation studies) as the most persistent research gap of the field [1,18,34].

The degradation modes are strongly linked to stack architecture. Silver can corrode through pinholes or edge defects, agglomerate during high-temperature processing, or migrate into adjacent TiO_2-rich layers during annealing. Chloride- and sulphur-containing atmospheres accelerate tarnishing, while humidity promotes electrochemical pathways that are difficult to reproduce with a single accelerated test. For this reason, durability assessment increasingly combines environmental exposure with post-mortem SEM/TEM, XPS, SIMS, and optical measurements rather than relying only on a pass/fail visual inspection [17,36,41].

The durability question is especially important for first-surface and retrofit applications. A soft coating protected inside an IGU experiences a much milder environment than a coating exposed to cleaning, abrasion, condensation, salt aerosols, or polymer-substrate deformation. Edge seals, spacers, and desiccants therefore become part of the low-E performance chain: failure of the IGU perimeter can expose the silver stack to moisture and destroy its optical and thermal function, even if the original coating was well designed [17,30,159].

3.3.1. Failure Mechanisms in Ag-Based Soft-Coat Stacks

From a materials perspective, corrosion and weather resistance in low-E glazing should be treated as coupled phenomena rather than as independent pass/fail tests. The Ag layer, the seed layer, the blocker, and the surrounding dielectrics form an electrochemically and mechanically coupled stack. Moisture, oxygen, chloride aerosols, sulphur-containing gases, and thermal cycling can act through different entry points, but they often converge on the same failure signatures: Ag_2S/AgCl-like tarnish products, localized pinholes, white-dot defects, interfacial delamination, Ag diffusion into oxide layers, and roughness-driven optical scattering. This is why recent durability reviews emphasize not only the initial optical performance of soft low-E products but also the stack design, the defect density of the capping layers, and the coating's resistance to first-surface exposure or accidental moisture ingress [17,42,118,165,166].

A critical feature of degradation in low-E glazing (frequently underestimated in single-test qualification) is that the relevant stressors rarely act in isolation. UV exposure, cyclic temperature loading, moisture ingress, atmospheric pollutants, and mechanical wear are coupled: photothermal energy raises interfacial reactivity and accelerates moisture- and pollutant-driven electrochemistry; thermal cycling opens diffusion and delamination pathways that subsequently admit water and reactive gases, and mechanical damage to the capping layers removes the very barriers that suppress all of the above. Because the active layer is a Ag film that is only a few nanometres thick, a single localized breach of the protective architecture can dominate the measurable loss in emissivity, sheet resistance, and haze, even when most of the coated area is intact. A systematic durability assessment must treat these stressors as a synergistic set rather than as independent pass/fail criteria and must correlate the resulting failure signatures with the specific microstructural pathway that produced them [17,118,166].

The most useful conceptual distinction is between intrinsic and extrinsic degradation, as presented in Figure 12. Intrinsic degradation is governed by the local microstructure of the stack: discontinuous Ag islands, grain-boundary diffusion, non-dense dielectrics, tensile/compressive stress gradients, and chemically reactive interfaces. Extrinsic degradation depends on the service environment: airborne $H_2S/SO_2/NO_2/Cl_2$, chloride-rich marine aerosols, cleaning agents, cyclic condensation, and edge-seal failure. Atmospheric-corrosion studies on Ag and mixed-flowing-gas testing show why sulphur and chlorine species are particularly important for noble-metal thin films, while low-E-specific work

demonstrates that the same chemistry becomes critical when Ag is only a few nanometers thick and must retain both high infrared reflectance and low visible absorption [48,119–124].

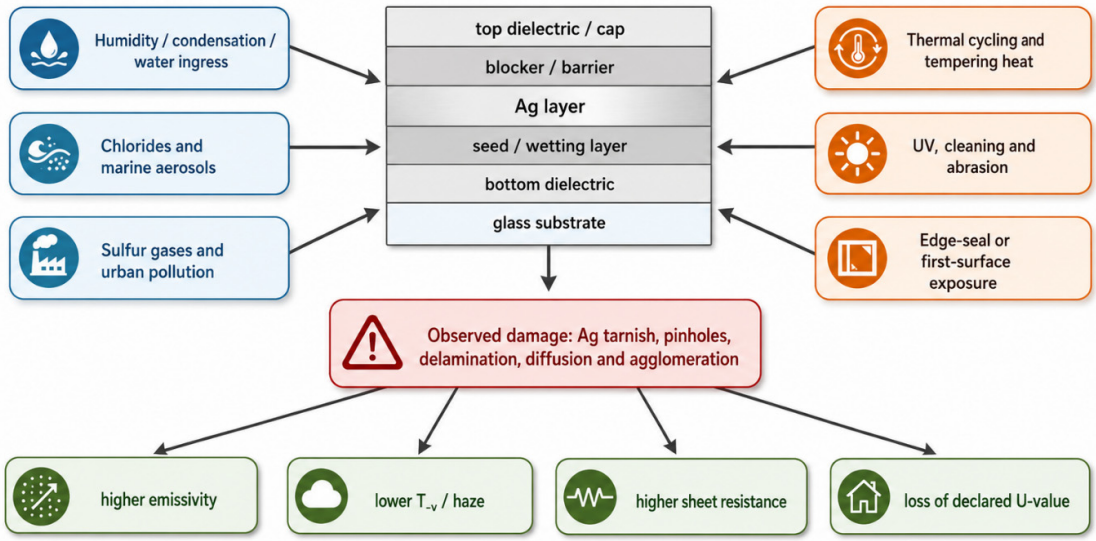


Figure 12. Conceptual map of corrosion and weathering pathways in Ag-based soft low-E stacks. The scheme links external stressors (humidity, chlorides, sulphur gases, UV/cleaning, and thermal cycling) to microstructural pathways (pinholes, diffusion, delamination, and agglomeration) and to measurable performance losses such as higher emissivity, haze, and increased sheet resistance, based on [17,42,48,118–124,165,166].

UV radiation is critical less for the encapsulated Ag stack itself than for the organic and first-surface components of the glazing system. In retrofit and polymer-substrate low-E products, UV photons drive chain scission, photo-oxidation, and yellowing of the carrier film and adhesive, and embrittlement of the encapsulant. Also, they cause loss of the hydrophobic or antireflective top functionality, and all degrade optical clarity and mechanical integrity before the metallic layer is directly attacked [17,166]. The mechanism is inherently photothermal: UV absorption is accompanied by local heating and coupled to moisture and oxygen availability; thus, accelerated protocols such as DIN EN ISO 4892-2 are most predictive when combined with condensation and thermal cycling rather than applied as a stand-alone exposure. This also motivates the use of inorganic or hybrid organic–inorganic barrier stacks (including ALD-grade moisture and UV barriers) for polymer-based and first-surface applications, where the barrier's defect density governs service life more strongly than its nominal thickness [132–134].

Cyclic temperature loading acts through two distinct regimes that must be separated when analyzing durability data. During high-temperature processing (tempering or bending at 620–700 °C), the dominant mechanisms are Ag agglomeration, dewetting, and diffusion into adjacent oxide dielectrics, as evidenced by cross-sectional and EDS-mapped annealing studies of TiO_2/Ag/TiO_2 systems [36,126,127]. During service, the coating suffers repeated moderate variations (typically −20 to +80 °C) in which the mismatch in thermal expansion between the metal, the dielectrics, and the glass generates cyclic interfacial stress. Over many cycles, this causes microcracking, delamination, and the gradual opening of diffusion paths, without any single event being catastrophic. The design consequence is that heat-treatable stacks rely on sacrificial blockers, diffusion barriers, and a controlled annealing window. Yet, these same measures introduce optical and process penalties, so thermal robustness cannot be maximized independently of visible transmittance and emissivity [36,42,126,127].

Moisture penetration is best understood as an electrochemical, defect-mediated process rather than as uniform corrosion. Water reaches the Ag film through pinholes, grain boundaries, and edge defects, where it enables ionic transport, localized galvanic activity, and the growth of white-dot defects that raise emissivity, sheet resistance, and haze [42,118]. At the product level, moisture control is inseparable from the insulating glass unit itself: the edge seal, spacer, and desiccant constitute the outer line of the coating-protection chain, and perimeter-seal failure converts an internally protected soft coating into a directly humidity-exposed surface, with consequent condensation, gas loss, and loss of the declared U-value [118,165]. This coupling explains why identical coatings can exhibit very different operational lifetimes depending on the sealing system, and why durability statements are meaningful only when the coating position and edge-seal context are reported alongside the exposure conditions.

Atmospheric pollutants attack the silver layer through chemically distinct pathways whose diagnostics do not overlap. Reduced sulphur species (H_2S) and SO_2/NO_2 mixtures drive Ag sulfidation with the growth of Ag_2S-type tarnish, producing surface darkening and reflectance loss, whereas chloride-rich marine aerosols promote AgCl-type corrosion and edge-driven, underfilm propagation [119–124]. Because these mechanisms are separated in the classical atmospheric-corrosion and mixed-flowing-gas literature (largely established on bulk or thicker Ag), their extrapolation to films only a few nanometres thick must be made critically: the reduced material inventory, the higher surface-to-volume ratio and the role of the capping layer mean that a given pollutant dose can be disproportionately damaging in a low-E stack. Consequently, sulphur- and chloride-resistant caps, low cap-defect density, and mixed-flowing-gas screening are complementary rather than interchangeable design and qualification responses [48,119–124].

Long-term mechanical durability is a first-order concern wherever the coating is not shielded inside an IGU cavity. First-surface, automotive, and retrofit configurations expose the stack to repeated cleaning, abrasion, handling, and substrate deformation, so that scratch resistance, adhesion, and resistance to interfacial delamination become as decisive as the intrinsic corrosion resistance [17,131]. The primary design responses (hard, abrasion-resistant top layers and denser and more continuous caps) illustrate a recurring trade-off in low-E engineering: increasing the cap thickness or hardness improves mechanical and chemical protection but can absorb visible light and shift the interference condition of the whole dielectric/metal/dielectric design, penalizing transmittance and colour [48]. Mechanical durability should therefore be qualified not by a stand-alone abrasion test but in combination with the optical and chemical metrics, and, where possible, benchmarked against outdoor-versus-accelerated durability curves so that the acceleration factor and failure-mode equivalence can be validated rather than assumed [128,129].

3.3.2. Protective Strategies: Blockers, Caps, Edge Seals, and First-Surface Barriers

Protection starts inside the stack. Thin blockers such as NiCr, Ti, TiO_x, or related sacrificial layers reduce Ag oxidation during reactive sputtering, while dense oxide/nitride caps slow water and ion diffusion during service. However, increasing the thickness of protective layers is not a free solution: metallic blockers can absorb visible light, and high-index oxide caps can shift the interference condition of the whole D/M/D design. Therefore, corrosion resistance must be optimized together with Tv, colour, emissivity, and heat-treatability, as synthesized in Table 9. Work on porous/compact multilayer oxides and protected Ag mirrors shows that the density, continuity, and defect population of the cap layer can matter as much as its nominal thickness [48,125–127].

At the glazing-product level, the first barrier is the insulating glass unit itself. When the low-E coating is applied to surface #2 or #3, the edge seal, spacer, and desiccant

become part of the coating-protection system. Failure of the perimeter seal can convert a protected soft coating into a humidity-exposed surface, leading to condensation, gas loss, optical haze, and degradation of the declared U-value. This also explains why VIG, triple glazing, and retrofitted films require different durability logic: in VIG, the seal and pump-out region dominate long-term vacuum integrity, whereas in retrofit films, the polymer, adhesive, and metallized layer are directly exposed to handling, cleaning, and solar ageing [118,130,131,165].

Table 9. Main corrosion/weathering stressors for low-E coatings and the corresponding diagnostic indicators.

Stress or Exposure	Likely Pathway	Typical Diagnostic Evidence	Design Response
Humidity/condensation	Electrochemical attack through pinholes, grain boundaries, or edge defects	White dots, haze, Ag migration, increased emissivity, and sheet resistance	Dense cap layers, improved edge seals, desiccants, IGU placement on protected surfaces [118,165]
Sulphur-containing gases	Ag sulfidation and dark tarnish products	Surface darkening, reflectance loss, Ag_2S-related surface chemistry	Sulphur-resistant caps, MFG/flowers-of-sulphur screening, lower cap defect density [119–124]
Chlorides/salt aerosols	AgCl-type corrosion and underfilm propagation	Localized spots, edge-driven defects, higher haze after salt/fog exposure	Barrier layers, edge protection, salt-fog, or cyclic corrosion testing [48,125]
Thermal cycling/tempering	Ag diffusion, agglomeration, and stress-driven delamination	SEM/TEM evidence of Ag redistribution, rougher interfaces, and optical drift	Heat-treatable blockers, diffusion barriers, controlled annealing window [36,126,127]
UV, cleaning and abrasion	Damage to polymer films, caps or first-surface coatings	Loss of hydrophobicity, scratches, coating removal, colour change	Hard topcoats, abrasion-resistant caps, cleaning protocols, and first-surface qualification [17,131,166]

3.3.3. Accelerated Weathering Protocols and Diagnostic Sequence

A robust corrosion protocol should not rely on a single accelerated test. Neutral salt spray is useful for chloride sensitivity, mixed-flowing-gas tests reproduce low-level industrial atmospheres, damp-heat or condensation cycles accelerate moisture ingress, and UV/thermal cycling exposes polymeric layers, adhesives, and first-surface coatings to photothermal ageing. For low-E coatings, these exposures are most informative when the optical and electrical properties are measured before, during, and after exposure. A practical sequence is: (i) baseline $T(\lambda)$, $R(\lambda)$, emissivity, colour, and sheet resistance; (ii) controlled exposure blocks; (iii) intermediate visual, haze, and resistance checks; and (iv) post-mortem SEM/AFM, XPS/AES, SIMS, or TEM on failed areas. This sequence separates harmless cosmetic changes from degradation that compromises the thermal function of the glazing [118–120,128,129,166].

Correlating accelerated tests with real service life remains difficult. The problem is not only the acceleration factor, but also the failure-mode equivalence: a damp-heat test may accelerate water diffusion, while sulphur/chlorine mixtures accelerate tarnish, and thermal cycling emphasizes stress and diffusion. If the accelerated test produces a failure mode that is not seen outdoors, the result is conservative or misleading rather than predictive. For this reason, field exposure, reference coupons, optical monitoring, and microstructural

post-analysis should be combined when proposing service-life statements for soft low-E stacks [121–124,128,129].

Figure 13 and the associated discussion draw on several groups of sources. Likins-White et al. [118] for a conceptual IGU degradation and measurement framework that links coating durability to whole-window energy performance; Memon et al. [165] for VIG-specific failure routes at the seal and pump-out region; Abbott and related atmospheric-corrosion studies [119–124] for the mixed-flowing-gas exposure matrix and the sulphur/chlorine tarnishing rationale, best used for test-method justification rather than as direct low-E product data; Kurapati, Bulkin, Fryauf and co-workers [125–127] for protected-silver stacks, ALD barrier schemes, and thermal-processing windows that are analogous to low-E architectures; Kennedy and Brogren [128,129] for outdoor-versus-accelerated reflectance-loss curves of reflective coatings, to be used with caution because solar mirrors and low-E stacks have different optical targets; and the barrier-film literature [132–134] for organic/inorganic barrier-stack schemes relevant to first-surface, retrofit, and polymer-substrate low-E applications.

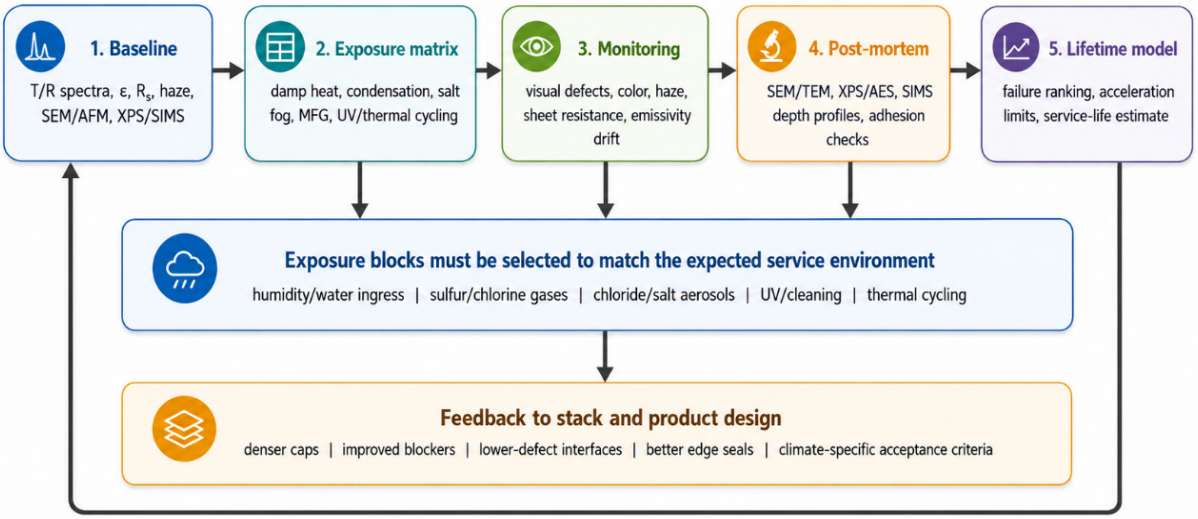


Figure 13. Suggested workflow for corrosion and weather-resistance assessment of low-E coatings. The scheme emphasizes baseline characterization, a multi-stressor exposure matrix, interim monitoring, post-mortem chemical/microstructural analysis, and feedback to stack design, according to [48,118–134,165,166].

3.3.4. Reporting Recommendations and Research Gaps

In the interest of comparability, the durability results should be reported together with the full stack architecture, coating position, edge-seal context, and exposure conditions (chemistry, temperature, relative humidity, duration, UV irradiance or cumulative dose, thermal-cycling range and number of cycles, cleaning/abrasion regime, and measurement interval). Also, the reported results should go beyond a simple pass/fail classification and capture the drift in Tv, RIR, emissivity, sheet resistance, haze, colour coordinates, and adhesion. Samples exhibiting failure should also be spatially mapped, since corrosion in Ag stacks is typically defect-driven and a single pinhole or edge path may dominate the observed optical loss, even when most of the coated area remains intact [48,118–124]. Regarding the main gap, there remains the absence of harmonized durability datasets linking accelerated laboratory exposures, outdoor weathering, microstructure, and building-level energy performance. A second, closely related gap is the scarcity of multi-stressor protocols: most published data isolate a single stressor (humidity) or an individual pollutant (UV or thermal cycling), whereas field degradation is governed by their synergistic action, so single-stressor testing tends to underestimate service-life risk and cannot establish

failure-mode equivalence with outdoor exposure. This is why future work should combine climate-specific, multi-stressor exposure libraries with in-line defect metrology, reference coupons, and chemical imaging—an approach that is especially important for thinner multi-silver stacks, first-surface automotive or polymer applications, and multifunctional glazing in which low-E performance is combined with electrochromic, thermochromic, photocatalytic, or retrofit functions [17,130–134,166].

3.4. Overall Low-E Efficiency

The end-point of every characterization chain is the demonstration of low-E function at product level. The primary quantity is the thermal-infrared emissivity: uncoated glass at $\varepsilon \approx 0.84$, pyrolytic FTO at $\varepsilon \approx 0.15$–0.2, single-silver sputtered stacks at $\varepsilon \approx 0.05$–0.10, and modern multi-silver products down to $\varepsilon \approx 0.02$ [1,5,6]. The emissivity enters the radiative conductance of the IGU cavity and hence the centre-of-glass U-value calculated according to EN 673; experimentally, U-values are verified in guarded hot boxes or by in situ heat-flux methods [14,163,167]. Representative figures illustrate the leverage of the coating: a standard air-filled double glazing at $U \approx 2.8$ W·m^{-2}·K^{-1} improves to ≈1.1 W·m^{-2}·K^{-1} with one low-E coating and argon filling, and below 0.5–0.7 W·m^{-2}·K^{-1} in triple glazing or vacuum glazing incorporating low-E surfaces [6–8,14,25]. Because the radiative and convective terms depend on the boundary conditions, declared and in-service U-values can differ appreciably (differences of up to ~17% were reported for double-glazed low-E units as a function of outdoor temperature), which motivates the reporting of design U-values for cold climates [167]. Alongside U, the solar factor g and the selectivity T_v/g complete the functional characterization, connecting the coating to heating- versus cooling-dominated energy balances (Section 4) [12–14].

A rigorous product-level assessment therefore combines emissivity with U-value, T_v, SHGC/g-value, and LSG rather than ranking coatings by a single metric. In heating-dominated climates, lower U-values and low cavity radiative exchange are decisive, and triple or vacuum glazing with low-E surfaces can substantially reduce heat loss. In cooling-dominated climates, however, SHGC may dominate annual energy performance: a coating with slightly higher U-value but much lower solar gain can outperform a highly insulating product if cooling loads and glare are the controlling design constraints [14,19–23].

This explains why laboratory characterization must be linked to building simulation and comfort metrics. EnergyPlus, TRNSYS, eQUEST, and related dynamic tools translate spectral data into annual heating, cooling, and lighting demand under specific weather, orientation, window-to-wall ratio, and shading assumptions. The same low-E stack can be optimal for a south-facing cold-climate façade and suboptimal for a west-facing façade in a hot climate, where solar-control coatings or external shading may yield greater benefits than additional reductions in emissivity [14,19–23,31].

3.4.1. Efficiency Metrics from Coating to Glazing Units

Overall low-E efficiency should be interpreted as a product-level balance rather than as a single coating property (see Figure 14). Emissivity remains the defining indicator for reducing thermal-infrared exchange, but it is only meaningful when evaluated together with a centre-of-glass U-value, visible transmittance (T_v), solar heat gain coefficient, or g-value, and the light-to-solar-gain ratio (LSG). In practice, a coating with the lowest emissivity is not automatically the best glazing option if it excessively reduces daylight, increases reflectance beyond architectural limits, or shifts SHGC away from the climate-specific optimum [58,67,168].

The spectral route to these metrics begins with wavelength-resolved transmittance and reflectance. Visible-weighted T_v is calculated over the photopic response of the human

eye, while the SHGC/g-value combines directly transmitted solar radiation with secondary inward heat transfer. The U-value, in contrast, is dominated by conduction, convection, and long-wave radiation in the glazing cavity; therefore, the effect of a low-E surface is strongest when the coating faces an air, argon, krypton, or vacuum gap. Reporting only T_v and emissivity can therefore overestimate low-E performance if spacer, edge-seal, frame and cavity effects are ignored [169–171].

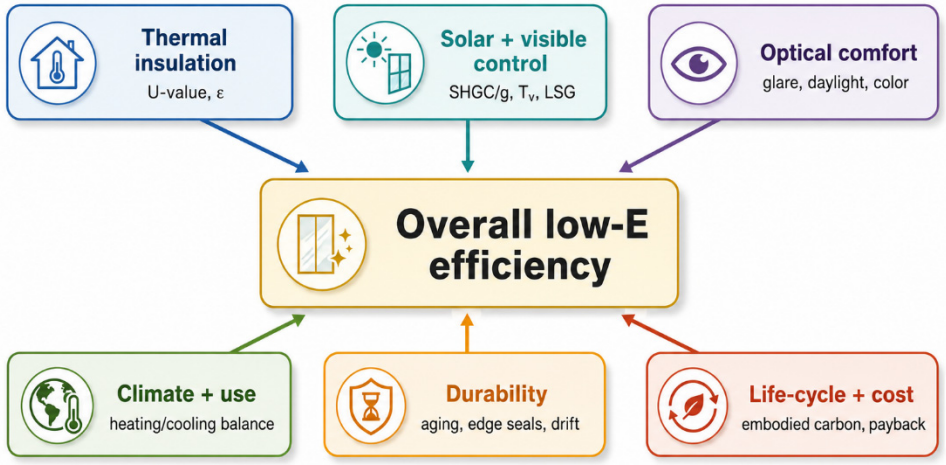


Figure 14. Framework for evaluating overall low-E efficiency. The figure links coating-level descriptors (emissivity, spectral transmittance, and durability) with product-level indicators (U-value, SHGC/g-value, T_v, and LSG), climatic boundary conditions, and life-cycle constraints. Original scheme prepared for this review.

3.4.2. Climate, Orientation, and Use-Profile Dependence

The same low-E coating may lead to different annual savings depending on the climate zone, façade orientation, window-to-wall ratio, shading schedule, and internal loads. In heating-dominated locations, the priority is usually to suppress long-wave heat loss through low U-values while preserving enough solar gain for south-facing façades when passive heating is useful. In cooling-dominated or hot-arid climates, a low SHGC and near-infrared rejection are often more decisive than further reductions in the U-value, because excessive solar gain drives cooling peaks and thermal discomfort [172–174].

Consequently, the efficiency of low-E glazing should be compared through annual building-performance scenarios rather than through isolated centre-pane values, as presented in Figure 15. Dynamic simulations can couple glazing spectra with weather files, occupancy, HVAC schedules, and lighting control, allowing a comparison of heating and cooling demand, daylight autonomy, glare risk, and peak load. This is particularly important for highly glazed offices and glass curtain walls, where modest changes in SHGC or façade orientation can dominate the benefit of lowering emissivity [175–177].

3.4.3. Comparative Product Assessment and Simulation Workflow

A useful assessment workflow starts with the measured optical constants or spectral T/R curves of the coated pane, converts them into standardized glazing indices, and then imports the glazing definition into building simulation. Simple double glazing with argon and a single low-E surface, triple glazing, vacuum glazing, aerogel glazing, solar-control coated glass, and smart glazing cannot be ranked by one universal number because each solution changes multiple boundary conditions at once, such as cavity heat transfer, daylight availability, solar gains, weight, cost, durability, and embodied emissions [55,178–181].

For this reason, recent studies increasingly apply the following multi-objective criteria: annual energy use, peak cooling or heating load, daylight and glare comfort, indoor

thermal comfort, payback time, and life-cycle carbon. The overall efficiency of a low-E solution is therefore maximized when the selected coating and glazing package give the best compromise for a specific building, not when a single material parameter is minimized in isolation (see Table 10) [182–184].

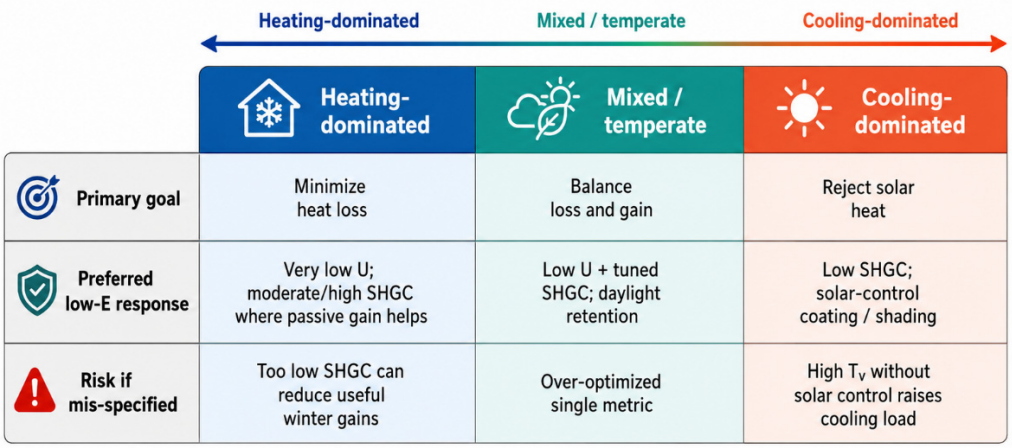


Figure 15. Climate-dependent interpretation of low-E efficiency. Heating-dominated, mixed, and cooling-dominated climates require different balances among U-value, SHGC/g-value, T_v, and shading strategy. Original scheme.

Table 10. Product-level indicators used to assess overall low-E efficiency.

Indicator	What It Captures	Why It Matters for Low-E Assessment	Typical Interpretation
Emissivity, ε	Long-wave infrared radiative exchange at the coated surface	Defines the low-E function but does not include solar/daylight balance	Lower ε reduces cavity radiation and U-value
U-value	Overall thermal transmittance through glazing	Connects coating emissivity with gas/vacuum gap, number of panes, and edge effects	Lower U improves insulation, especially in heating-dominated climates
SHGC/g-value	Fraction of incident solar energy admitted indoors	Determines cooling load and passive solar gain	Low SHGC benefits hot climates; moderate/high SHGC may help winter heating
T_v/VLT	Visible transmittance weighted by eye response	Controls daylight availability and visual acceptance	High T_v is desirable if glare and solar gain remain controlled
LSG	Ratio between T_v and SHGC	Measures daylight admission per unit solar heat gain	Higher LSG indicates stronger spectral selectivity
Annual energy/peak load	Building-level heating, cooling, and lighting response	Reveals climate, orientation, WWR, and HVAC interactions	Most reliable basis for product selection
Durability drift	Change in ε, T_v, SHGC, or haze after exposure	Links initial performance with service life	Stable coatings retain efficiency over years

Aste et al. [168] compare the performance of glazing features in office buildings across climates, connecting glazing selection with climate-specific techno-economic outcomes, while Ran et al. [67] compare solar-control and spectrally selective glass beyond the g-value, illustrating why a single index is insufficient for product ranking. Wang et al. [58] provide a coating-to-building performance example for a heat-blocking coating,

and Karimi et al. [169] analyze the window-to-wall ratio, orientation, and glazing-type scenarios that support facade-level optimization logic. Qahtan et al. [172] report smart-versus-conventional glazing indicators that are suitable for comparative assessments of temperature, daylight, and sustainability, Fang et al. [175] and Tiwari and Kim [177] document triple- and vacuum-glazing configurations and performance plots relevant to low-E surfaces in highly insulating packages, while Gao et al. [179] and Huang and Niu [180] quantify aerogel-glazing performance and its daylight/energy trade-offs, showing alternatives where a very low U-value can affect visible comfort. Finally, Raihan et al. [183] describe a smart-glazing simulation workflow that can serve as a methodological reference for simulation-based efficiency assessment.

3.4.4. Reporting Recommendations

For a complete low-E efficiency statement, the coating stack should be reported together with the pane position, cavity gas or vacuum level, spacer/edge-seal assumptions, spectral data source, calculation standard, weather file, and simulation boundary conditions. The most transparent form is a combined table containing ε, U-value, SHGC/g-value, T_v, LSG, annual heating/cooling/lighting energy, peak load, and at least one durability or ageing indicator. Such reporting avoids the common mistake of presenting a coating as universally efficient without specifying the climate and product context [67,168,182–184].

The practical conclusion is that overall low-E efficiency is a coupled material–optical–thermal–building problem. Coating design controls spectral selectivity and emissivity; glazing design controls heat transfer and condensation risk; building design controls solar exposure, daylight, and user comfort. The highest-performing solution is therefore the one that minimizes total energy and comfort penalties over the service life of the window, not necessarily the one with the lowest initial emissivity.

4. Functional Performance in Glazing Systems

In insulating glass units, the low-E surface suppresses the dominant radiative term of the cavity heat transfer; combined with low-conductivity gas fills (argon, krypton) and warm-edge spacers, this technology defines the mainstream of energy-efficient fenestration [6,14,159]. In heating-dominated climates, the optimum is a low U-value together with a comparatively high solar factor, so that passive solar gains offset heating demand; parametric studies for Central European conditions show that appropriate glazing selection can reduce annual heating demand by ~27% and cooling demand by ~40% in office buildings [21,22]. In cooling-dominated climates, by contrast, reducing the SHGC is significantly more effective than lowering U: lowering SHGC from ~0.69 to ~0.26 can cut cooling loads by about 40% in hot-arid regions, and spectrally selective double-silver products (high T_v, low g) are the technology of choice [20,21]. Dynamic building simulation (EnergyPlus, TRNSYS, and related tools) is the standard method for translating glazing properties into annual energy balances across climate zones [14,19–22].

Low-E coatings are also the enabling component of advanced glazing. In vacuum-insulated glazing (VIG), where conduction and convection are suppressed by the evacuated gap, the residual radiative transfer is controlled entirely by the low-E surfaces, allowing centre-of-glass U-values around 0.3–0.5 $W \cdot m^{-2} \cdot K^{-1}$ [7,8,25]. In electrochromic (EC) smart windows, TCO electrodes (functionally low-E layers) drive a reversible modulation of visible transmittance (e.g., from ~0.60 to ~0.05) and of SHGC over ranges of roughly 0.09–0.47, with simulated reductions in annual cooling demand of 17–50% depending on the climate and control strategy [19,26,27,29]. Thermochromic VO_2-based coatings modulate near-infrared transmission passively at the phase-transition temperature, though durability under prolonged solar exposure and the position of the transition temperature

remain challenges [24,28]. Multifunctional combinations (low-E plus self-cleaning TiO_2, low-E plus semi-transparent photovoltaics, and EC vacuum glazing) represent the current frontier of multilayer glazing systems [3,4,6,29].

At room level, glazing choice propagates into occupant comfort. Low-E and solar-control coatings raise the interior surface temperature of the glass in winter and reduce radiant asymmetry and downdraught near facades, improving the predicted mean vote (PMV) and predicted percentage of dissatisfied (PPD) indices; window-to-wall ratios in the range of roughly 40–60%, combined with appropriate glazing and shading, are generally reported as optimal compromises between daylight, comfort, and energy [19,23]. Comparative simulations in Mediterranean offices found that a low-E solar-control window reduced air-conditioning costs by about 15% while providing the best indoor thermal comfort among the tested options [19,23].

Climate specificity is therefore central to functional performance. In cold and temperate climates, low-E surfaces positioned to retain long-wave heat, together with argon/krypton fills and warm-edge spacers, raise the interior glass-surface temperature and lower the risk of condensation. In hot-arid or hot-humid climates, the same low-E concept is redesigned as solar-control glazing: multi-silver stacks or retrofit films are selected to reject near-infrared solar radiation, reduce cooling load, and maintain usable daylight [19–23,30].

Advanced systems broaden the performance envelope but also introduce new trade-offs. Vacuum-insulated glazing can achieve very low U-values because the evacuated cavity suppresses gas conduction and convection, leaving radiation and edge/pillar conduction as the dominant residual terms. Aerogel glazing provides very low thermal conductivity but may introduce haze and daylight-quality penalties. Electrochromic and thermochromic glazing can dynamically adjust T_v and SHGC, yet cost, response time, switching durability, and user comfort remain barriers to universal adoption [7,8,25–29,31].

At building scale, the most useful comparison is not between isolated products but between glazing packages matched to façade orientation and occupancy. A high-transmittance low-E IGU may be optimal where daylight and winter gains are valuable; a low-SHGC spectrally selective product may be preferable where cooling and glare dominate; and a smart or retrofit solution may be justified when replacement of the whole fenestration system is impractical. This system view is consistent with recent reviews that frame glazing as an interaction among materials, optical design, HVAC demand, daylighting, and occupant comfort rather than as a single-layer coating problem [3,14,19,30,31].

4.1. System-Level Performance Variables

Functional performance in glazing systems is governed by the interaction between the low-E coating and the complete window package, as presented in Figure 16. The coating sets emissivity, spectral selectivity, and colour neutrality, but the delivered performance depends on the pane position, cavity gas, gap width, edge spacer, frame thermal bridge, shading strategy, and control logic. For this reason, a low-E layer that is excellent at the coating level may perform differently when installed in a double IGU, a triple IGU, vacuum-insulated glazing, a retrofit secondary system, or a switchable glazing assembly [61–64,185].

At the product level, the main pathways are as follows: (i) reducing long-wave radiative exchange inside the cavity; (ii) limiting solar heat gain where cooling or glare dominates; (iii) maintaining daylight admission and view quality; and (iv) preserving performance over time through gas retention, seal durability, and surface stability. In practice, these pathways are assessed through coupled indicators such as the U-value, SHGC/g-value, visible transmittance, daylight autonomy, glare probability, interior glass-surface temperature, annual energy use, and peak HVAC load [185–190].

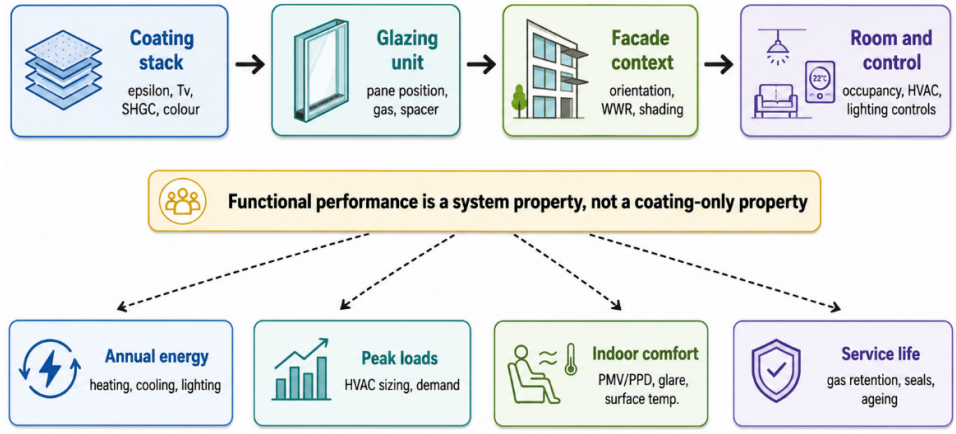


Figure 16. System-level performance chain for low-E glazing. The scheme shows how coating properties propagate through the IGU, facade, and occupied zone, where energy, peak-load, comfort, and service-life outcomes are ultimately assessed [61–64,185–190].

4.2. Climate, Orientation, and Control Strategy

The same low-E technology supports different functional objectives depending on the climate. In heating-dominated contexts, a low U-value, high interior surface temperature, and controlled condensation risk are usually more important than the lowest possible SHGC. In cooling-dominated climates, low SHGC, low peak cooling load, and acceptable daylight quality prevail; a product with very low emissivity but excessive solar gain can be less useful than a spectrally selective solar-control glazing. Mixed climates require a compromise, and the optimum shifts with orientation, window-to-wall ratio, internal gains, occupancy schedule, and the presence of exterior shading or automated controls [62–65,186–188,191–195].

The control strategy is especially important for dynamic or responsive systems. Electrochromic, thermochromic, photochromic, and switchable-U-value windows extend the performance range by changing optical or thermal behaviour over time, but their benefits depend on sensor placement, switching thresholds, occupant override, response time, and integration with HVAC and lighting controls, as synthesized in Figure 17. For static low-E systems, functional robustness is achieved by selecting the coating family and pane position so that the installed glazing remains useful under realistic seasonal and daily solar angles [61–65,189,190].

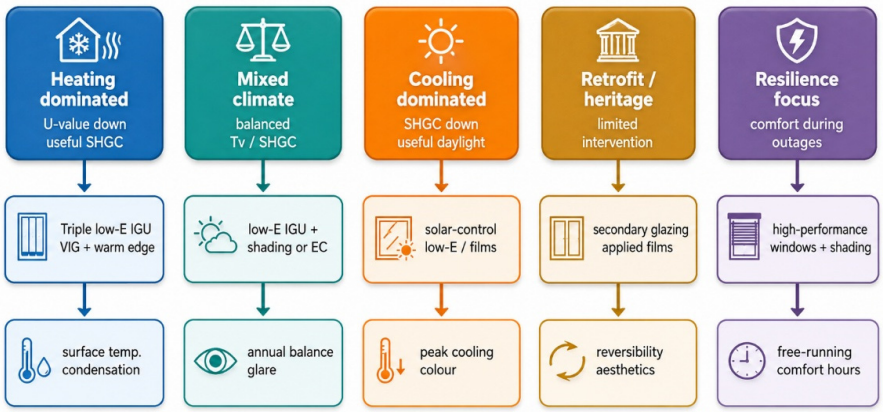


Figure 17. The figure summarizes how heating-dominated, mixed, cooling-dominated, retrofit, and resilience-oriented applications lead to different low-E or advanced-glazing choices [185,186,191–199].

At the product and building scales, the main system options can be compared based on their dominant functional advantages, key design variables, main risks, and the indicators

by which they should be evaluated. The double low-E IGU remains the mainstream route for reducing heat loss at a moderate cost; its performance is governed by pane position, gas fill, gap width, spacer, and coating type; it may be insufficient for very cold climates or high-performance targets, and it is best evaluated through U-value, SHGC, T_v, interior glass-surface temperature, and annual energy. The triple low-E IGU adds insulation and comfort near the facade at the price of reduced daylight, higher embodied material, and a thicker, heavier unit that the frame must accommodate; the number and location of low-E surfaces and the gas cavities are the key variables, and daylight autonomy, condensation risk, and frame compatibility should complement the U-value. Vacuum-insulated glazing reaches very low centre-of-glass U-values within a thin assembly, but edge and pillar thermal bridges, manufacturing costs, and durability validation remain its main risks, so centre- and edge-of-glass U-values, mechanical stress, and service-life indicators should be reported together.

Solar-control low-E glazing reduces the cooling load and glare while retaining useful daylight; silver-stack design, target SHGC, colour, and the interplay with external shading are its design variables; over-selection can introduce daylight or colour penalties, and the SHGC/g-value, T_v, LSG, glare probability, and peak cooling load are the relevant indicators. Electrochromic and other smart glazing provide dynamic control of heat gain and daylight, with the switching range, control algorithm, TCO durability, and user override as design variables; cost, response time, cycling durability, and controls complexity are the main barriers, and annual cooling/heating/lighting energy, glare, comfort hours, and switching-cycle counts are the appropriate metrics. Finally, retrofit films and secondary glazing improve existing windows without full replacement; film type, adhesive, mounting surface, reversibility, and esthetics must be selected with care because long-term adhesion, optical distortion, and maintenance are the typical risks, while payback time, cooling reduction, visual comfort, and surface durability are the natural evaluation indicators [61–65,185–199].

4.3. From Product Testing to Building-Scale Evidence

Functional claims should not rely on a single property measured under ideal conditions. A credible assessment combines product-level data (U-value, SHGC, optical spectra, gas fill, and spacer details), climate files, facade geometry, shading/control assumptions, and an occupied-zone model. Dynamic simulation is therefore needed to distinguish a coating that is thermally excellent from a glazing system that actually reduces annual energy, peak demand, and discomfort. This is particularly important for highly glazed facades, schools, offices, and retrofits, where occupant comfort, daylight quality, and resilience during heat waves or cold snaps may dominate over simple centre-of-glass performance [185,191–199].

For experimental or case-study papers, the most useful evidence includes side-by-side comparisons of glazing alternatives, measured indoor temperature or surface temperature, daylight/glare indicators, HVAC energy or peak-load outputs, and user-comfort metrics.

For a system-level review diagram that places glazing within the building envelope, Akram et al. [185] and Cannavale et al. [199] are suitable starting points. Electrochromic performance plots covering cooling/lighting energy and visual comfort can be adapted from DeForest et al. [61], Lee and Tavil [63], and Sibilio et al. [65], while the effect of shading and overhang design on cooling and lighting demand is documented by Tzempelikos and Athienitis [62] and Favoino et al. [64]. Climate, window-to-wall-ratio and orientation comparisons across building contexts are provided by Calautit et al. [191], Chen et al. [192], and Er et al. [195]; occupant-centred comfort and resilience evidence, including cooling energy and survivability metrics, are available in Oliveira and Corvacho [189], Fang et al. [193], and Krelling et al. [198]; and the visual/thermal performance of retrofit window films

and luminous factors is covered by Khaled and Berardi [188], Shaker [196], and Tabbah et al. [197].

Figures 18 and 19 illustrate two complementary types of system-level evidence: climate-resolved energy performance and occupant-centred thermal resilience. Figure 18 exemplifies the climate-resolved performance evidence for dynamic (electrochromic) glazing recommended above, while Figure 19 exemplifies the occupant-centred resilience evidence that complements conventional energy metrics [61,198].

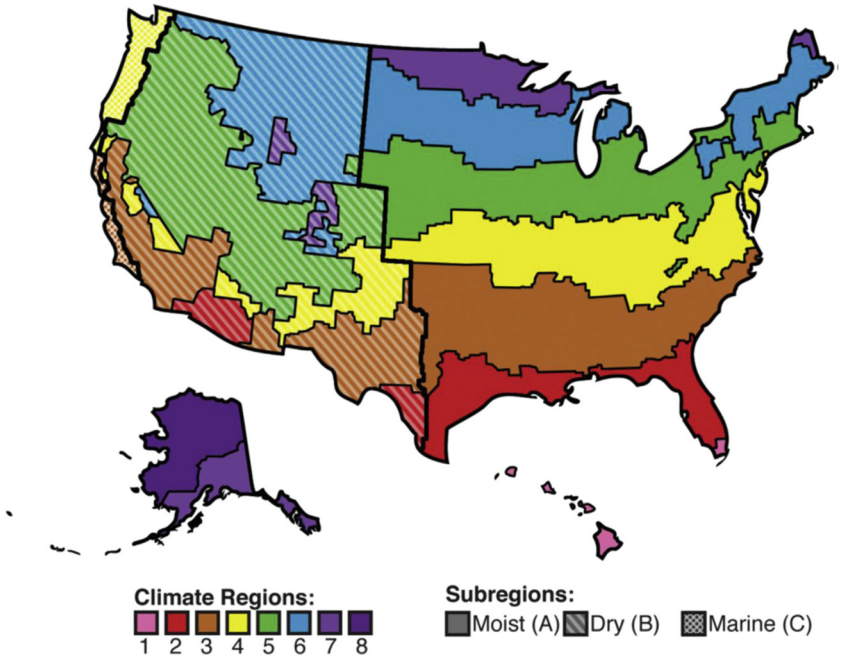


Figure 18. Climate-zone-resolved energy and CO_2 savings potential of near-infrared switching electrochromic glazing across U.S. building stock, illustrating how the benefit of dynamic low-E systems depends on climate, building type, and control strategy. Adapted from DeForest et al. [61].

4.4. Life-Cycle Environmental Performance, Material Criticality, and Circularity

The environmental value of low-E glazing cannot be established from operational energy savings alone. A complete assessment must compare the additional embodied impacts of the coating and glazing assembly with the heating, cooling, and lighting energy avoided throughout the service life of the product. Published window LCAs remain difficult to compare because they use different system boundaries, reference service periods, climates, electricity mixes, replacement scenarios, and functional units [200–203]. For low-E glazing, an appropriate functional unit should therefore specify one square metre of installed glazing delivering defined values of the U-factor, SHGC/g-value, visible transmittance, and durability over a stated reference service life and under a defined climatic and operational context. Cradle-to-gate results should be distinguished from cradle-to-grave assessments that include transport, installation, maintenance, replacement, operational energy, and end-of-life treatment.

The glass substrate normally dominates the mass of the glazing system, and float-glass manufacturing requires high-temperature melting and refining. Reported estimates place the energy intensity of glass production at approximately 7–8 GJ per tonne, although the value varies with furnace technology, fuel mix, cullet content, and production yield [204,205]. The low-E coating represents only a small fraction of the total product mass, but its environmental contribution is not necessarily proportional to its thickness. Magnetron sputtering requires vacuum pumping, plasma power, cooling, target manufacture, chamber cleaning, and management of material deposited outside the useful substrate

area. APCVD avoids a separate vacuum stage and can use the thermal state of the float ribbon, but requires precursor production, delivery, conversion, and exhaust treatment. ALD can minimize layer thickness and provide highly conformal barriers, although precursor synthesis, purging, vacuum operation, and long cycle times may become important environmental hotspots. Consequently, the deposition yield, line speed, target or precursor utilization, rejected-pane rate, and electricity source should accompany a nominal film thickness when coating routes are compared [95,96].

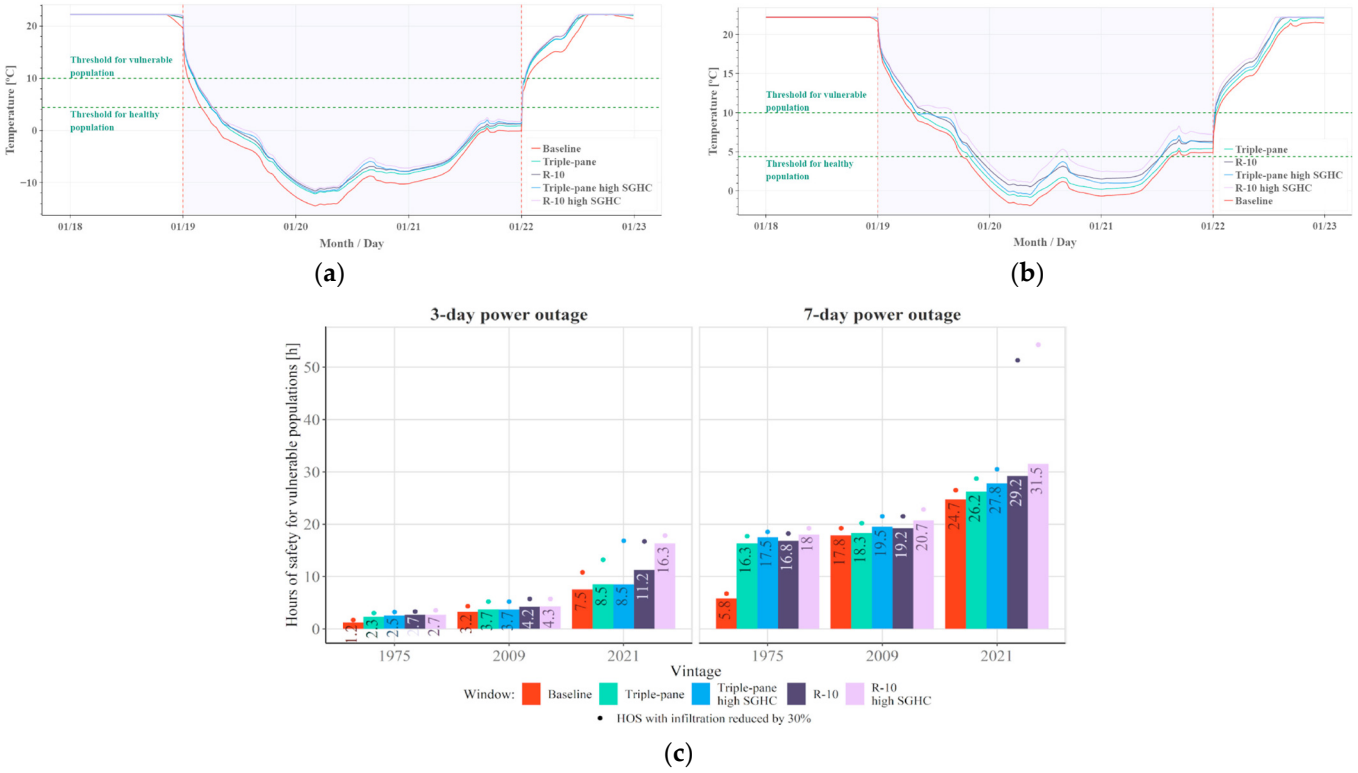


Figure 19. Effect of high-performance windows on indoor-temperature decay and occupant thermal survivability during a power-interrupted cold snap. Simulated indoor-air-temperature histories during a three-day power outage in Chicago for residential buildings constructed in (**a**) 1975 and (**b**) 2021, equipped with baseline, triple-pane, and R-10 window configurations; (**c**) corresponding hours of safety for vulnerable occupants across building vintages, window technologies, and three- and seven-day outage scenarios. The dashed lines in panels (**a**,**b**) represent indicative thermal-survivability threshold for healthy and vulnerable populations. They are included solely as visual guides and should not be interpreted as strict or universally applicable limits. Adapted from Ref. [198].

Material criticality introduces a second dimension. Ag-based soft coatings contain only nanometre-thick metallic layers, but double- and triple-Ag architectures increase noble-metal demand and the importance of target utilization, chamber losses, production scrap, and recovery. Reducing Ag thickness through improved seed-layer control may lower material intensity, but only if continuity, emissivity, colour, and durability are preserved. ITO presents a different concern because indium is obtained mainly as a byproduct of other metal-processing chains and is exposed to supply concentration and competing demands from electronic, photovoltaic, and display applications. AZO, GZO, and FTO reduce or eliminate indium dependence, but substitution cannot be evaluated from elemental abundance alone. A replacement coating may require greater thickness, a higher deposition temperature, additional processing, or more frequent replacement if its conductivity or environmental stability is inferior. LCA studies comparing ITO and

AZO in transparent-electrode applications demonstrate that the result depends on both production burdens and functional performance, although those device-specific findings cannot be transferred quantitatively to architectural glazing without dedicated coating and service-life inventories [206].

Durability is therefore an environmental parameter as well as a technical one. A coating with slightly higher embodied impact may provide a lower whole-life burden if it maintains its optical and thermal properties for a substantially longer period. Conversely, premature corrosion of Ag, degradation of an electrochromic layer, loss of cavity gas, edge-seal failure, or irreversible optical drift can eliminate the expected operational benefit by requiring early replacement of the complete insulating glass unit. Whole-life assessments should consequently include realistic service-life distributions, maintenance and replacement scenarios, performance degradation, local climate, façade orientation, control strategies, and the future carbon intensity of the energy supply [200–203]. This is particularly important for advanced multilayer, vacuum, and dynamic glazing, whose greater manufacturing complexity must be justified by verified operational savings and service-life performance.

Flat glass is technically recyclable, but practical closed-loop recycling remains limited. End-of-life insulating glass units must be collected, separated from frames, and disassembled to remove spacers, sealants, laminated interlayers, fittings, and other contaminants before the glass can return to a float-glass furnace. The nanometre-scale low-E coating is not necessarily the principal barrier; the difficulty arises from the complete multi-material assembly, contamination requirements, fragmented collection systems, and the absence of reliable product and coating traceability [204,205]. Increased cullet use reduces demand for primary raw materials and melting energy; reported industrial estimates indicate an energy reduction of approximately 3% for each 10% increase in cullet content, although the achievable benefit depends on cullet quality and furnace operation [205]. Direct reuse or remanufacture of intact panes or insulating glass units can avoid remelting altogether, but requires the reliable identification of glass type, coating position, as well as the performance, dimensions, heat-treatment history, and residual service life.

Future assessments should therefore report, in addition to emissivity, sheet resistance, and SHGC, at least the following environmental indicators: embodied energy and global-warming potential per square metre; Ag, In, and other relevant material intensities per square metre; deposition yield and target or precursor utilization; expected service life; recycled content; replacement frequency; end-of-life recovery assumptions; and climate-specific operational carbon savings. These indicators should not be combined into a universal ranking without explicit weighting. Instead, they should support a transparent comparison between coatings that deliver equivalent optical, thermal, durability, and product-level functions.

4.5. Reporting Recommendations for Functional Performance

A functional-performance statement should identify the complete glazing assembly, not only the low-E coating. Minimum reporting should include the coating type, pane position, number of panes, cavity gas or vacuum level, spacer/edge details, frame assumptions, orientation, WWR, shading, climate file, simulation tool, control schedule, and comfort metric. For smart or responsive systems, the switching logic and the boundary between automatic control and occupant override should be stated explicitly. For retrofit systems, reversibility, installation surface, adhesive ageing, cleaning/maintenance, and compatibility with existing glass should also be reported [64,185,188,191–199].

The main conclusion for glazing systems is that low-E coatings become functionally valuable only when they are integrated into a product and facade strategy. The preferred

option in a cold residential retrofit, a highly glazed hot-climate office, a school classroom, a heritage facade, or an outage-resilient building can be different, even when the same low-E material family is available. Thus, chapter-level performance evaluations should connect laboratory optical/thermal data with system configuration, user comfort, and building-operation evidence [61–65,185–199].

5. Future Trends in Low-E Glass Manufacturing

Across four decades of reviews, a consistent set of open problems still emerges. First, durability and ageing: the literature repeatedly notes the absence of long-term, microstructure-resolved degradation data for low-E stacks and the lack of harmonized accelerated-test protocols that correlate with service life; recent dedicated reviews of soft low-E durability, corrosion mechanisms, and protective strategies confirm that this remains the field's critical gap [1,18,34,164]. Second, materials sustainability: indium scarcity keeps the pressure on scalable indium-free TCOs (AZO, GZO, doped SnO_2) that match ITO without performance penalties, particularly at the small thicknesses relevant to low-E stacks [24,33,38]. Third, design methodology: transfer-matrix modelling combined with multi-objective numerical optimization is increasingly used to design thinner multi-silver stacks that maximize selectivity, neutral colour, and mechanical robustness simultaneously, and machine learning methods have begun to assist both stack design and the interpretation of complex characterization data such as ToF-SIMS hyperspectral volumes [18,41].

Application-driven trends include first-surface low-E coatings for polymer glazing and automotive applications, where the coating must survive direct environmental exposure without an IGU cavity [18]; radio-frequency transparency (5G compatibility) of metallic stacks, addressed through laser patterning or TCO-based designs; seasonal adaptivity through the integration of static low-E with thermochromic, electrochromic, and photovoltaic functions [24,26–29]; and the coupling of low-E surfaces with vacuum insulated glazing to approach opaque-wall insulation levels in transparent components [7,8,25]. Finally, the harmonization of optical-thermal metrology (EN 410/EN 673/ISO 9050 with angle-dependent and ageing-aware extensions) and open access to manufacturer performance data are prerequisites for a reliable whole-building assessment of the coming generation of multifunctional glazing [13,14,19].

A major manufacturing trend is digital, model-based optimization of the stack. Multi-objective algorithms can search layer thicknesses and material combinations that simultaneously maximize visible transmittance, minimize emissivity and SHGC, preserve neutral colour, reduce the number of layers, and keep the solution tolerant to process drift. Such methods are particularly valuable for multi-silver stacks, where a small thickness error can shift colour or raise absorption, and for indium-free designs where the material property space is less mature [14,18,38].

Another trend is the integration of low-E function with adaptive and energy-generating glazing. Electrochromic, thermochromic, luminescent, photovoltaic, BIPV/PCM, and vacuum-hybrid concepts all rely on transparent conductors, selective reflectors, or barrier layers that overlap technically with low-E coating science. The manufacturing challenge is to combine these functions without excessive cost, haze, colour shift, embodied carbon, or durability penalties. As a result, future low-E research will likely be judged not only by ε and T_v, but also by service-life evidence, repairability, recyclability, and compatibility with high-throughput coating lines [3,17,19,24,26–31].

5.1. Digital, Model-Based and AI-Assisted Manufacturing

The next generation of low-E glass manufacturing is expected to move from empirical stack tuning toward data-rich, model-based designs. In conventional practice, a multilayer

design is iterated through optical modelling, deposition trials, optical testing, and durability checks. Future production lines can compress this loop by combining transfer-matrix models, measured optical constants, process history, and machine-learning-assisted inverse design, as presented in Figure 20. Such workflows are especially relevant for complex D/Ag/D, multi-silver, and hybrid oxide/metal/oxide stacks because many combinations of layer sequence and thickness can lead to similar visible transmittance but different emissivity, colour, angular behaviour, and ageing responses [207–211].

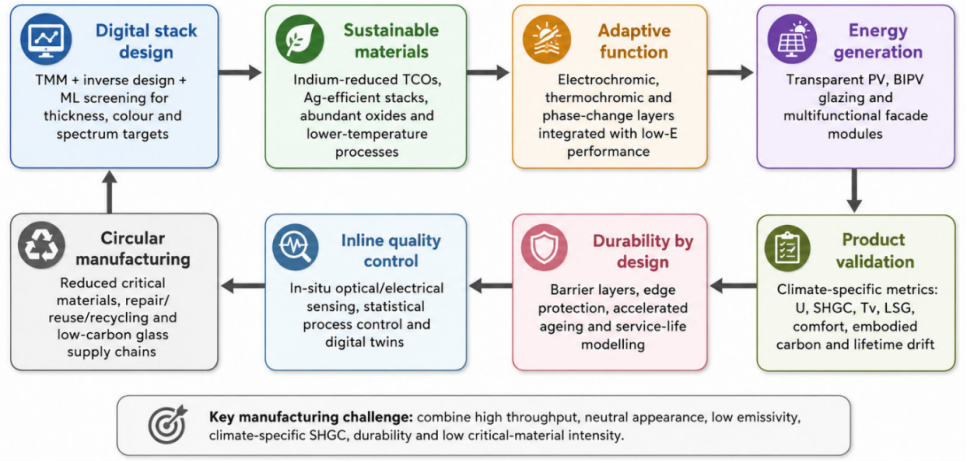


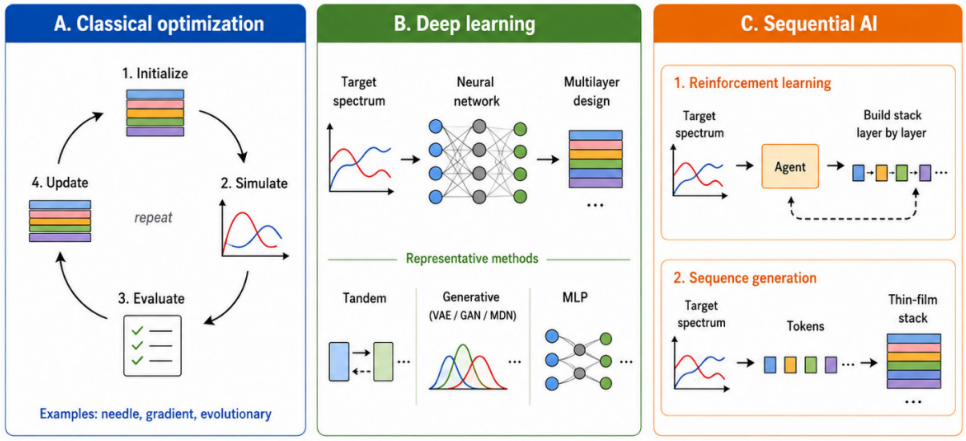
Figure 20. Future-oriented roadmap for low-E glass manufacturing, linking digital designs, sustainable materials, adaptive functions, inline quality control, durability-by-design, and system-level validation. Original scheme prepared for this review.

A practical industrial target is therefore not only the minimization of emissivity, but the simultaneous optimization of a neutral colour, high visible transmittance, controlled SHGC, low sheet resistance, process robustness, and service-life stability. In this context, artificial intelligence is not a replacement for physical modelling; rather, it acts as a screening and inverse-design layer that proposes feasible candidates, while optical modelling and experimental deposition validate manufacturability (Figure 21). This is particularly useful when new materials or reduced-silver stack designs must be explored under constraints imposed by target availability, line speed, residual stress, coating colour, and edge-deletion compatibility [82,207–211].

It is useful to frame the computational toolbox for low-E coatings as a multiscale materials-science pipeline rather than as a set of isolated techniques. At the atomistic scale, first-principles calculations predict intrinsic materials properties. At the thermodynamic and kinetic scales, computational phase-stability methods describe interdiffusion, segregation, and phase formation during processing and ageing. At the device scale, optical transfer-matrix and dispersion models translate the resulting optical constants into measurable spectra (as detailed in Section 3.2.2). Spanning all scales, machine learning and AI-assisted optimization accelerate screening and inverse design (Figures 20 and 21). Viewing these as complementary tiers of a single workflow clarifies where each method adds predictive value and where experimental validation remains indispensable [207–211].

At the atomistic scale, first-principles methods (mostly density functional theory; DFT) are increasingly used to predict the properties that govern low-E behaviour before deposition. For transparent conducting oxides, DFT informs band structure, carrier effective mass, defect and dopant formation energies, and the dielectric function that sets the plasma frequency and thus the onset of infrared reflection. For the metallic layer and its interfaces, first-principles calculations address the work function, surface energy and adhesion, and energetics of oxygen incorporation and of silver transport across seed,

blocker, and dielectric boundaries. DFT therefore supplies materials-level inputs (optical constants, defect chemistry, and interface stability) that downstream transfer-matrix and thermodynamic models consume, offering a rational basis for selecting dopants, seed layers, and diffusion barriers rather than relying solely on empirical screening [212].



	Optimization	Deep learning	RL	Sequence generation
Speed	Low	High	Medium	High
Global design	Yes	Yes	No	Yes
Diversity	Medium	High	Medium	High
Flexibility	High	Medium	Medium	High

Figure 21. Overview of inverse-design approaches for optical multilayer thin-film structures and their relevance to AI-assisted low-E coating design. Classical methods use an iterative loop of optical simulation, objective-function evaluation, and structural updating through needle, gradient-based, or evolutionary optimization. Deep-learning approaches learn mappings between target optical responses and multilayer structures using tandem networks or probabilistic generative models such as VAE, GAN, and MDN. Sequential approaches additionally construct the multilayer stack layer by layer through reinforcement learning or conditional sequence generation. Redrawn and adapted from Ref [207].

At the mesoscale, computational thermodynamics (most notably the CALculation of PHAse Diagrams; CALPHAD) addresses the phase-stability and interdiffusion phenomena that dominate the durability of silver-based stacks. By combining assessed Gibbs-energy databases with kinetic descriptions, CALPHAD-type modelling can rationalize the driving forces for silver agglomeration and dewetting, the tendency toward oxide/metal interdiffusion during tempering, and the stability windows of seed and blocker layers, thereby complementing the experimental annealing and corrosion studies discussed in Section 3.3. Coupling thermodynamic and kinetic modelling with atomistic inputs from DFT and optical outputs from transfer-matrix simulation would allow degradation-relevant processing windows to be predicted rather than discovered empirically, which is particularly valuable for heat-treatable and reduced-silver designs [213].

An AI-generated architecture should therefore be treated as a candidate hypothesis rather than as a directly transferable production recipe. Industrial validation requires a staged workflow. First, candidate designs should be filtered using hard manufacturing constraints, including available target or precursor materials, cathode and chamber sequences, minimum stable layer thicknesses, deposition temperatures, reactive-gas compatibility, line speeds, heat-treatment requirements, and the permitted material and cost envelope. Second, selected designs should be deposited on witness coupons and verified by spectral transmittance and reflectance, ellipsometry, emissivity, sheet resistance, colour, and structural and interfacial analyses. Third, design-of-experiment or machine-learning-assisted

process optimization should determine the process window and its sensitivity to deviations in thickness, composition, pressure, gas flow, and power rather than merely reproducing a single nominal optimum. Fourth, transfer to pilot and production coaters must demonstrate dynamic-deposition equivalence, across-pane uniformity, batch reproducibility, and tolerance to target erosion, gas-distribution gradients, chamber history, and substrate-speed variations. Finally, the complete coated product must retain its performance after tempering, bending, lamination, edge deletion, IGU assembly, and accelerated ageing (Figure 22). The resulting production and durability data should be returned to the model so that subsequent inverse-design cycles penalize fragile or unmanufacturable solutions. Only candidates that satisfy the optical targets within sufficiently wide process, yield, and service-life windows should be considered industrially validated [75–94,118–134,165,166,207–211].

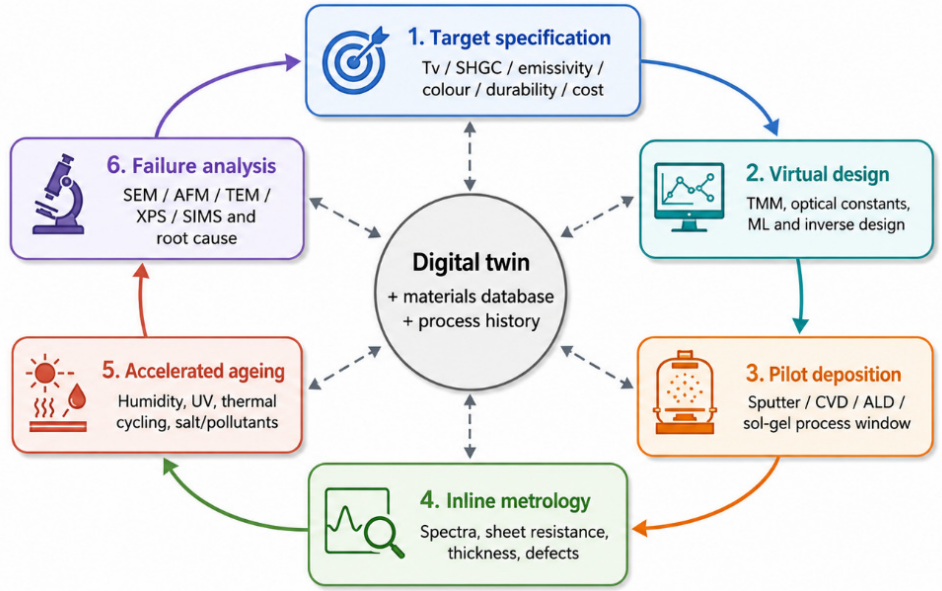


Figure 22. Closed-loop workflow for future low-E stack manufacturing, showing feedback between optical targets, virtual design, pilot deposition, inline metrology, accelerated ageing, and failure analysis. Original scheme prepared for this review.

5.2. Sustainable and Lower-Criticality Material Platforms

A second trend concerns the material basis of low-E stacks. Commercial soft-coat products remain dominated by ultrathin Ag layers, while hard-coat products rely strongly on robust TCOs such as FTO. Future manufacturing will probably emphasize reduced critical-material intensity rather than a single universal replacement. Relevant directions include thinner Ag layers stabilized by seed/wetting layers, indium-reduced or indium-free transparent conductors, abundant oxide dielectrics, lower-temperature deposition for polymer-compatible glazing, and barrier concepts that enable first-surface applications without sacrificing durability [82,214–218].

Sustainability also changes the meaning of performance. A coating that gives a slightly better laboratory spectrum may not be the best future product if it requires scarce elements, high thermal budgets, poor repairability, or short service life. For this reason, future low-E manufacturing should report not only optical and thermal metrics, but also the material intensity per square metre, expected lifetime, process yield, recyclability of coated cullet, and compatibility with laminated, vacuum-insulated, or BIPV glazing architectures [214–220].

These future trends can be summarized together with their manufacturing implications and their main barriers. AI-assisted inverse design promises faster stack screening, fewer pilot runs, and better multi-objective trade-offs, but depends on reliable materials databases

and on constraints that keep the proposed designs manufacturable [207–211]. Ag-efficient soft-coats aim to lower noble-metal use while retaining low emissivity, with layer continuity, island growth, corrosion, and colour control as the critical barriers [18,35–37,42,43,45–48]. Indium-reduced or indium-free transparent conductors decrease the exposure to critical-material supply chains, although carrier mobility, stability, haze, and low-temperature process limits still constrain them [82,214].

Adaptive-glazing integration combines the low-E function with electrochromic, thermochromic, or phase-change responses, with cycling durability, switching uniformity, and cost as the limiting factors [215–219]. Transparent-PV and BIPV glazing turn the window into both an envelope and energy generator, provided that transparency, colour, electrical yield, and thermal performance can be balanced [220–227]. Inline metrology and digital twins enable closed-loop process control in sputter, CVD, and ALD lines, but require data integration across tools, batches, and ageing tests [207–211,227]. Finally, circular manufacturing (design for reuse, recycling, and low-carbon production) must solve the separation of coated-glass streams and the management of complex laminates [223–227].

5.3. Adaptive, Reconfigurable, and Energy-Generating Low-E Glazing

Future low-E glazing is also likely to become more multifunctional. Instead of a static coating designed for a single spectral balance, adaptive glazing can change the optical response with voltage, temperature, or phase state. Electrochromic and plasmonic nanocrystal approaches can tune visible and/or near-infrared transmission, thermochromic VO_2-related systems can respond passively to temperature, and phase-change thin films can modify the infrared response while maintaining a relatively neutral visible appearance [215–219].

Another convergence path is the combination of low-E functionality with energy generation. Transparent and semi-transparent photovoltaics, as well as wavelength-selective organic PV, BIPV glazing, and facade-integrated PV systems can transform the window from a passive thermal weak point into an active envelope component, as presented in Figure 23. For manufacturing, this raises new stack-compatibility questions: optical coatings must be coordinated with electrodes, absorbers, encapsulants, lamination materials, edge seals, and building-code requirements [220–227].

5.4. Fundamental Scientific Challenges Limiting Commercialisation

The identification of a promising material or optical architecture does not, by itself, establish commercial readiness. Low-E products operate through nanometre-scale layers whose optical, electrical, chemical, and mechanical functions are strongly coupled. A laboratory coating may therefore exhibit excellent initial transmittance or emissivity while remaining unsuitable for large-area production, heat treatment, lamination, edge deletion, environmental exposure, or long-term operation. The fundamental challenges that currently separate emerging coating concepts from commercially qualified products are summarized in Table 11.

These challenges should be interpreted as coupled rather than independent. Reducing the thickness of Ag decreases noble-metal consumption and may improve visible transmission, but it simultaneously increases sensitivity to nucleation, roughness, contamination, and local thickness variations. Adding blockers or diffusion barriers can stabilize Ag, but may introduce visible absorption, residual stress, additional interfaces, and longer process sequences. Similarly, incorporating adaptive or energy-generating functions increases the number of material combinations that must remain chemically, mechanically, optically, and electrically compatible throughout manufacturing and service.

Table 11. Fundamental scientific challenges limiting the commercialisation of emerging low-E and multifunctional glazing technologies.

Scientific and Technological Challenge	Underlying Scientific Problem	Consequence for Commercialisation	Required Research or Qualification Milestone	Ref.
Continuity of ultrathin Ag layers	Ag deposited at very small thickness initially nucleates as isolated three-dimensional islands. The transition to a continuous film depends on substrate chemistry, seed-layer structure, surface energy, roughness, contamination, deposition rate, and energetic-particle bombardment. Below or close to the percolation threshold, small local variations cause large changes in absorption, sheet resistance, and IR reflectance	Coatings may exhibit haze, colour variations, unstable sheet resistance, increased emissivity, and poor across-pane reproducibility. A laboratory optimum obtained on a small substrate may not remain continuous over square-metre panes	Reproducible formation of smooth, continuous Ag at reduced thickness, with narrow across-pane and batch-to-batch distributions of sheet resistance, emissivity, haze, and colour under industrial line conditions	[18,35–37,42,43,45,46]
Chemical and mechanical stability of buried interfaces	Seed/Ag, Ag/blocker, and blocker/dielectric interfaces are chemically metastable and may change during reactive deposition, tempering, lamination, thermal cycling, or humidity exposure. Interfacial reactions, roughening, oxidation, stress accumulation, and adhesion loss can occur even when the bulk layers remain nominally unchanged	Initial optical performance may deteriorate after heat treatment or ageing. Local interfacial failure can generate delamination, white-dot defects, increased absorption, colour shift, and loss of electrical continuity	Interface-specific stability maps linking composition, thickness, roughness, residual stress, and thermal history to optical and adhesion changes after realistic manufacturing and accelerated-ageing sequences	[35,36,41,42,44–48,118–129,165,166]
Suppression of Ag diffusion, agglomeration, and corrosion	Ag atoms can diffuse through grain boundaries, defects, porous oxides, or chemically reactive interfaces. Moisture, sulphur- and chlorine-containing species, oxygen, elevated temperature, and stress accelerate migration, agglomeration, and tarnishing. Protective layers must remain dense despite being only a few nanometres thick	Incomplete barriers can lead to gradual emissivity increase, loss of reflectance, isolated corrosion defects, and premature failure. Thicker barriers may improve protection but penalize transmittance, colour, stress, and production time	Pinhole-free diffusion and corrosion barriers with negligible optical penalty, demonstrated after combined humidity, pollutant, UV, thermal-cycling, and heat-treatment exposure. For ALD-derived barriers, large-area precursor utilization, cycle time, adhesion, and compatibility with the underlying Ag stack must also be demonstrated	[36,42,44,47,87–89,121–134]

Table 11. *Cont.*

Scientific and Technological Challenge	Underlying Scientific Problem	Consequence for Commercialisation	Required Research or Qualification Milestone	Ref.
Stable indium-free transparent conducting oxides	AZO, GZO, and doped SnO_2 must combine high carrier concentration and mobility with low visible absorption. Dopant activation, grain-boundary scattering, oxygen stoichiometry, thickness, surface roughness, and environmental exposure can affect conductivity and optical responses, particularly in ultrathin or low-temperature-processed films	Laboratory resistivity values may not translate into stable large-area electrodes. Conductivity loss, haze, non-uniform dopant incorporation, or humidity sensitivity can prevent replacement of ITO in demanding low-E or smart-window architectures	ITO-comparable optical and electrical performance at application-relevant thickness, combined with stable properties after heat treatment, humidity exposure, thermal cycling, and large-area deposition using abundant and industrially available precursors or targets	[24,32–34,38,50,52,81–85]
Multifunctional coating architectures	Low emissivity, solar control, UV protection, self-cleaning, antireflection, electrical conductivity, switching, and energy generation frequently require materials with incompatible refractive indices, deposition conditions, chemical environments, or preferred thicknesses. Adding functionality can also increase haze, parasitic absorption, stress, and interface density	Improvements in one function may degrade another. A coating with good switching amplitude or photovoltaic output may no longer provide neutral colour, low emissivity, adequate daylight transmission, or acceptable durability	Simultaneous optimization and experimental validation of all required functions within the complete product architecture, followed by durability testing that monitors each function rather than only initial optical performance	[24,26–29,40,51–60,67,214–227]
Scalable deposition of complex multilayers	Complex stacks accumulate thickness errors, interfacial roughness, stress, contamination, and optical-constant variations across successive deposition zones. Reactive-process drift and target ageing can alter layer properties, even when nominal deposition times remain unchanged	Designs optimized numerically may be too sensitive to realistic process fluctuations. Small deviations can cause visible colour non-uniformity, increased emissivity, local defects, or unacceptable batch-to-batch variation over jumbo panes	Robust stack designs with quantified process tolerances, combined with calibrated in-line optical and electrical metrology, closed-loop process control, and demonstrated capability indices for across-pane and batch-to-batch uniformity	[16,75–80,87–94,207–211]
Integration with smart-window and energy-generating technologies	Electrochromic, thermochromic, phase-change, and photovoltaic layers introduce ion transport, electrical contacts, switching kinetics, thermal expansion mismatch, encapsulation, and control-system requirements. Transparent electrodes may themselves alter the low-E spectral response, while switching layers can change colour and emissivity non-uniformly	Full-size panes may switch slowly or non-uniformly, lose modulation after cycling, require complex busbars and controls, or suffer incompatibility with laminants, edge seals, IGU cavities, and building electrical systems	Uniform full-pane switching, stable optical and thermal modulation over a product-relevant number of cycles, low standby energy, fail-safe operation, compatible edge sealing and electrical connection, and qualification of the complete glazing assembly rather than the active film alone	[26–29,53–57,214–225]

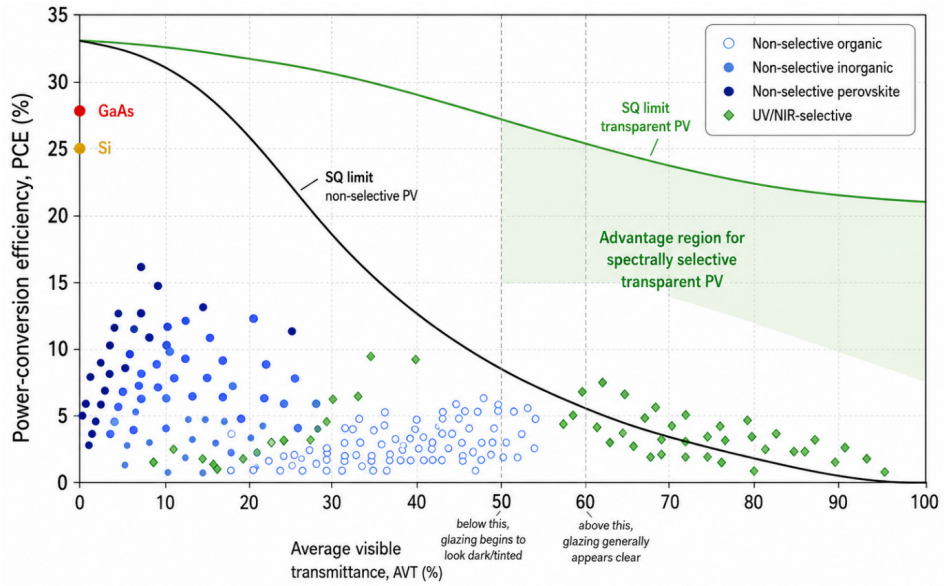


Figure 23. Efficiency–transmittance trade-off for transparent and semi-transparent photovoltaics. Power-conversion efficiency is compared with average visible transmittance for non-selective and UV/NIR-selective photovoltaic technologies. Open light-blue, blue, and dark-blue circles denote non-selective organic, inorganic, and perovskite devices, respectively; green diamonds denote UV/NIR-selective devices, while the GaAs and Si symbols indicate reference efficiency benchmarks. The transition towards a darker or tinted appearance is indicated separately by AVT values below approximately 50%. Conceptual redraw based on Ref. [220].

The central commercialisation problem is therefore not the isolated optimization of emissivity, visible transmittance, conductivity, switching amplitude, or photovoltaic efficiency. It is the preservation of an acceptable combination of these properties within a sufficiently wide manufacturing and service-life window. Technologies approaching commercial readiness must demonstrate not only peak laboratory performance, but also tolerance to thickness and composition variations, large-area uniformity, stable interfaces, compatibility with downstream processing, and predictable degradation under combined environmental stresses. This requirement is particularly important for adaptive glazing and ALD-assisted barrier concepts. ALD can provide exceptional conformality and thickness control, but its commercial value for architectural glass depends on whether pinhole-free protection can be obtained over square-metre substrates with acceptable precursor utilization and process time. Similarly, smart-window integration must be evaluated at a glazing-system level because switching layers, transparent electrodes, electrical connections, encapsulants, and edge seals form a coupled device whose failure modes cannot be inferred from measurements on the active material alone.

5.5. Industrial Deployment, Qualification, and Circularity

The main manufacturing challenge is not the lack of isolated high-performance materials, but the translation of promising thin films into reliable square-metre-scale products. Future production routes must demonstrate uniformity over large substrates, stable colour between batches, low defect density, compatibility with heat treatment or lamination, robust edge deletion, and predictable ageing after years of humidity, UV, temperature cycling, and pollutant exposure. Consequently, the most important trend may be the integration of durability testing into the design phase rather than as a final pass/fail step [82,207–211,214–227].

For industrial adoption, new low-E technologies should be evaluated with a qualification package that combines coating-level data, product-level glazing metrics, and

building-level simulation. Recommended reporting includes spectral T/R data, emissivity, sheet resistance, colour coordinates, angular responses, haze, thermal stress behaviour, accelerated ageing results, embodied-carbon indicators, recyclability assumptions, and climate-specific energy simulations. This integrated reporting would make it easier to compare static low-E, solar-control, dynamic, and BIPV glazing options on a common basis [218–227].

For further details, Ma et al., Luce et al., Jung et al., and Schaible et al. [207–210] describe inverse-design workflows for multilayer optical thin films and support the discussion of AI-assisted stack design and closed-loop optimization. Molesky et al. [211] provide the broader inverse-design framework in nanophotonics that supports optimization beyond trial-and-error coating development. Youngblood et al. [214] introduce the phase-change low-emissivity coating concept that is relevant to reconfigurable infrared control and seasonal adaptability; Llordes et al. and Runnerstrom et al. [215,216] cover NIR-selective electrochromic nanocrystal glazing for the adaptive solar-control/low-E convergence; Kamalisarvestani et al., Ke et al., and Casini [217–219] review thermochromic and dynamic smart windows, supporting the trends in passive and active adaptive glazing; Traverse et al. and Lunt and Bulovic [220,221] establish the performance and theoretical limits of transparent photovoltaic windows for multifunctional glazing and BIPV convergence; Tripathy et al. and Yang and Zou [222,223] analyze BIPV products, risks, barriers, and façade-integration strategies; and Davy et al., Yu et al., and Intal and Ebong [224–227] map PV-window potential and thin-film PV trends for energy-generating glazing and future facade manufacturing.

6. Conclusions

Low-E thin-film technology has evolved from single-layer “heat mirrors” and pyrolytic transparent conducting oxides to highly engineered multi-silver dielectric/metal/dielectric architectures in which every nanometre and every interface is functional. The mature materials palette, such as ultrathin Ag stabilized by ZnO seed layers and NiCr/Ti blockers, protective and antireflective dielectrics, hard-coat FTO, and indium-free TCO alternatives such as AZO and GZO, is deposited predominantly by large-area magnetron sputtering for soft-coat products and by on-line pyrolytic/APCVD routes for hard-coat products, with ALD, evaporation, and sol–gel serving research and niche functions. The choice between these routes remains a structured trade-off between emissivity, spectral selectivity, mechanical and chemical durability, processability (heat treatment, bending, lamination), and cost rather than a search for a single universally optimal coating.

A mature, multi-technique analysis chain now connects processing to performance in a quantitative way. Structural and compositional methods (XRD/XRR, SEM/TEM, AFM, EDS/XPS/AES/SIMS/RBS) resolve crystallinity, silver continuity, interface roughness, and depth-dependent chemistry; optical methods (spectrophotometry, spectroscopic ellipsometry, FTIR emissometry) feed dispersion models and transfer-matrix fits; and standardized metrology, according to EN 410, EN 673 and ISO 9050, translates spectra into the U-value, g/SHGC, T_v, and ε metrics used by industry. The review shows that correlative workflows (combining structure, chemistry, optics, and thermal metrology on the same stack, with explicit reporting of angular behaviour, uncertainty, and ageing state) are the decisive methodological trend, while corrosion diagnostics and accelerated weathering protocols remain the least harmonized link of the chain.

Functionally, low-E surfaces underpin the entire hierarchy of energy-efficient glazing, but they create value only when integrated into a complete system. The delivered performance depends on pane position, cavity gas or vacuum level, spacer and edge seal, frame, orientation, window-to-wall ratio, shading, and control strategy, so that the same

coating family can be optimal in one application and inadequate in another. The climate dependence is systematic: heating-dominated regions favour low U-values with useful solar gains, whereas cooling-dominated regions favour low SHGC and high light-to-solar-gain ratios; argon-filled insulating units reach $U \approx 1.1\ W \cdot m^{-2} \cdot K^{-1}$, vacuum glazing approaches 0.3–0.5 $W \cdot m^{-2} \cdot K^{-1}$, and electrochromic and thermochromic systems extend the envelope dynamically. Equally important, occupant-centred indicators, such as the interior surface temperature, radiant asymmetry, daylight and glare quality, condensation risk, and thermal survivability during power outages, have become first-class performance criteria alongside annual energy, with dynamic building simulation as the standard translation layer from product data to building-scale evidence.

Looking forward, low-E glass manufacturing is moving from empirical stack tuning toward digital, model-based and AI-assisted design, in which transfer-matrix modelling, multi-objective optimization, and machine learning inverse designs compress the loop between optical targets, pilot deposition, inline metrology, and accelerated ageing. Material development is expected to emphasize reduced critical-material intensity, such as thinner, seed-stabilized Ag layers and indium-reduced or indium-free transparent conductors, together with sustainability metrics such as material intensity per square metre, embodied carbon, recyclability of coated cullet, and repairability. In parallel, the low-E concept is converging with adaptive (electrochromic, thermochromic, phase-change) and energy-generating (transparent photovoltaic and BIPV) glazing, transforming the window from a passive thermal weak point into an active, multifunctional envelope component whose qualification requires coating-, product- and building-level evidence on a common basis.

The principal barrier to the commercialisation of emerging low-E technologies is no longer the absence of high-performance candidate materials, but the difficulty of preserving their combined functions within stable, reproducible, and manufacturable multilayer architectures. The decisive scientific challenges are the controlled formation of continuous ultrathin Ag, long-term stability of buried interfaces, suppression of diffusion and corrosion without an optical penalty, reliable indium-free transparent conductors, compatibility between multiple active functions, and large-area deposition with sufficiently wide process tolerances. For adaptive and energy-generating glazing, these requirements extend to switching uniformity, cycling durability, electrical integration, encapsulation, and qualification of the complete glazing assembly.

The decisive challenges for the next decade are therefore fourfold: (i) long-term durability and its standardized assessment, including accelerated-test protocols demonstrably correlated with service life; (ii) scalable, low-criticality material platforms that match silver- and indium-based performance at industrially relevant thicknesses and line speeds; (iii) harmonized, angle- and ageing-aware optical-thermal metrology with open access to manufacturer performance data; and (iv) the multi-objective co-design of thinner, more selective, more robust, radio-frequency-compatible and increasingly multifunctional stacks. Progress along these axes will determine how quickly the coming generation of low-E, adaptive and energy-generating glazing can contribute to building decarbonization targets across climates.

Author Contributions: Conceptualization, A.T., F.B., and V.P.; methodology, A.T., C.M., and F.B.; formal analysis, V.P. and B.I.; investigation, B.I. and F.-C.L.; writing—original draft preparation, D.-L.C. and F.-C.L.; writing—review and editing, B.I. and F.-C.L.; visualization, F.B., V.P., and D.-L.C.; supervision, A.T., C.M., and V.P.; project administration, D.-L.C.; funding acquisition, A.T. and C.M. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded through PR/NE/2024/P1/RSO1.1_RSO1.3/1—R&D projects and investments in SMEs, Project code 338197, Contract no. 731/10.07.2025, titled "Development and realization of an innovative installation intended for the implementation of an advanced thin-film coating technology for manufacturing windows with increased energy efficiency—TEGAS".

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Jelle, B.P.; Kalnæs, S.E.; Gao, T. Low-emissivity materials for building applications: A state-of-the-art review and future research perspectives. *Energy Build.* **2015**, *96*, 329–356. [CrossRef]
2. Rezaei, S.D.; Shannigrahi, S.; Ramakrishna, S. A review of conventional, advanced, and smart glazing technologies and materials for improving indoor environment. *Sol. Energy Mater. Sol. Cells* **2017**, *159*, 26–51. [CrossRef]
3. Michael, M.; Favoino, F.; Jin, Q.; Luna-Navarro, A.; Overend, M. A systematic review and classification of glazing technologies for building façades. *Energies* **2023**, *16*, 5357. [CrossRef]
4. Wheeler, V.M.; Kim, J.-H.; Daligault, T.; Rosales, B.A.; Engtrakul, C.; Tenent, R.C.; Wheeler, L.M. Photovoltaic windows cut energy use and CO_2 emissions by 40% in highly glazed buildings. *One Earth* **2022**, *5*, 1271–1285. [CrossRef]
5. Granqvist, C.G. Transparent conductors as solar energy materials: A panoramic review. *Sol. Energy Mater. Sol. Cells* **2007**, *91*, 1529–1598. [CrossRef]
6. Jelle, B.P.; Hynd, A.; Gustavsen, A.; Arasteh, D.; Goudey, H.; Hart, R. Fenestration of today and tomorrow: A state-of-the-art review and future research opportunities. *Sol. Energy Mater. Sol. Cells* **2012**, *96*, 1–28. [CrossRef]
7. Eames, P.C. Vacuum glazing: Current performance and future prospects. *Vacuum* **2008**, *82*, 717–722. [CrossRef]
8. Cuce, E.; Cuce, P.M. Vacuum glazing for highly insulating windows: Recent developments and future prospects. *Renew. Sustain. Energy Rev.* **2016**, *54*, 1345–1357. [CrossRef]
9. *EN 673:2011*; Glass in Building—Determination of Thermal Transmittance (U Value)—Calculation Method. CEN: Brussels, Belgium, 2011.
10. *EN 410:2011*; Glass in Building—Determination of Luminous and Solar Characteristics of Glazing. CEN: Brussels, Belgium, 2011.
11. *ISO 9050:2003*; Glass in Building—Determination of Light Transmittance, Solar Direct Transmittance, Total Solar Energy Transmittance, Ultraviolet Transmittance and Related Glazing Factors. ISO: Geneva, Switzerland, 2003.
12. Lee, E.S. *Spectrally Selective Glazings*; Federal Technology Alert; Lawrence Berkeley National Laboratory: Berkeley, CA, USA, 1998. [CrossRef] [PubMed]
13. Hutchins, M.G.; Topping, A.J.; Anderson, C.; Olive, F.; van Nijnatten, P.; Polato, P.; Roos, A.; Rubin, M. Measurement and prediction of angle-dependent optical properties of coated glass products: Results of an inter-laboratory comparison of spectral transmittance and reflectance. *Thin Solid Films* **2001**, *392*, 269–275. [CrossRef]
14. Sun, Y.; Wu, Y.; Wilson, R. A review of thermal and optical characterisation of complex window systems and their building performance prediction. *Appl. Energy* **2018**, *222*, 729–747. [CrossRef]
15. Chopra, K.L.; Major, S.; Pandya, D.K. Transparent conductors—A status review. *Thin Solid Films* **1983**, *102*, 1–46. [CrossRef]
16. Schaefer, C.; Bräuer, G.; Szczyrbowski, J. Low emissivity coatings on architectural glass. *Surf. Coat. Technol.* **1997**, *93*, 37–45. [CrossRef]
17. Wan Shamsuddin, W.N.S.; Zuber, K.; Murphy, P.J.; Jane, M.L. Environmental durability of soft low-e coatings: A review. *Sol. Energy Mater. Sol. Cells* **2024**, *267*, 112673. [CrossRef]
18. Dalapati, G.K.; Kushwaha, A.K.; Sharma, M.; Suresh, V.; Shannigrahi, S.; Zhuk, S.; Masudy-Panah, S. Transparent heat regulating (THR) materials and coatings for energy saving window applications: Impact of materials design, micro-structural, and interface quality on the THR performance. *Prog. Mater. Sci.* **2018**, *95*, 42–131. [CrossRef]
19. Teixeira, H.; Moret Rodrigues, A.; Aelenei, D.; Gomes, M.G. Literature review of solar control smart building glazing: Technologies, performance, and research insights. *Build. Environ.* **2025**, *274*, 112784. [CrossRef]
20. Aldawoud, A. Assessing the energy performance of modern glass facade systems. *MATEC Web Conf.* **2017**, *120*, 08001. [CrossRef]
21. Harmathy, N. Determination of glazing material's influence on the energy performance of office buildings using dynamic simulation techniques. *MATEC Web Conf.* **2016**, *73*, 02026. [CrossRef]
22. Brdnik, A.P. Thermal performance optimization of double and triple glazing systems for Slovenian climate conditions. *Sustainability* **2021**, *13*, 11857. [CrossRef]

23. Rezaei, F.; Sangin, H.; Heiranipour, M.; Attia, S. A multi-objective optimization of window and light shelf design in office buildings to improve occupants' thermal and visual comfort. *J. Daylighting* **2024**, *11*, 55–68. [CrossRef]
24. Anderson, A.-L.; Chen, S.; Romero, L.; Top, I.; Binions, R. Thin films for advanced glazing applications. *Buildings* **2016**, *6*, 37. [CrossRef]
25. Jung, W.; Kim, D.; Ko, S.H. Recent progress in high-efficiency transparent vacuum insulation technologies for carbon neutrality. *Int. J. Precis. Eng. Manuf.-Green Technol.* **2024**, *11*, 1681–1702. [CrossRef]
26. Baetens, R.; Jelle, B.P.; Gustavsen, A. Properties, requirements and possibilities of smart windows for dynamic daylight and solar energy control in buildings: A state-of-the-art review. *Sol. Energy Mater. Sol. Cells* **2010**, *94*, 87–105. [CrossRef]
27. Granqvist, C.G.; Arvizu, M.A.; Bayrak Pehlivan, İ.; Qu, H.-Y.; Wen, R.-T.; Niklasson, G.A. Electrochromic materials and devices for energy efficiency and human comfort in buildings: A critical review. *Electrochim. Acta* **2018**, *259*, 1170–1182. [CrossRef]
28. Cao, X.; Jin, P. Solar modulation utilizing VO_2-based thermochromic coatings for energy-saving applications. In *Emerging Solar Energy Materials*; IntechOpen: London, UK, 2018. [CrossRef]
29. Liu, X.; Wu, Y. A review of advanced architectural glazing technologies for solar energy conversion and intelligent daylighting control. *Archit. Intell.* **2022**, *1*, 10. [CrossRef]
30. Pereira, J.; Teixeira, H.; Gomes, M.G.; Moret Rodrigues, A. Performance of solar control films on building glazing: A literature review. *Appl. Sci.* **2022**, *12*, 5923. [CrossRef]
31. Hee, W.J.; Alghoul, M.A.; Bakhtyar, B.; Elayeb, O.; Shameri, M.A.; Alrubaih, M.S.; Sopian, K. The role of window glazing on daylighting and energy saving in buildings. *Renew. Sustain. Energy Rev.* **2015**, *42*, 323–343. [CrossRef]
32. Ginley, D.S.; Bright, C. Transparent conducting oxides. *MRS Bull.* **2000**, *25*, 15–18. [CrossRef]
33. Minami, T. Present status of transparent conducting oxide thin-film development for Indium-Tin-Oxide (ITO) substitutes. *Thin Solid Films* **2008**, *516*, 5822–5828. [CrossRef]
34. Stadler, A. Transparent conducting oxides—An up-to-date overview. *Materials* **2012**, *5*, 661–683. [CrossRef] [PubMed]
35. Álvarez, R.; González, J.C.; Espinós, J.P.; González-Elipe, A.R.; Cueva, A.; Villuendas, F. Growth of silver on ZnO and SnO_2 thin films intended for low emissivity applications. *Appl. Surf. Sci.* **2013**, *268*, 507–515. [CrossRef]
36. Kulczyk-Malecka, J.; Kelly, P.J.; West, G.T.; Clarke, G.C.B.; Ridealgh, J.A.; Almtoft, K.P.; Greer, A.L.; Barber, Z.H. Investigation of silver diffusion in TiO_2/Ag/TiO_2 coatings. *Acta Mater.* **2014**, *66*, 396–404. [CrossRef]
37. Kim, S.H.; Kim, H.; Song, P.K. Near infrared reflecting properties of TiO_2/Ag/TiO_2 multilayers prepared by DC/RF magnetron sputtering. *Korean J. Met. Mater.* **2017**, *55*, 581–586. [CrossRef]
38. Lungwitz, F.; Escobar Galindo, R.; Janke, D.; Schumann, E.; Wenisch, R.; Gemming, S.; Krause, M. Transparent conductive tantalum doped tin oxide as selectively solar-transmitting coating for high temperature solar thermal applications. *Sol. Energy Mater. Sol. Cells* **2019**, *196*, 84–93. [CrossRef]
39. Heidari, A.; Popov, G.; Hatanpää, T.; Weiß, A.; Chundak, M.; Mizohata, K.; Ritala, M.; Kemell, M. Atomic layer deposition of AgX (X = Cl, Br, I) thin films. *ACS Omega* **2025**, *10*, 55682–55691. [CrossRef] [PubMed]
40. Eltohamy, M.; Etman, I.M.; Geioushy, R.A.; El Kazazz, H.A.; Badawy, K.; Fouad, O.A. Energy-saving low-E with dual IR-reflective and UV-protective TiO_2–ZnO–SiO_2 coated glass. *Res. Sq.* **2025**. [CrossRef] [PubMed]
41. Bamford, S.E.; Jones, R.; Gardner, W.; Muir, B.W.; Winkler, D.A.; Pigram, P.J. Profiling a low emissivity glass coating with ToF-SIMS and machine learning. *Adv. Mater. Interfaces* **2024**, *11*, 2300645. [CrossRef]
42. Depla, D. Chemical stability of sputter deposited silver thin films. *Coatings* **2022**, *12*, 1915. [CrossRef]
43. Ding, G.; Clavero, C. Silver-Based Low-Emissivity Coating Technology for Energy-saving Window Applications. In *Modern Technologies for Creating the Thin-Film Systems and Coatings*; InTechOpen: London, UK, 2017. [CrossRef] [PubMed]
44. Chernysheva, E.; Philippe, B.; Rensmo, H.; Karis, O.; Félix, R.; Gorgoi, M.; Grachev, S.; Burov, E.; Montigaud, H.; Lazzari, R. Redox reaction at buried ZnO/Ti thin film interface as seen by hard X-ray photoemission and thermal desorption spectroscopy. *Appl. Surf. Sci.* **2024**, *680*, 161409. [CrossRef]
45. Guillén, C.; Herrero, J. SnOx/Ag/SnOx heat-reflector coatings prepared by DC sputtering. *SN Appl. Sci.* **2020**, *2*, 1717. [CrossRef]
46. Hammarberg, E.; Roos, A. Antireflection treatment of low-emitting glazings for energy efficient windows with high visible transmittance. *Thin Solid Films* **2003**, *442*, 222–226. [CrossRef]
47. García-Segura, A.; Fernández-García, A.; Ariza, M.J.; Sutter, F.; Watermeyer, P.; Schmücker, M.; Valenzuela, L. Corrosion on silvered-glass solar reflectors exposed to accelerated aging tests with polluting gases: A microscopic study. *Corros. Sci.* **2020**, *176*, 108928. [CrossRef]
48. García-Valenzuela, A.; López-Santos, C.; Álvarez, R.; Rico, V.; Cotrino, J.; González-Elipe, A.R.; Palmero, A. Structural control in porous/compact multilayer systems grown by magnetron sputtering. *Nanotechnology* **2017**, *28*, 465605. [CrossRef] [PubMed]
49. Zhao, H.; Zhou, C.M.; Cai, Y.; Zhang, F.C. Nanostructure analysis of graded oxidation coated glass prepared by APCVD on-line. *Mater. Lett.* **2008**, *62*, 140–142. [CrossRef]

50. Chen, S.; Carraro, G.; Barreca, D.; Sapelkin, A.; Chen, W.; Huang, X.; Cheng, Q.; Zhang, F.; Binions, R. Aerosol assisted chemical vapour deposition of Ga-doped ZnO films for energy efficient glazing: Effects of doping concentration on the film growth behaviour and opto-electronic properties. *J. Mater. Chem. A* **2015**, *3*, 13039–13050. [CrossRef]
51. Ullah, M.B.; Akter, Y.; Afrin, K.; Quddus, M.S. Development of modified glasses by transparent, functional hybrid sol-gel nano-ceramic coatings: A comparative study. *World J. Eng. Technol.* **2024**, *12*, 170–188. [CrossRef]
52. Mak, A.K.; Öztürk, O.; Karabulut, M. Investigation of nanoscale Ta:TiO_2 transparent conductive layers for smart glass applications. *J. Inst. Sci. Technol.* **2025**, *15*, 470–478.
53. Niklasson, G.A.; Granqvist, C.G. Electrochromics for smart windows: Thin films of tungsten oxide and nickel oxide, and devices based on these. *J. Mater. Chem.* **2007**, *17*, 127–156. [CrossRef]
54. Granqvist, C.G.; Arvizu, M.A.; Qu, H.; Wen, R.; Niklasson, G.A. Advances in electrochromic device technology: Multiple roads towards superior durability. *Surf. Coat. Technol.* **2018**, *357*, 619–625. [CrossRef]
55. Liang, R.; Sun, Y.; Aburas, M.; Wilson, R.; Wu, Y. An exploration of the combined effects of NIR and VIS spectrally selective thermochromic materials on building performance. *Energy Build.* **2019**, *201*, 149–162. [CrossRef]
56. Fahland, M.; Szelwicka, J.; Langgemach, W. Energy-saving potential of thermochromic coatings in transparent building envelope components. *J. Facade Des. Eng.* **2023**, *11*, 197. [CrossRef]
57. Bianco, L.; Cascone, Y.; Goia, F.; Perino, M.; Serra, V. Responsive glazing systems: Characterisation methods, summer performance and implications on thermal comfort. *Sol. Energy* **2017**, *158*, 819–836. [CrossRef]
58. Wang, D.; Lu, L.; Zhang, W. Overall energy performance assessment of a new heat blocking coating. *J. Sustain. Dev. Energy Water Environ. Syst.* **2018**, *7*, 1–12. [CrossRef]
59. Pérez, G.; Allegro, V.R.; Alonso, C.; Martín-Consuegra, F.; Oteiza, I.; Vázquez, B.F.; Guerrero, A. Selection of suitable materials for the development of an innovative thermochromic Trombe wall. *Adv. Build. Energy Res.* **2019**, *15*, 146–169. [CrossRef]
60. Wang, P.; Li, X.; Zhang, N.; Zhai, Y.; Wang, Z.; Liu, S.; Ho, W.; Huang, Y.; Lee, S. Room-temperature fabrication of dual-functional SiO_2/TiO_2 coating for photovoltaic efficiency enhancement: From laboratory development to field validation. *Res. Sq.* **2025**, *9*, 1–13. [CrossRef] [PubMed]
61. DeForest, N.; Shehabi, A.; O'Donnell, J.; Garcia, G.; Greenblatt, J.; Lee, E.S.; Selkowitz, S.; Milliron, D.J. United States energy and CO_2 savings potential from deployment of near-infrared electrochromic window glazings. *Build. Environ.* **2015**, *89*, 107–117. [CrossRef]
62. Tzempelikos, A.; Athienitis, A.K. The impact of shading design and control on building cooling and lighting demand. *Sol. Energy* **2007**, *81*, 369–382. [CrossRef]
63. Lee, E.S.; Tavil, A. Energy and visual comfort performance of electrochromic windows with overhangs. *Build. Environ.* **2007**, *42*, 2439–2449. [CrossRef]
64. Favoino, F.; Overend, M.; Jin, Q. The optimal thermo-optical properties and energy saving potential of adaptive glazing technologies. *Appl. Energy* **2015**, *156*, 1–15. [CrossRef]
65. Sibilio, S.; Rosato, A.; Scorpio, M.; Iuliano, G.; Ciampi, G.; Vanoli, G.P.; de Rossi, F. A review of electrochromic windows for residential applications. *Int. J. Heat Technol.* **2016**, *34*, S481–S488. [CrossRef]
66. Chiu, P.-K.; Lee, C.-T.; Chiang, D.; Cho, W.-H.; Hsiao, C.-N.; Chen, Y.; Huang, B.-M.; Yang, J.-R. Conductive and transparent multilayer films for low-temperature $TiO_2/Ag/SiO_2$ electrodes by E-beam evaporation with IAD. *Nanoscale Res. Lett.* **2014**, *9*, 35. [CrossRef] [PubMed]
67. Ran, J.; Nan, Y.C.; Gao, Q.; Si, P.; Zhong, H.; Feng, Y. Beyond the g-value: A comparative study of solar control coated glass and spectrally selective glass in terms of building energy efficiency. *Energy Build.* **2024**, *325*, 114977. [CrossRef]
68. Akin Sonmez, N.; Donmez, M.; Comert, B.; Ozcelik, S. Ag/M-seed/AZO/glass structures for low-E glass: Effects of metal seeds. *Int. J. Appl. Glass Sci.* **2018**, *9*, 383–391. [CrossRef]
69. Cinali, M.B.; Coskun, O.D. Optimization of physical properties of sputtered silver films by change of deposition power for low emissivity applications. *J. Alloys Compd.* **2021**, *853*, 157073. [CrossRef]
70. Sun, K.; Zhang, D.; Yin, H.; Cheng, L.; Yuan, H.; Yang, C. Preparation of AZO/Cu/AZO films with low infrared emissivity, high conductivity and high transmittance by adjusting the AZO layer. *Appl. Surf. Sci.* **2022**, *578*, 152051. [CrossRef]
71. Zhang, D.; Sun, K.; Yin, H.; Cheng, L.; Yuan, H.; Yang, C. Optimization of photoelectric properties and temporal stability of AZO/Ti/Cu/AZO films by insertion of Ti layer for low emissivity applications. *Mater. Sci. Eng. B* **2023**, *293*, 116471. [CrossRef]
72. *ISO 9211-3:2024*; Optics and Photonics—Optical Coatings—Part 3: Environmental Durability. ISO: Geneva, Switzerland, 2024.
73. *ISO 9211-4:2022*; Optics and Photonics—Optical Coatings—Part 4: Specific Test Methods: Abrasion, Adhesion and Resistance to Water. ISO: Geneva, Switzerland, 2022.
74. Mouedden, Y.E.; Alghamedi, R.; Nur-E-Alam, M.; Vasiliev, M.; Alameh, K. Thin film coatings for solar and thermal radiation control prepared by physical vapour deposition. In Proceedings of the 9th International Conference on High-Capacity Optical Networks and Enabling Technologies (HONET), Istanbul, Turkey, 12–14 December 2012; pp. 83–86. [CrossRef]

75. Kelly, P.J.; Arnell, R.D. Magnetron sputtering: A review of recent developments and applications. *Vacuum* **2000**, *56*, 159–172. [CrossRef]
76. Berg, S.; Nyberg, T. Fundamental understanding and modeling of reactive sputtering processes. *Thin Solid Films* **2005**, *476*, 215–230. [CrossRef]
77. Anders, A. A structure zone diagram including plasma-based deposition and ion etching. *Thin Solid Films* **2010**, *518*, 4087–4090. [CrossRef]
78. Helmersson, U.; Lattemann, M.; Bohlmark, J.; Ehiasarian, A.P.; Gudmundsson, J.T. Ionized physical vapor deposition (IPVD): A review of technology and applications. *Thin Solid Films* **2006**, *513*, 1–24. [CrossRef]
79. Gudmundsson, J.T.; Brenning, N.; Lundin, D.; Helmersson, U. High power impulse magnetron sputtering discharge. *J. Vac. Sci. Technol. A* **2012**, *30*, 030801. [CrossRef]
80. Sarakinos, K.; Alami, J.; Konstantinidis, S. High power pulsed magnetron sputtering: A review on scientific and engineering state of the art. *Surf. Coat. Technol.* **2010**, *204*, 1661–1684. [CrossRef]
81. Ellmer, K. Past achievements and future challenges in the development of optically transparent electrodes. *Nat. Photonics* **2012**, *6*, 809–817. [CrossRef]
82. Fortunato, E.; Ginley, D.; Hosono, H.; Paine, D.C. Transparent conducting oxides for photovoltaics. *MRS Bull.* **2007**, *32*, 242–247. [CrossRef]
83. Gordon, R.G. Criteria for choosing transparent conductors. *MRS Bull.* **2000**, *25*, 52–57. [CrossRef]
84. Minami, T. Transparent conducting oxide semiconductors for transparent electrodes. *Semicond. Sci. Technol.* **2005**, *20*, S35–S44. [CrossRef]
85. Granqvist, C.G.; Hultaker, A. Transparent and conducting ITO films: New developments and applications. *Thin Solid Films* **2002**, *411*, 1–5. [CrossRef]
86. Greene, J.E. Tracing the recorded history of thin-film sputter deposition: From the 1800s to 2017. *J. Vac. Sci. Technol. A* **2017**, *35*, 05C204. [CrossRef]
87. Puurunen, R.L. Surface chemistry of atomic layer deposition: A case study for the trimethylaluminum/water process. *J. Appl. Phys.* **2005**, *97*, 121301. [CrossRef]
88. George, S.M. Atomic layer deposition: An overview. *Chem. Rev.* **2010**, *110*, 111–131. [CrossRef] [PubMed]
89. Marichy, C.; Bechelany, M.; Pinna, N. Atomic layer deposition of nanostructured materials for energy and environmental applications. *Adv. Mater.* **2012**, *24*, 1017–1032. [CrossRef] [PubMed]
90. Hench, L.L.; West, J.K. The sol-gel process. *Chem. Rev.* **1990**, *90*, 33–72. [CrossRef]
91. Brinker, C.J.; Frye, G.C.; Hurd, A.J.; Ashley, C.S. Fundamentals of sol-gel dip coating. *Thin Solid Films* **1991**, *201*, 97–108. [CrossRef]
92. Gordon, R.G. Chemical vapor deposition of coatings on glass. *J. Non-Cryst. Solids* **1997**, *218*, 81–91. [CrossRef]
93. Choy, K.L. Chemical vapour deposition of coatings. *Prog. Mater. Sci.* **2003**, *48*, 57–170. [CrossRef]
94. Chandrasekhar, R.; Choy, K.L. Electrostatic spray assisted vapour deposition of fluorine doped tin oxide. *J. Cryst. Growth* **2001**, *231*, 215–221. [CrossRef]
95. Fiameni, S.; Battiston, S.; Castellani, V.; Barison, S.; Armelao, L. Implementing sustainability in laboratory activities: A case study on aluminum titanium nitride based thin-film magnetron sputtering deposition onto commercial laminated steel. *J. Clean. Prod.* **2021**, *285*, 124869. [CrossRef]
96. Wang, E.; Yuan, C. A hybrid life cycle assessment of atomic layer deposition process. *J. Clean. Prod.* **2014**, *74*, 145–154. [CrossRef]
97. Ali, K.; Khan, S.A.; Mat Jafri, M.Z. Structural and optical properties of ITO/TiO_2 anti-reflective films for solar cell applications. *Nanoscale Res. Lett.* **2014**, *9*, 175. [CrossRef] [PubMed]
98. Smith, J.R.; Larson, C.; Campbell, S.A. Recent applications of SEM and AFM for assessing topography of metal and related coatings—A review. *Trans. IMF* **2011**, *89*, 18–27. [CrossRef]
99. Hilfiker, J.N.; Bungay, C.L.; Synowicki, R.A.; Tiwald, T.E.; Herzinger, C.M.; Johs, B.; Pribil, G.K.; Woollam, J.A. Progress in spectroscopic ellipsometry: Applications from vacuum ultraviolet to infrared. *J. Vac. Sci. Technol. A* **2003**, *21*, S1103–S1108. [CrossRef]
100. Binnig, G.; Quate, C.F.; Gerber, C. Atomic force microscope. *Phys. Rev. Lett.* **1986**, *56*, 930–933. [CrossRef] [PubMed]
101. Necas, D.; Klapetek, P. Gwyddion: An open-source software for SPM data analysis. *Open Phys.* **2012**, *10*, 181–188. [CrossRef]
102. Villarrubia, J.S. Algorithms for scanned probe microscope image simulation, surface reconstruction, and tip estimation. *J. Res. Natl. Inst. Stand. Technol.* **1997**, *102*, 425–454. [CrossRef] [PubMed]
103. Canestrari, R.; Spiga, D.; Pareschi, G. Analysis of microroughness evolution in X-ray astronomical multilayer mirrors by surface topography with the MPES program and by X-ray scattering. In *Space Telescopes and Instrumentation II: Ultraviolet to Gamma Ray*; SPIE: Bellingham, WA, USA, 2006; Volume 6266, p. 626613. [CrossRef]
104. Newbury, D.E.; Ritchie, N.W.M. Is scanning electron microscopy/energy dispersive X-ray spectrometry (SEM/EDS) quantitative? *Scanning* **2013**, *35*, 141–168. [CrossRef] [PubMed]
105. Parratt, L.G. Surface studies of solids by total reflection of X-rays. *Phys. Rev.* **1954**, *95*, 359–369. [CrossRef]

106. Nevot, L.; Croce, P. Caracterisation des surfaces par reflexion rasante de rayons X. Application a l'etude du polissage de quelques verres silicates. *Rev. Phys. Appl.* **1980**, *15*, 761–779. [CrossRef]
107. Holy, V.; Kubena, J.; Ohlidal, I.; Lischka, K.; Plotz, W. X-ray reflection from rough layered systems. *Phys. Rev. B* **1993**, *47*, 15896–15903. [CrossRef] [PubMed]
108. Windt, D.L. IMD—Software for modeling the optical properties of multilayer films. *Comput. Phys.* **1998**, *12*, 360–370. [CrossRef]
109. Egerton, R.F. Electron energy-loss spectroscopy in the TEM. *Rep. Prog. Phys.* **2009**, *72*, 016502. [CrossRef]
110. Midgley, P.A.; Dunin-Borkowski, R.E. Electron tomography and holography in materials science. *Nat. Mater.* **2009**, *8*, 271–280. [CrossRef] [PubMed]
111. Haider, M.; Uhlemann, S.; Schwan, E.; Rose, H.; Kabius, B.; Urban, K. Electron microscopy image enhanced. *Nature* **1998**, *392*, 768–769. [CrossRef]
112. Muller, D.A. Structure and bonding at the atomic scale by scanning transmission electron microscopy. *Nat. Mater.* **2009**, *8*, 263–270. [CrossRef] [PubMed]
113. Pazzaglia, A.; Maffini, A.; Dellasega, D.; Lamperti, A.; Passoni, M. Reference-free evaluation of thin films mass thickness and composition through energy dispersive X-ray spectroscopy. *Mater. Charact.* **2019**, *153*, 92–102. [CrossRef]
114. Seah, M.P.; Dench, W.A. Quantitative electron spectroscopy of surfaces: A standard data base for electron inelastic mean free paths in solids. *Surf. Interface Anal.* **1979**, *1*, 2–11. [CrossRef]
115. Scofield, J.H. Hartree-Slater subshell photoionization cross-sections at 1254 and 1487 eV. *J. Electron Spectrosc. Relat. Phenom.* **1976**, *8*, 129–137. [CrossRef]
116. Greczynski, G.; Hultman, L. Compromising science by ignorant instrument calibration—Need to revisit half a century of published XPS data. *Angew. Chem.* **2020**, *59*, 5002–5006. [CrossRef] [PubMed]
117. McDonnell, L.A.; Heeren, R.M.A. Imaging mass spectrometry. *Mass Spectrom. Rev.* **2007**, *26*, 606–643. [CrossRef] [PubMed]
118. Likins-White, M.; Tenent, R.C.; Zhai, Z. Degradation of insulating glass units: Thermal performance, measurements and energy impacts. *Buildings* **2023**, *13*, 551. [CrossRef]
119. Abbott, W.H. The development and performance characteristics of mixed flowing gas test environment. *IEEE Trans. Compon. Hybrids Manuf. Technol.* **1988**, *11*, 22–35. [CrossRef]
120. Abbott, W.H. The corrosion of copper and porous gold in flowing mixed gas environments. *IEEE Trans. Compon. Hybrids Manuf. Technol.* **1990**, *13*, 40–45. [CrossRef]
121. Rice, D.W.; Peterson, P.; Rigby, E.B.; Phipps, P.B.P.; Cappell, R.J.; Tremoureux, R. Atmospheric corrosion of copper and silver. *J. Electrochem. Soc.* **1981**, *128*, 275–284. [CrossRef]
122. Graedel, T.E. Corrosion mechanisms for silver exposed to the atmosphere. *J. Electrochem. Soc.* **1992**, *139*, 1963–1970. [CrossRef]
123. Volpe, L.; Peterson, P.J. The atmospheric sulfidation of silver in a tubular corrosion reactor. *Corros. Sci.* **1989**, *29*, 1179–1196. [CrossRef]
124. Franey, J.P.; Kammlott, G.W.; Graedel, T.E. The corrosion of silver by atmospheric sulfurous gases. *Corros. Sci.* **1985**, *25*, 133–143. [CrossRef]
125. Kurapati, R.; Maurice, V.; Seyeux, A.; Klein, L.; Mercier, D.; Chauveau, G.; Grezes-Besset, C.; Berthod, L.; Marcus, P. Advanced protection against environmental degradation of silver mirror stacks for space application. *arXiv* **2020**, arXiv:2004.06590.
126. Bulkin, P.; Gaiaschi, S.; Chapon, P.; Daineka, D.; Kundikova, N. Protective coatings for front surface silver mirrors by atomic layer deposition. *arXiv* **2019**, arXiv:1912.01000.
127. Fryauf, D.M.; Phillips, A.C.; Kobayashi, N.P. Critical processing temperature for high performance protected silver thin film mirrors. *J. Astron. Telesc. Instrum. Syst.* **2021**, *7*, 034002. [CrossRef]
128. Kennedy, C.E.; Terwilliger, K. Optical durability of candidate solar reflectors. *J. Sol. Energy Eng.* **2005**, *127*, 262–269. [CrossRef]
129. Brogren, M.; Karlsson, B.; Roos, A.; Werner, A. Analysis of the effects of outdoor and accelerated ageing on the optical properties of reflector materials for solar energy applications. *Sol. Energy Mater. Sol. Cells* **2004**, *82*, 491–515. [CrossRef]
130. Milardi, M.; Mandaglio, M.; Mazzucchelli, E.S.; Rigone, P.; Giussani, P. New assessment protocols to improve building envelope resilience. *TECHNE-J. Technol. Archit. Environ.* **2024**, *27*, 270–279. [CrossRef]
131. Li, X.; Wu, Y. A review of complex window-glazing systems for building energy saving and daylight comfort: Glazing technologies and their building performance prediction. *J. Build. Phys.* **2024**, *48*, 496–526. [CrossRef]
132. Arancibia-Bulnes, C.A.; Peñá-Cruz, M.I.; Mutuberria, A.; Díaz-Uribe, R.; Sánchez-González, M. A survey of methods for the evaluation of reflective solar concentrator optics. *Renew. Sustain. Energy Rev.* **2017**, *69*, 673–684. [CrossRef]
133. Czanderna, A.W.; Pern, F.J. Encapsulation of PV modules using ethylene vinyl acetate copolymer as a pottant: A critical review. *Sol. Energy Mater. Sol. Cells* **1996**, *43*, 101–181. [CrossRef]
134. Dameron, A.A.; Davidson, S.D.; Burton, B.B.; Carcia, P.F.; McLean, R.S.; George, S.M. Gas diffusion barriers on polymers using multilayers fabricated by Al_2O_3 and rapid SiO_2 atomic layer deposition. *J. Phys. Chem. C* **2008**, *112*, 4573–4580. [CrossRef]
135. Manifacier, J.C.; Gasiot, J.; Fillard, J.P. A simple method for the determination of the optical constants n, k and the thickness of a weakly absorbing thin film. *J. Phys. E Sci. Instrum.* **1976**, *9*, 1002–1004. [CrossRef]

136. Swanepoel, R. Determination of the thickness and optical constants of amorphous silicon. *J. Phys. E Sci. Instrum.* **1983**, *16*, 1214–1222. [CrossRef]
137. Poelman, D.; Smet, P.F. Methods for the determination of the optical constants of thin films from single transmission measurements: A critical review. *J. Phys. D Appl. Phys.* **2003**, *36*, 1850–1857. [CrossRef]
138. Forouhi, A.R.; Bloomer, I. Optical dispersion relations for amorphous semiconductors and amorphous dielectrics. *Phys. Rev. B* **1986**, *34*, 7018–7026. [CrossRef] [PubMed]
139. Jellison, G.E.; Modine, F.A. Parameterization of the optical functions of amorphous materials in the interband region. *Appl. Phys. Lett.* **1996**, *69*, 371–373. [CrossRef]
140. Harbecke, B. Coherent and incoherent reflection and transmission of multilayer structures. *Appl. Phys. B* **1986**, *39*, 165–170. [CrossRef]
141. Katsidis, C.C.; Siapkas, D.I. General transfer-matrix method for optical multilayer systems with coherent, partially coherent, and incoherent interference. *Appl. Opt.* **2002**, *41*, 3978–3987. [CrossRef] [PubMed]
142. Centurioni, E. Generalized matrix method for calculation of internal light energy flux in mixed coherent and incoherent multilayers. *Appl. Opt.* **2005**, *44*, 7532–7539. [CrossRef] [PubMed]
143. Baenas, T.; Machado, M. Optical model for multilayer glazing systems: Application to laminated glass and photovoltaic modules. *Sol. Energy* **2016**, *125*, 256–266. [CrossRef]
144. Fujiwara, H.; Kondo, M. Effects of carrier concentration on the dielectric function of ZnO:Ga and In2O3:Sn studied by spectroscopic ellipsometry: Analysis of free-carrier and band-edge absorption. *Phys. Rev. B—Condens. Matter Mater. Phys.* **2005**, *71*, 075109. [CrossRef]
145. Ferlauto, A.S.; Ferreira, G.M.; Pearce, J.M.; Wronski, C.R.; Collins, R.W.; Deng, X.; Ganguly, G. Analytical model for the optical functions of amorphous semiconductors from the near-infrared to ultraviolet: Applications in thin film photovoltaics. *J. Appl. Phys.* **2002**, *92*, 2424–2436. [CrossRef]
146. Brendel, R.; Bormann, D. An infrared dielectric function model for amorphous solids. *J. Appl. Phys.* **1992**, *71*, 1–6. [CrossRef]
147. Rakic, A.D.; Djurisic, A.B.; Elazar, J.M.; Majewski, M.L. Optical properties of metallic films for vertical-cavity optoelectronic devices. *Appl. Opt.* **1998**, *37*, 5271–5283. [CrossRef] [PubMed]
148. Rubin, M. Optical properties of soda-lime silica glasses. *Sol. Energy Mater.* **1985**, *12*, 275–288. [CrossRef]
149. Jelle, B.P. Solar radiation glazing factors for window panes, glass structures and electrochromic windows in buildings—Measurement and calculation. *Sol. Energy Mater. Sol. Cells* **2013**, *116*, 291–323. [CrossRef]
150. Goia, F.; Zinzi, M.; Carnielo, E.; Serra, V. Spectral and angular solar properties of a PCM-filled double glazing unit. *Energy Build.* **2015**, *87*, 302–312. [CrossRef]
151. Ichinose, M.; Ishino, H.; Nagata, A. A study on spectral sensitivity of solar transmission and reflection performance of architectural materials. *J. Environ. Eng.* **2004**, *69*, 15–22. [CrossRef] [PubMed]
152. Mohelnikova, J.; Altan, H.; Tower, A. Evaluation of optical and thermal properties of window glazing. *WSEAS Trans. Environ. Dev.* **2009**, *5*, 86–93.
153. Alghamedi, R.; Vasiliev, M.; Nur-E-Alam, M.; Alameh, K. Spectrally-selective all-inorganic scattering luminophores for solar energy-harvesting clear glass windows. *Sci. Rep.* **2014**, *4*, 6632. [CrossRef] [PubMed]
154. Gorantla, K.K.; Shaik, S.; Setty, A.B.T.P. Day lighting and thermal analysis using various double reflective window glasses for green energy buildings. *Int. J. Heat Technol.* **2018**, *36*, 1121–1128. [CrossRef]
155. Smilgies, D.-M. Scherrer grain-size analysis adapted to grazing-incidence scattering with area detectors. *J. Appl. Crystallogr.* **2009**, *42*, 1030–1034. [CrossRef] [PubMed]
156. Scherrer, P. Bestimmung der Groesse und der inneren Struktur von Kolloidteilchen mittels Roentgenstrahlen. Nachrichten von der Gesellschaft der Wissenschaften zu Goettingen. *Math.-Phys. Kl.* **1918**, *1918*, 98–100.
157. *ASTM E424-71(2023)*; Standard Test Methods for Solar Energy Transmittance and Reflectance (Terrestrial) of Sheet Materials. ASTM International: West Conshohocken, PA, USA, 2023.
158. *EN 12898:2019*; Glass in Building—Determination of the Emissivity. CEN: Brussels, Belgium, 2019.
159. Van den Bergh, S.; Hart, R.; Jelle, B.P.; Gustavsen, A. Window spacers and edge seals in insulating glass units: A state-of-the-art review and future perspectives. *Energy Build.* **2013**, *58*, 263–280. [CrossRef]
160. *DIN EN ISO 4892-2:2021-11*; Plastics—Methods of Exposure to Laboratory Light Sources—Part 2: Xenon-Arc Lamps (ISO 4892-2:2013 + Amd 1:2021). DIN: Berlin, Germany, 2021.
161. *ISO 9142:2003*; Adhesives—Guide to the Selection of Standard Laboratory Ageing Conditions for Testing Bonded Joints. ISO: Geneva, Switzerland, 2003.
162. *EN 1096-1:2012*; Glass in Building—Coated Glass—Part 1: Definitions and Classification. CEN: Brussels, Belgium, 2012.
163. Soares, N.; Martins, C.; Gonçalves, M.; Santos, P.; Simões da Silva, L.; Costa, J.J. Laboratory and in-situ non-destructive methods to evaluate the thermal transmittance and behavior of walls, windows, and construction elements with innovative materials: A review. *Energy Build.* **2019**, *182*, 88–110. [CrossRef]

164. Grosjean, A.; Le Baron, E. Longtime solar performance estimations of low-E glass depending on local atmospheric conditions. *Sol. Energy Mater. Sol. Cells* **2022**, *240*, 111730. [CrossRef]
165. Memon, S.; Fang, Y.; Eames, P. The influence of low-temperature surface induction on evacuation, pump-out hole sealing and thermal performance of composite edge-sealed vacuum insulated glazing. *Renew. Energy* **2019**, *135*, 450–464. [CrossRef]
166. Paolini, R.; Zani, A.; Poli, T.; Antretter, F.; Zinzi, M. Natural aging of cool walls: Impact on solar reflectance, sensitivity to thermal shocks and building energy needs. *Energy Build.* **2017**, *153*, 287–296. [CrossRef]
167. Burlingis, A.; Samajauskienė, B.; Ramanauskas, J. An influence of outdoor temperature on thermal performance of windows with a low-emissivity glazing. *J. Civ. Eng. Manag.* **2003**, *9*, 115–121. [CrossRef]
168. Aste, N.; Buzzetti, M.; Del Pero, C.; Leonforte, F. Glazing's techno-economic performance: A comparison of window features in office buildings in different climates. *Energy Build.* **2018**, *159*, 123–135. [CrossRef]
169. Karimi, M.; Chikamoto, T.; Lee, M. Influence of window-to-wall ratio, orientation, and glazing type on building energy performance across climatic zones in Afghanistan. *J. Build. Constr. Plan. Res.* **2025**, *13*, 151–178. [CrossRef]
170. Makhloufi, A.W.; Louafi, S. The impact of glazing types and window-to-wall ratios on energy consumption in semi-arid, Mediterranean and arid climates. *Prostor* **2022**, *30*, 140–151. [CrossRef] [PubMed]
171. Jaouaf, S.-E.; Bensaad, B.; Dorbane, A. Energy efficiency of a house in Mediterranean region: Insulation and glazing impact. *J. Eng. Exact. Sci.* **2024**, *10*, 17038. [CrossRef]
172. Qahtan, A.M.; Shaik, S.; Alwadai, Y.A.; Alhamami, S.; Alawlaki, A.S.; Alyami, M.H.; Kodati, J.A. Smart glazing systems vs conventional glazing: A comprehensive study on temperature control, daylighting, and sustainability. *Int. J. Sustain. Dev. Plan.* **2024**, *19*, 939–948. [CrossRef]
173. Sayed, M.A.A.E.D.A.; Fikry, M. Impact of glass facades on internal environment of buildings in hot arid zone. *Alex. Eng. J.* **2019**, *58*, 1063–1075. [CrossRef]
174. Mahmoud, A. Investigating the impact of different glazing types on the energy performance in hot arid climate. *J. Adv. Eng. Trends* **2021**, *42*, 69–83. [CrossRef]
175. Fang, Y.; Memon, S.; Peng, J.; Tyrer, M.; Ming, T. Solar thermal performance of two innovative configurations of air-vacuum layered triple glazed windows. *Renew. Energy* **2020**, *150*, 167–175. [CrossRef]
176. Memon, S.; Eames, P. Solar energy gain and space-heating energy supply analyses for solid-wall dwelling retrofitted with the experimentally achievable U-value of novel triple vacuum glazing. *J. Daylighting* **2017**, *4*, 15–25. [CrossRef]
177. Tiwari, R.; Kim, J. Multi-cell, triple pane, vacuum insulated glazing. *Sol. Energy* **2022**, *245*, 340–352. [CrossRef]
178. Lolli, N.; Andresen, I. Aerogel vs. argon insulation in windows: A greenhouse gas emissions analysis. *Build. Environ.* **2016**, *101*, 64–76. [CrossRef]
179. Gao, T.; Ihara, T.; Grynning, S.; Jelle, B.P.; Lien, A.G. Perspective of aerogel glazings in energy efficient buildings. *Build. Environ.* **2016**, *95*, 405–413. [CrossRef]
180. Huang, Y.; Niu, J. Energy and visual performance of the silica aerogel glazing system in commercial buildings of Hong Kong. *Constr. Build. Mater.* **2015**, *94*, 57–72. [CrossRef]
181. Olivieri, L.; Caamaño-Martín, E.; Moralejo-Vázquez, F.J.; Martín-Chivelet, N.; Olivieri, F.; Neila-González, F.J. Energy saving potential of semi-transparent photovoltaic elements for building integration. *Energy* **2014**, *76*, 572–583. [CrossRef]
182. Walsh, A.; Cóstola, D.; Labaki, L.C. Validation of the climatic zoning defined by ASHRAE Standard 169-2013. *Energy Policy* **2019**, *135*, 111016. [CrossRef]
183. Raihan, M.A.; Chattopadhyay, K.; Bhatia, A.; Garg, V.; Hussain, A.M. SGLSim: Tool for smart glazing energy performance analysis. *Energy Inform.* **2022**, *5*, 38. [CrossRef]
184. Rubel, A.; Joarder, M.A.R. Performance of shading against west glass facades to optimise daylight, thermal comfort and energy efficiency of office buildings. *J. Daylighting* **2024**, *11*, 131–147. [CrossRef]
185. Akram, M.W.; Hasannuzaman, M.; Cüce, E.; Cüce, P.M. Global technological advancement and challenges of glazed window, facade system and vertical greenery-based energy savings in buildings: A comprehensive review. *Energy Built Environ.* **2023**, *4*, 206–226. [CrossRef]
186. Khattabi, E.M.E.; Mharzi, M.; Zouini, M.; Valancius, K. Comparative study of various thermal analyses of smart windows in cubic building design. *J. Eng. Sci. Technol. Rev.* **2018**, *11*, 86–91. [CrossRef]
187. Magri, E.; Buhagiar, V.; Overend, M. The potential of smart glazing for occupant well-being and reduced energy load in a Central-Mediterranean climate. *KnE Soc. Sci.* **2019**, 534–546. [CrossRef]
188. Khaled, K.; Berardi, U. Characterization and energy performance of flexible photochromic window films. *E3S Web Conf.* **2023**, *396*, 04019. [CrossRef]
189. Oliveira, A.M.F.F.; Corvacho, H. Application of thermal comfort assessment models to indoor areas near glazed walls—Experimental evaluation. *Rev. Constr.* **2021**, *20*, 106–119. [CrossRef]
190. Pflug, T.; Nestle, N.; Kuhn, T.E.; Siroux, M.; Maeurer, C. Modeling of facade elements with switchable U-value. *Energy Build.* **2018**, *164*, 1–13. [CrossRef]

191. Calautit, J.K.; Sun, H.; Li, J.; Dik, A.; Mohammadi, M. Keeping it simple: Field testing and techno-economic assessment of a low-cost secondary quad glazing for enhanced energy efficiency in buildings. *Energy Built Env.* **2025**, *7*, 766–777. [CrossRef]
192. Chen, H.; Nie, J.; Liu, Y.; Li, Y. Theoretical study on impact of solar radiation heat gain on thermal comfort and energy efficiency in glass curtain wall buildings based on PMV index. *Buildings* **2025**, *15*, 2228. [CrossRef]
193. Fang, P.-L.; Chen, W.-A.; Hwang, R. An experimental investigation for the influence of glass type on indoor thermal comfort and cooling energy consumption. *CIB Conf.* **2025**, *1*, 1664. [CrossRef]
194. Gabriel, E.; Grigoletti, G.C.; Meller, G.; Zambonato, B. Daylight and energy performance of side lighting systems in an office room in a subtropical climate. *PARC Pesq. Arquit. Constr.* **2024**, *15*, e024003. [CrossRef]
195. Er, A.; Cakici, F.Z.; Selcuk, S.A. Increasing the energy performance of existing buildings in cold climate regions. *Period. Polytech. Archit.* **2023**, *54*, 120–132. [CrossRef]
196. Shaker, L.M. Evaluating UV-protective low-emissivity window films: Implications for thermal and visual comfort in luminous office buildings. *AUIQ Complement. Biol. Syst.* **2024**, *1*, 46–60. [CrossRef]
197. Tabbah, A.; Johansson, P.; Aries, M.M. Evaluating visual and beyond-vision light effects and energy consumption for luminous and temporal light factors: A single office case. *J. Daylighting* **2025**, *12*, 306–331. [CrossRef]
198. Krelling, A.F.; Jiang, Y.; Sun, K.; Lafrance, M.; Hong, T. High-performance windows improve thermal survivability of occupants during cold snaps. *Energy Built Env.* **2025**, *in press*. [CrossRef]
199. Cannavale, A.; Carlucci, F.; Fiorito, F.; Martellotta, F.; Ayr, U.; Berardi, U. Thermal enhancement of windows performance by means of innovative technologies. *E3S Web Conf.* **2021**, *312*, 02015. [CrossRef]
200. Souviron, J.; Van Moeseke, G.; Khan, A.Z. Analysing the environmental impact of windows: A review. *Build. Environ.* **2019**, *161*, 106268. [CrossRef]
201. Asdrubali, F.; Roncone, M.; Grazieschi, G. Embodied energy and embodied GWP of windows: A critical review. *Energies* **2021**, *14*, 3788. [CrossRef]
202. O'Neill, R.; Window, A.; Kenway, S.; Dargusch, P. Integrated operational and life-cycle modelling of energy, carbon and cost for building façades. *J. Clean. Prod.* **2021**, *286*, 125370. [CrossRef]
203. Gutai, M.; Mok, B.; Cavana, G.; Kheybari, A.G. Global carbon viability of glass technologies: Life-cycle assessment of standard, advanced and water-filled glass building envelopes. *Appl. Energy* **2024**, *367*, 123281. [CrossRef]
204. Westbroek, C.D.; Bitting, J.; Craglia, M.; Azevedo, J.M.C.; Cullen, J.M. Global material flow analysis of glass: From raw materials to end of life. *J. Ind. Ecol.* **2021**, *25*, 333–343. [CrossRef]
205. Geboes, E.; Galle, W.; De Temmerman, N. Make or break the loop: A cross-practitioners review of glass circularity. *Glass Struct. Eng.* **2023**, *8*, 193–210. [CrossRef]
206. Kawajiri, K.; Tahara, K.; Uemiya, S. Lifecycle assessment of critical material substitution: Indium tin oxide and aluminum zinc oxide in transparent electrodes. *Resour. Environ. Sustain.* **2022**, *7*, 100047. [CrossRef]
207. Ma, T.; Ma, M.; Guo, L.J. Optical Multilayer Thin Film Structure Inverse Design: From Optimization to Deep Learning. *arXiv* **2024**, arXiv:2409.17199.
208. Luce, A.; Mahdavi, A.; Wankerl, H.; Marquardt, F. Investigation of inverse design of multilayer thin-films with conditional invertible neural networks. *Mach. Learn. Sci. Technol.* **2023**, *4*, 015014. [CrossRef]
209. Schaible, J.; Oezdemir, A.K.; Debus, C.; Burger, S.; Streit, A.; Becker, C.; Jaeger, K.; Goetz, M. Inverse Design of Optical Multilayer Thin Films using Robust Masked Diffusion Models. *arXiv* **2026**, arXiv:2604.01106.
210. Jung, U.; Jang, D.; Kim, S.; Kim, J. Improving the performance of optical inverse design of multilayer thin films using CNN-LSTM tandem neural networks. *arXiv* **2025**, arXiv:2506.10044.
211. Molesky, S.; Lin, Z.; Piggott, A.Y.; Jin, W.; Vuckovic, J.; Rodriguez, A.W. Inverse design in nanophotonics. *Nat. Photonics* **2018**, *12*, 659–670. [CrossRef]
212. Cornil, D.; Wiame, H.; Lecomte, B.; Cornil, J.; Beljonne, D. Which Oxide for Low-Emissivity Glasses? First-Principles Modeling of Silver Adhesion. *ACS Appl. Mater. Interfaces* **2017**, *9*, 18346–18354. [CrossRef] [PubMed]
213. Wang, B.; Xu, J.; Lee, B.; Huang, K.; Wang, W.; Peng, H.; Xu, J. Machine learning assisted CALPHAD framework for thermodynamic analysis of CVD SiOxNy thin films. *Calphad* **2025**, *88*, 102806. [CrossRef]
214. Youngblood, N.; Talagrand, C.; Porter, B.; Galante, C.G.; Kneepkens, S.; Sarwat, S.G.; Yarmolich, D.; Bonilla, R.S.; Hosseini, P.; Taylor, R.A.; et al. Reconfigurable Low-Emissivity Optical Coating Using Ultrathin Phase Change Materials. *ACS Photonics* **2022**, *9*, 3337–3344. [CrossRef]
215. Llordes, A.; Garcia, G.; Gazquez, J.; Milliron, D.J. Tunable near-infrared and visible-light transmittance in nanocrystal-in-glass composites. *Nature* **2013**, *500*, 323–326. [CrossRef] [PubMed]
216. Runnerstrom, E.L.; Llordes, A.; Lounis, S.D.; Milliron, D.J. Nanostructured electrochromic smart windows: Traditional materials and NIR-selective plasmonic nanocrystals. *Chem. Commun.* **2014**, *50*, 10555–10572. [CrossRef] [PubMed]
217. Kamalisarvestani, M.; Saidur, R.; Mekhilef, S.; Javadi, F.S. Performance, materials and coating technologies of thermochromic thin films on smart windows. *Renew. Sustain. Energy Rev.* **2013**, *26*, 353–364. [CrossRef]

218. Ke, Y.; Zhou, C.; Zhou, Y.; Wang, S.; Chan, S.H.; Long, Y. Emerging thermal-responsive materials and integrated techniques targeting the energy-efficient smart window application. *Adv. Funct. Mater.* **2018**, *28*, 1800113. [CrossRef]
219. Casini, M. Active dynamic windows for buildings: A review. *Renew. Energy* **2018**, *119*, 923–934. [CrossRef]
220. Traverse, C.J.; Pandey, R.; Barr, M.C.; Lunt, R.R. Emergence of highly transparent photovoltaics for distributed applications. *Nat. Energy* **2017**, *2*, 849–860. [CrossRef]
221. Lunt, R.R.; Bulovic, V. Transparent, near-infrared organic photovoltaic solar cells for window and energy-scavenging applications. *Appl. Phys. Lett.* **2011**, *98*, 113305. [CrossRef]
222. Tripathy, M.; Sadhu, P.K.; Panda, S.K. A critical review on building integrated photovoltaic products and their applications. *Renew. Sustain. Energy Rev.* **2016**, *61*, 451–465. [CrossRef]
223. Yang, R.J.; Zou, P.X.W. Building integrated photovoltaics (BIPV): Costs, benefits, risks, barriers and improvement strategy. *Int. J. Constr. Manag.* **2016**, *16*, 39–53. [CrossRef]
224. Davy, N.C.; Sezen-Edmonds, M.; Gao, J.; Lin, X.; Liu, A.; Yao, N.; Kahn, A.; Loo, Y.-L. Pairing of near-ultraviolet solar cells with electrochromic windows for smart management of the solar spectrum. *Nat. Energy* **2017**, *2*, 17104. [CrossRef]
225. Chen, L.; Yang, J.; Li, P. Modelling the effect of BIPV window in the built environment: Uncertainty and sensitivity. *Build. Environ.* **2022**, *208*, 108605. [CrossRef]
226. Yu, Q.; Dong, K.; Guo, Z.; Li, J.; Tan, H.; Jin, Y.; Yuan, J.; Zhang, H.; Liu, J.; Chen, Q.; et al. Global Estimation of Building-Integrated Facade and Rooftop Photovoltaic Potential by Integrating 3D Building Footprint and Spatio-Temporal Datasets. *arXiv* **2024**, arXiv:2412.01291.
227. Intal, D.; Ebong, A.U. Thin-Film Solar Photovoltaics: Trends and Future Directions. *Renew. Sustain. Energy Rev.* **2026**, *226*, 116464. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.